

Code

AUTHOR

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Survival Analysis of Socialism Countries

```
# 1. Load Required Packages
```

```
library(survival)
library(survminer)
```

Loading required package: ggplot2

Loading required package: ggpubr

Attaching package: 'survminer'

The following object is masked from 'package:survival':

myeloma

```
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr	1.2.0	✓ readr	2.2.0
✓ forcats	1.0.1	✓ stringr	1.6.0
✓ lubridate	1.9.5	✓ tibble	3.3.1
✓ purrr	1.2.1	✓ tidyr	1.3.2

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

! Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
library(broom)
library(readr)
library(lme4)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Registered S3 method overwritten by 'lme4':

method	from
na.action.merMod	car

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

```
select
```

```
# 2. Data Loading and Preprocessing
```

```
static_data <- read.csv("D:/paper/data.csv", stringsAsFactors = FALSE) %>%  
  rename(country = Countries) %>%  
  mutate(  
    country = as.character(country),  
  
    country = case_when(  
      country == "Czech Republic" ~ "Czech",  
      country == "Bosnia and Herzegovina" ~ "BAH",  
      country == "North Korea" ~ "NK",  
      country == "East Germany" ~ "Eg",  
      TRUE ~ country  
    ),  
  
    Survivaltime_numeric = as.numeric(gsub("\\+", "", Survivaltime)),  
  
    event = ifelse(!is.na(Endpoint), 1, 0),  
  
    survival_time = ifelse(event == 1, Endpoint - 1985, Survivaltime_numeric)  
  )  
  
cat("=== Static Data Structure ===\n")
```

```
=== Static Data Structure ===
```

```
print(str(static_data))
```

```
'data.frame': 39 obs. of 16 variables:  
 $ country      : chr  "China" "NK" "Mongolia" "Kyrgyzstan" ...  
 $ Region       : chr  " East Asia" " East Asia" " East Asia" "Central Asia" ...  
 $ Endpoint     : int  NA NA 1992 1991 1991 1991 1991 1991 1989 NA ...  
 $ Survivaltime : chr  "37+" "37+" "7" "6" ...  
 $ Regionfmt    : int  5 5 5 4 4 4 4 4 6 6 ...  
 $ WTO          : int  0 0 0 1 1 1 1 1 0 0 ...  
 $ Distance     : num  5796 6423 4265 3003 3169 ...  
 $ Capital      : chr  "Beijing" "Pyongyang" "Ulan Bator" "Bishkek" ...  
 $ Reform       : int  1 0 0 1 1 1 1 1 0 1 ...  
 $ Capitalism   : int  1912 NA NA 1861 1861 1861 1861 1861 1953 1953 ...  
 $ Socialism    : int  1949 1949 1921 1921 1924 1924 1921 1921 1975 1975 ...  
 $ Capitalismtime : int  37 0 0 60 63 63 60 60 22 22 ...  
 $ Union        : int  0 0 0 1 1 1 1 1 0 0 ...  
 $ Survivaltime_numeric: num  37 37 7 6 6 6 6 6 4 37 ...
```

```
$ event          : num  0 0 1 1 1 1 1 1 1 0 ...
$ survival_time  : num  37 37 7 6 6 6 6 6 4 37 ...
NULL
```

```
cat("Number of countries:", nrow(static_data), "\n")
```

Number of countries: 39

```
cat("Number of events:", sum(static_data$event, na.rm = TRUE), "\n")
```

Number of events: 34

```
time_varying_data <- read.csv("D:/paper/dataset.csv", stringsAsFactors = FALSE)
```

```
cat("\n=== Time-varying Data Structure ===\n")
```

=== Time-varying Data Structure ===

```
print(str(time_varying_data[, 1:50]))
```

```
'data.frame':  39 obs. of  50 variables:
 $ Year          : int   1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 ...
 $ Survivaltime  : int    0  1  2  3  4  5  6  7  8  9 ...
 $ Chinapop     : int  1051040 1066790 1084035 1101630 1118650 1135185 1149979 1163363
1176727 1189962 ...
 $ NKpop        : int  18831 19137 19449 19769 20103 20451 20813 21182 21563 21852 ...
 $ Mongoliapop  : int  1859 1903 1947 1992 2070 2121 2170 2210 NA NA ...
 $ Kyrgyzstanpop : int  4006 4089 4176 4244 4302 4382 4446 NA NA NA ...
 $ Kazakhstanpop : int  15790 15946 16109 16207 16265 16398 16522 NA NA NA ...
 $ Tajikistanpop : int  4569 4716 4865 5006 5148 5272 5407 NA NA NA ...
 $ Turkmenistanpop : int  3240 3324 3411 3492 3572 3658 3745 NA NA NA ...
 $ Uzbekistanpop : int  18215 18718 19219 19625 20033 20530 21059 NA NA NA ...
 $ Cambodiapop  : int  7822 8079 8380 8692 9014 NA NA NA NA NA ...
 $ Laospop      : int  3657 3762 3869 3980 4094 4210 4331 4454 4581 4712 ...
 $ Myanmarpop   : int  36346 36998 37625 38196 NA NA NA NA NA NA ...
 $ Vietnampop   : int  60093 61440 62826 64211 65868 67283 68675 70032 71357 72654 ...
 $ Armeniapop   : int  3374 3422 3469 3516 3326 3377 3434 NA NA NA ...
 $ Azerbaijanpop : int  6682 6776 6874 6976 7102 7200 7300 NA NA NA ...
 $ Georgiapop   : int  5193 5219 5240 5320 5398 5426 5426 NA NA NA ...
 $ Iraqpop      : int  15694 16247 16543 17038 17568 18135 17474 17863 18404 18968 ...
 $ Syriapop     : int  10481 10857 11243 11632 12018 12436 12867 13243 13615 13989 ...
 $ Bulgariapop  : int  8944 8959 8972 8982 8990 NA NA NA NA NA ...
 $ Belaruspop   : int  9982 10044 10097 10150 10184 10201 10206 NA NA NA ...
 $ Czechpop     : int  10310 10309 10312 10314 10314 NA NA NA NA NA ...
 $ Hungarypop   : int  10649 10631 10613 10443 10398 NA NA NA NA NA ...
 $ Moldovapop   : int  4148 4183 4217 4290 4359 4394 4401 NA NA NA ...
 $ Polandpop    : int  37226 37504 37741 37867 37970 NA NA NA NA NA ...
 $ Romaniapop   : int  22521 22600 22686 22769 22852 NA NA NA NA NA ...
 $ Russiapop    : int  143938 144967 145959 146866 147352 147973 148398 NA NA NA ...
 $ Slovakiapop  : int  5145 5172 5199 5223 5245 NA NA NA NA NA ...
 $ Ukrainepop   : int  50944 51095 51218 51423 51501 51622 51730 NA NA NA ...
```

```

$ Estoniapop      : int  1538 1550 1562 1569 1571 1569 1560 NA NA NA ...
$ Lithuaniapop    : int  3587 3620 3653 3674 3681 3695 NA NA NA NA ...
$ Latviapop       : int  2606 2627 2649 2664 2667 2664 2651 NA NA NA ...
$ Albaniapop      : int  2957 3015 3075 3137 3196 3251 3187 3078 NA NA ...
$ BAHpop          : int  4275 4308 4339 4370 4398 4424 4437 4233 NA NA ...
$ Serbiapop       : int  7720 7740 7756 7769 7780 7786 7685 7646 NA NA ...
$ Croatiapop      : int  4458 4472 4484 4494 4501 4508 NA NA NA NA ...
$ Montenegropop   : int  561 565 569 574 579 583 591 622 NA NA ...
$ Macedoniapop    : int  1845 1851 1856 1859 1863 1861 1868 NA NA NA ...
$ Sloveniapop     : int  1914 1932 1946 1962 1976 NA NA NA NA NA ...
$ Egpop           : int  16660 16630 16590 16560 16430 NA NA NA NA NA ...
$ Cubapop         : int  10084 10183 10279 10382 10489 10597 10697 10780 10850 10893 ...
$ Chinagdppc      : int  2619 2748 2931 3045 3022 2982 3051 3228 3506 3730 ...
$ NKgdppc         : int  2415 2689 2796 2553 2710 2456 2316 2121 1996 1926 ...
$ Mongoliagdppc   : int  2004 2142 2168 2227 2232 2123 1976 1841 NA NA ...
$ Kyrgyzstangdppc : int  5616 5413 5241 5679 5719 5742 5162 NA NA NA ...
$ Kazakhstangdppc : int  11545 11576 11389 12256 12153 11888 10557 NA NA NA ...
$ Tajikistangdppc : int  5364 5319 5040 5515 4962 4803 4270 NA NA NA ...
$ Turkmenistangdppc : int  6186 6204 6194 6601 5915 5796 5458 NA NA NA ...
$ Uzbekistangdppc : int  7181 6904 6680 7101 7093 6790 6486 NA NA NA ...
$ Cambodiagdppc   : int  1540 1578 1490 1473 1455 NA NA NA NA NA ...

```

NULL

```
cat("Year range:", range(time_varying_data$Year, na.rm = TRUE), "\n")
```

Year range: 1985 2022

3. Improved Data Reshaping Function

```

get_country_codes <- function(df) {
  pop_columns <- names(df)[grepl("pop$", names(df))]
  country_codes <- gsub("pop$", "", pop_columns)
  return(country_codes)
}

country_codes <- get_country_codes(time_varying_data)
cat("Detected country codes:", paste(country_codes, collapse = ", "), "\n")

```

Detected country codes: China, NK, Mongolia, Kyrgyzstan, Kazakhstan, Tajikistan, Turkmenistan, Uzbekistan, Cambodia, Laos, Myanmar, Vietnam, Armenia, Azerbaijan, Georgia, Iraq, Syria, Bulgaria, Belarus, Czech, Hungary, Moldova, Poland, Romania, Russia, Slovakia, Ukraine, Estonia, Lithuania, Latvia, Albania, BAH, Serbia, Croatia, Montenegro, Macedonia, Slovenia, Eg, Cuba

```

reshape_to_long_format <- function(df, country_codes, static_info) {
  long_data <- data.frame()

  for (country_code in country_codes) {
    cat("Processing country:", country_code, "\n")

    country_static <- static_info %>% filter(country == country_code)

    if (nrow(country_static) == 0) {

```

```

    cat(" Warning: Country", country_code, "not found in static data\n")
  next
}

end_year <- ifelse(!is.na(country_static$Endpoint),
                  country_static$Endpoint,
                  max(df$Year, na.rm = TRUE))

country_df <- df %>%
  dplyr::select(Year, Suvivaltime,
               pop = paste0(country_code, "pop"),
               gdpir = paste0(country_code, "gdpir"),
               gdp = paste0(country_code, "gdp")) %>%

  filter(Year <= end_year) %>%
  mutate(
    country = country_code,

    pop = as.numeric(pop),

    gdpir = as.numeric(gsub("%", "", gdpir)) / 100,
    gdp = as.numeric(gdp)
  ) %>%

  filter(!is.na(pop) | !is.na(gdpir) | !is.na(gdp))

if (nrow(country_df) > 0) {
  long_data <- bind_rows(long_data, country_df)
} else {
  cat(" Warning:", country_code, "has no valid data\n")
}
}

return(long_data)
}

long_data <- reshape_to_long_format(time_varying_data, country_codes, static_data)

```

Processing country: China
 Processing country: NK
 Processing country: Mongolia
 Processing country: Kyrgyzstan
 Processing country: Kazakhstan
 Processing country: Tajikistan
 Processing country: Turkmenistan
 Processing country: Uzbekistan
 Processing country: Cambodia
 Processing country: Laos
 Processing country: Myanmar
 Processing country: Vietnam
 Processing country: Armenia
 Processing country: Azerbaijan
 Processing country: Georgia
 Processing country: Iraq

Processing country: Syria
Processing country: Bulgaria
Processing country: Belarus
Processing country: Czech
Processing country: Hungary
Processing country: Moldova
Processing country: Poland
Processing country: Romania
Processing country: Russia
Processing country: Slovakia
Processing country: Ukraine
Processing country: Estonia
Processing country: Lithuania
Processing country: Latvia
Processing country: Albania
Processing country: BAH
Processing country: Serbia
Processing country: Croatia
Processing country: Montenegro
Processing country: Macedonia
Processing country: Slovenia
Processing country: Eg
Processing country: Cuba

```
cat("Transformed long-format data:\n")
```

Transformed long-format data:

```
print(str(long_data))
```

```
'data.frame':  431 obs. of  6 variables:
 $ Year      : int  1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 ...
 $ Suvaltime: int   0  1  2  3  4  5  6  7  8  9 ...
 $ pop       : num 1051040 1066790 1084035 1101630 1118650 ...
 $ gdpir     : num  0.0863 0.065 0.0838 0.0558 0.0078 0.0013 0.0366 0.0701 0.0988 0.0758 ...
 $ gdp       : num  2.75e+09 2.93e+09 3.18e+09 3.35e+09 3.38e+09 ...
 $ country   : chr  "China" "China" "China" "China" ...
NULL
```

```
cat("Total observations:", nrow(long_data), "\n")
```

Total observations: 431

```
cat("Countries included:", paste(unique(long_data$country), collapse = ", "), "\n")
```

Countries included: China, NK, Mongolia, Kyrgyzstan, Kazakhstan, Tajikistan, Turkmenistan, Uzbekistan, Cambodia, Laos, Myanmar, Vietnam, Armenia, Azerbaijan, Georgia, Iraq, Syria, Bulgaria, Belarus, Czech, Hungary, Moldova, Poland, Romania, Russia, Slovakia, Ukraine, Estonia, Lithuania, Latvia, Albania, BAH, Serbia, Croatia, Montenegro, Macedonia, Slovenia, Eg, Cuba

```
# 4. Data Merging and Cleaning
```

```
merged_data <- long_data %>%  
  left_join(static_data, by = "country") %>%  
  
  arrange(country, Year) %>%  
  group_by(country) %>%  
  mutate(  
  
    log_pop = log(pop + 1),  
  
    time_from_start = Year - 1985,  
  
    event_year = ifelse(Year == Endpoint & !is.na(Endpoint), 1, 0)  
  ) %>%  
  ungroup() %>%  
  
  filter(!is.na(survival_time))
```

```
# 4.1 Detailed Data Quality Check
```

```
cat("=== Detailed Data Quality Check ===\n")
```

```
=== Detailed Data Quality Check ===
```

```
missing_summary <- merged_data %>%  
  summarise(across(everything(), ~sum(is.na(.)))) %>%  
  pivot_longer(everything(), names_to = "Variable", values_to = "Missing_Count") %>%  
  filter(Missing_Count > 0)  
  
print(missing_summary)
```

```
# A tibble: 2 × 2
```

	Variable	Missing_Count
	<chr>	<int>
1	Endpoint	190
2	Capitalism	46

```
cat("\n=== Key Variable Distributions ===\n")
```

```
=== Key Variable Distributions ===
```

```
cat("Survival time summary:\n")
```

```
Survival time summary:
```

```
print(summary(merged_data$survival_time))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
3.00	6.00	15.00	20.36	37.00	37.00

```
cat("\nEvent variable distribution:\n")
```

Event variable distribution:

```
print(table(merged_data$event))
```

```
  0   1
190 241
```

```
cat("\nReform status distribution:\n")
```

Reform status distribution:

```
print(table(merged_data$Reform))
```

```
  0   1
213 218
```

```
cat("\nUnion status distribution:\n")
```

Union status distribution:

```
print(table(merged_data$Union))
```

```
  0   1
275 156
```

```
cat("\nGDP Growth Rate (gdpir) distribution:\n")
```

GDP Growth Rate (gdpir) distribution:

```
print(summary(merged_data$gdpir))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.62650	-0.01225	0.02350	0.01625	0.06335	0.37730

```
cat("\nNumber of countries:", n_distinct(merged_data$country), "\n")
```

Number of countries: 39


```
cat("Total observations:", nrow(merged_data), "\n")
```

Total observations: 431

```
model_vars <- c("WTO", "Distance", "Reform", "Union",  
               "pop", "gdpir", "log_pop",  
               "survival_time", "event")  
  
cat("\n=== Missing Values for Modeling Variables ===\n")
```

=== Missing Values for Modeling Variables ===

```
for (var in model_vars) {  
  missing_count <- sum(is.na(merged_data[[var]]))  
  cat(sprintf("%-15s: %d missing values (%.1f%%)\n",  
              var, missing_count, missing_count/nrow(merged_data)*100))  
}
```

```
WTO           : 0 missing values (0.0%)  
Distance      : 0 missing values (0.0%)  
Reform        : 0 missing values (0.0%)  
Union         : 0 missing values (0.0%)  
pop           : 0 missing values (0.0%)  
gdpir         : 0 missing values (0.0%)  
log_pop       : 0 missing values (0.0%)  
survival_time : 0 missing values (0.0%)  
event         : 0 missing values (0.0%)
```

```
complete_cases <- complete.cases(merged_data[, model_vars])  
cat(sprintf("\nComplete observations: %d/%d (%.1f%%)\n",  
            sum(complete_cases), nrow(merged_data),  
            sum(complete_cases)/nrow(merged_data)*100))
```

Complete observations: 431/431 (100.0%)

```
merged_data_complete <- merged_data[complete_cases, ]  
  
cat("Cleaned data:\n")
```

Cleaned data:

```
cat("Sample size:", nrow(merged_data_complete), "\n")
```

Sample size: 431

```
cat("Number of countries:", n_distinct(merged_data_complete$country), "\n")
```

Number of countries: 39

```
cat("Number of events:", sum(merged_data_complete$event), "\n")
```

Number of events: 241

```
# 4.2 Intraclass Correlation Coefficient (ICC) Calculation
```

```
cat("\n=== Intraclass Correlation Coefficient (ICC) Analysis ===\n")
```

=== Intraclass Correlation Coefficient (ICC) Analysis ===

```
if (n_distinct(merged_data_complete$country) > 1) {

  tryCatch({
    gdpir_model <- lmer(gdpir ~ 1 + (1|country), data = merged_data_complete)

    var_components <- VarCorr(gdpir_model)
    country_variance <- as.data.frame(var_components)$vcov[1]
    residual_variance <- sigma(gdpir_model)^2

    icc_gdpir <- country_variance / (country_variance + residual_variance)

    cat("ICC for GDP Growth Rate (gdpir):\n")
    cat("Between-country variance:", round(country_variance, 4), "\n")
    cat("Within-country variance:", round(residual_variance, 4), "\n")
    cat("ICC:", round(icc_gdpir, 4), "\n")

    if (icc_gdpir < 0.05) {
      cat("Interpretation: ICC < 0.05 indicates weak within-cluster correlation.\n")
    } else if (icc_gdpir < 0.10) {
      cat("Interpretation: ICC between 0.05-0.10 indicates moderate within-cluster correlation")
    } else {
      cat("Interpretation: ICC ≥ 0.10 indicates strong within-cluster correlation. Clustered s")
    }
  }, error = function(e) {
    cat("Failed to calculate ICC for gdpir:", e$message, "\n")
  })

  tryCatch({
    log_pop_model <- lmer(log_pop ~ 1 + (1|country), data = merged_data_complete)

    var_components <- VarCorr(log_pop_model)
    country_variance <- as.data.frame(var_components)$vcov[1]
    residual_variance <- sigma(log_pop_model)^2

    icc_log_pop <- country_variance / (country_variance + residual_variance)

    cat("\nICC for Log Population (log_pop):\n")
    cat("Between-country variance:", round(country_variance, 4), "\n")
    cat("Within-country variance:", round(residual_variance, 4), "\n")
    cat("ICC:", round(icc_log_pop, 4), "\n")
  })
}
```

```

if (icc_log_pop < 0.05) {
  cat("Interpretation: ICC < 0.05 indicates weak within-cluster correlation.\n")
} else if (icc_log_pop < 0.10) {
  cat("Interpretation: ICC between 0.05-0.10 indicates moderate within-cluster correlation\n")
} else {
  cat("Interpretation: ICC ≥ 0.10 indicates strong within-cluster correlation. Clustered s
}
}, error = function(e) {
  cat("Failed to calculate ICC for log_pop:", e$message, "\n")
})

cat("\n=== ICC Analysis Summary ===\n")
icc_vars <- c()
if (exists("icc_gdpir")) icc_vars <- c(icc_vars, icc_gdpir)
if (exists("icc_log_pop")) icc_vars <- c(icc_vars, icc_log_pop)

if (length(icc_vars) > 0) {
  avg_icc <- mean(icc_vars, na.rm = TRUE)
  cat("Average ICC across key variables:", round(avg_icc, 4), "\n")

  if (avg_icc >= 0.05) {
    cat("Conclusion: ICC ≥ 0.05 indicates non-negligible within-country correlation.\n")
    cat("Recommendation: Use clustered Cox models with country-level clustering.\n")
  } else {
    cat("Conclusion: ICC < 0.05 indicates negligible within-country correlation.\n")
    cat("Note: While independent assumption might be acceptable, we proceed with clustered m
  }
}
}

```

ICC for GDP Growth Rate (gdpir):

Between-country variance: 7e-04

Within-country variance: 0.0048

ICC: 0.1198

Interpretation: ICC ≥ 0.10 indicates strong within-cluster correlation. Clustered standard errors are recommended.

ICC for Log Population (log_pop):

Between-country variance: 1.9571

Within-country variance: 0.0104

ICC: 0.9947

Interpretation: ICC ≥ 0.10 indicates strong within-cluster correlation. Clustered standard errors are recommended.

=== ICC Analysis Summary ===

Average ICC across key variables: 0.5573

Conclusion: ICC ≥ 0.05 indicates non-negligible within-country correlation.

Recommendation: Use clustered Cox models with country-level clustering.

```
cat("\n=== Descriptive Statistics at Baseline (1985) by Reform Status ===\n")
```

=== Descriptive Statistics at Baseline (1985) by Reform Status ===

```
baseline_data <- merged_data_complete %>%  
  filter(Year == 1985) %>%  
  group_by(country) %>%  
  slice(1) %>%  
  ungroup()  
  
cat("\nNumber of countries at baseline:", nrow(baseline_data), "\n")
```

Number of countries at baseline: 39

```
cat("\n--- Categorical Variables ---\n")
```

--- Categorical Variables ---

```
cat("\nWTO Membership (WTO):\n")
```

WTO Membership (WTO):

```
wto_table <- table(Reform = baseline_data$Reform, WTO = baseline_data$WTO)  
print(wto_table)
```

	WTO	
Reform	0	1
0	14	7
1	3	15

```
cat("\nWTO Membership percentages by Reform (row %):\n")
```

WTO Membership percentages by Reform (row %):

```
wto_prop <- prop.table(wto_table, margin = 1) * 100  
print(round(wto_prop, 1))
```

	WTO	
Reform	0	1
0	66.7	33.3
1	16.7	83.3

```
cat("\nOverall WTO Membership:\n")
```

Overall WTO Membership:

```
print(table(baseline_data$WTO))
```

```
0 1  
17 22
```

```
cat("\nUnion Status (Union):\n")
```

Union Status (Union):

```
union_table <- table(Reform = baseline_data$Reform, Union = baseline_data$Union)  
print(union_table)
```

```
      Union  
Reform 0 1  
0 13 8  
1 3 15
```

```
cat("\nUnion percentages by Reform (row %):\n")
```

Union percentages by Reform (row %):

```
union_prop <- prop.table(union_table, margin = 1) * 100  
print(round(union_prop, 1))
```

```
      Union  
Reform 0 1  
0 61.9 38.1  
1 16.7 83.3
```

```
cat("\nOverall Union Status:\n")
```

Overall Union Status:

```
print(table(baseline_data$Union))
```

```
0 1  
16 23
```

```
cat("\n--- Continuous Variables: Mean (SD), Median [Min, Max] ---\n")
```

--- Continuous Variables: Mean (SD), Median [Min, Max] ---

```
continuous_vars <- c("Distance", "gdpir", "pop", "log_pop")

for (var in continuous_vars) {
  cat("\n", var, ":\n", sep = "")

  stats_by_group <- baseline_data %>%
    group_by(Reform) %>%
    summarise(
      N = n(),
      Mean = mean(.data[[var]], na.rm = TRUE),
      SD = sd(.data[[var]], na.rm = TRUE),
      Median = median(.data[[var]], na.rm = TRUE),
      Min = min(.data[[var]], na.rm = TRUE),
      Max = max(.data[[var]], na.rm = TRUE)
    ) %>%
    mutate(across(where(is.numeric), ~ round(., 3)))
  print(stats_by_group)

  cat("Overall:\n")
  stats_overall <- baseline_data %>%
    summarise(
      N = n(),
      Mean = mean(.data[[var]], na.rm = TRUE),
      SD = sd(.data[[var]], na.rm = TRUE),
      Median = median(.data[[var]], na.rm = TRUE),
      Min = min(.data[[var]], na.rm = TRUE),
      Max = max(.data[[var]], na.rm = TRUE)
    ) %>%
    mutate(across(where(is.numeric), ~ round(., 3)))
  print(stats_overall)
}
```

Distance:

A tibble: 2 × 7

	Reform	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	21	2902.	2312.	1878.	1156.	9088.
2	1	18	2441.	2007.	1902.	0	6525.

Overall:

A tibble: 1 × 6

	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	39	2690.	2161.	1878.	0	9088.

gdpir:

A tibble: 2 × 7

	Reform	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	21	0.019	0.035	0.013	-0.03	0.13
2	1	18	0.013	0.043	0.011	-0.098	0.086

Overall:

A tibble: 1 × 6

	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	39	0.016	0.039	0.011	-0.098	0.13

pop:

A tibble: 2 × 7

	Reform	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	21	11252.	10367.	8944	561	37226
2	1	18	77367.	245517.	4881	1538	1051040

Overall:

A tibble: 1 × 6

	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	39	41767.	167744.	7720	561	1051040

log_pop:

A tibble: 2 × 7

	Reform	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	21	8.88	1.07	9.10	6.33	10.5
2	1	18	9.25	1.68	8.49	7.34	13.9

Overall:

A tibble: 1 × 6

	N	Mean	SD	Median	Min	Max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	39	9.05	1.38	8.95	6.33	13.9

```
surv_obj <- Surv(merged_data_complete$survival_time, merged_data_complete$event)
cat("=== Survival Data Summary ===\n")
```

=== Survival Data Summary ===

```
print(surv_obj)
```

```
[1] 7 7 7 7 7 7 7 7 6 6 6 6 6 6 6 6 6
[19] 6 6 6 6 7 7 7 7 7 7 7 6 6 6 6 6 6
[37] 6 4 4 4 4 4 4 4 4 4 4 37+ 37+ 37+ 37+ 37+ 37+
[55] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
[73] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 5 5 5 5 5
[91] 5 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
[109] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
[127] 37+ 37+ 37+ 4 4 4 4 4 4 4 4 4 4 6 6 6 6 6
[145] 6 6 6 6 6 6 6 6 6 4 4 4 4 4 18 18 18 18
[163] 18 18 18 18 18 18 18 18 18 18 18 18 18 18 6 6 6
[181] 6 6 6 6 6 6 6 6 6 6 6 37+ 37+ 37+ 37+ 37+ 37+
[199] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
[217] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 6 6 6 6 6
[235] 6 6 5 5 5 5 5 5 6 6 6 6 6 6 6 6 6
[253] 6 6 6 6 7 7 7 7 7 7 7 7 7 7 7 7 7
[271] 7 7 3 3 3 3 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
```

```
[289] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
[307] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 4 4 4 4 4 4 4 4 4
[325] 6 6 6 6 6 6 6 7 7 7 7 7 7 7 4 4 4
[343] 4 4 4 4 4 4 4 15 15 15 15 15 15 15 15 15 15
[361] 15 15 15 15 15 6 6 6 6 6 6 6 6 6 6 6 6
[379] 6 6 6 6 6 6 6 6 6 6 6 6 6 6 6 37+ 37+ 37+
[397] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
[415] 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+ 37+
```

```
if (n_distinct(merged_data_complete$country) > 0) {

  survival_plot_data <- merged_data_complete %>%
    group_by(country) %>%
    slice_tail(n = 1) %>%
    ungroup()

  fit_reform <- survfit(Surv(survival_time, event) ~ Reform, data = survival_plot_data)

  cat("=== Survival Curves by Reform Status ===\n")
  print(summary(fit_reform))
}
```

=== Survival Curves by Reform Status ===

Call: survfit(formula = Surv(survival_time, event) ~ Reform, data = survival_plot_data)

```
Reform=0
time n.risk n.event survival std.err lower 95% CI upper 95% CI
3    21      1  0.9524  0.0465  0.8655  1.000
4    20      9  0.5238  0.1090  0.3484  0.788
5    11      1  0.4762  0.1090  0.3041  0.746
6    10      1  0.4286  0.1080  0.2615  0.702
7     9      5  0.1905  0.0857  0.0789  0.460
15    4      1  0.1429  0.0764  0.0501  0.407
18    3      1  0.0952  0.0641  0.0255  0.356
```

```
Reform=1
time n.risk n.event survival std.err lower 95% CI upper 95% CI
5    18      1  0.944  0.0540  0.8443  1.000
6    17     14  0.167  0.0878  0.0593  0.468
```

6. Cox Univariate Analysis

```
cat("=== Cox Univariate Analysis ===\n")
```

=== Cox Univariate Analysis ===

```
if (nrow(merged_data_complete) > 0 && sum(merged_data_complete$event) > 0) {

  univariate_vars <- c("Reform", "WTO", "Distance", "Union", "gdpir", "log_pop")

  univariate_results <- data.frame()

  for (var in univariate_vars) {
```



```

cat("\nPerforming univariate analysis - Variable:", var, "\n")

tryCatch({

  formula_str <- paste("Surv(survival_time, event) ~", var)
  cox_univariate <- coxph(as.formula(formula_str), data = merged_data_complete)

  univariate_summary <- summary(cox_univariate)

  result_row <- data.frame(
    Variable = var,
    HR = round(univariate_summary$coefficients[1, 2], 3),
    CI_lower = round(univariate_summary$conf.int[1, 3], 3),
    CI_upper = round(univariate_summary$conf.int[1, 4], 3),
    p_value = round(univariate_summary$coefficients[1, 5], 4),
    stringsAsFactors = FALSE
  )

  result_row$significance <- case_when(
    result_row$p_value < 0.001 ~ "***",
    result_row$p_value < 0.01 ~ "**",
    result_row$p_value < 0.05 ~ "*",
    TRUE ~ ""
  )

  result_row$HR_CI <- sprintf("%.3f (%.3f-%.3f)%s",
                              result_row$HR, result_row$CI_lower,
                              result_row$CI_upper, result_row$significance)

  univariate_results <- bind_rows(univariate_results, result_row)

  cat("Univariate analysis result:\n")
  print(result_row)

}, error = function(e) {
  cat("Univariate analysis failed:", e$message, "\n")
})

cat("\n=== Complete Univariate Analysis Results ===\n")
print(univariate_results)

} else {
  cat("Insufficient data or no events, cannot perform univariate analysis\n")
}

```

Performing univariate analysis - Variable: Reform

Univariate analysis result:

Variable	HR	CI_lower	CI_upper	p_value	significance	HR_CI
----------	----	----------	----------	---------	--------------	-------

```
1 Reform 0.708 0.548 0.914 0.0081 ** 0.708 (0.548-0.914)**
```

Performing univariate analysis - Variable: WTO

Univariate analysis result:

```
Variable    HR CI_lower CI_upper p_value significance
1    WTO 31.306 19.914 49.215 0 ***
      HR_CI
1 31.306 (19.914-49.215)***
```

Performing univariate analysis - Variable: Distance

Univariate analysis result:

```
Variable    HR CI_lower CI_upper p_value significance HR_CI
1 Distance 0.999 0.999 0.999 0 *** 0.999 (0.999-0.999)***
```

Performing univariate analysis - Variable: Union

Univariate analysis result:

```
Variable    HR CI_lower CI_upper p_value significance
1    Union 15.394 10.757 22.03 0 ***
      HR_CI
1 15.394 (10.757-22.030)***
```

Performing univariate analysis - Variable: gdpir

Univariate analysis result:

```
Variable    HR CI_lower CI_upper p_value significance HR_CI
1    gdpir 0.048 0.017 0.135 0 *** 0.048 (0.017-0.135)***
```

Performing univariate analysis - Variable: log_pop

Univariate analysis result:

```
Variable    HR CI_lower CI_upper p_value significance HR_CI
1 log_pop 0.618 0.56 0.683 0 *** 0.618 (0.560-0.683)***
```

=== Complete Univariate Analysis Results ===

```
Variable    HR CI_lower CI_upper p_value significance
1 Reform 0.708 0.548 0.914 0.0081 **
2    WTO 31.306 19.914 49.215 0.0000 ***
3 Distance 0.999 0.999 0.999 0.0000 ***
4    Union 15.394 10.757 22.030 0.0000 ***
5    gdpir 0.048 0.017 0.135 0.0000 ***
6 log_pop 0.618 0.560 0.683 0.0000 ***
      HR_CI
1 0.708 (0.548-0.914)**
2 31.306 (19.914-49.215)***
3 0.999 (0.999-0.999)***
4 15.394 (10.757-22.030)***
5 0.048 (0.017-0.135)***
6 0.618 (0.560-0.683)***
```

```
# 7. Build Clustered Cox Proportional Hazards Models
```

```
cat("=== Building Clustered Cox Proportional Hazards Models ===\n")
```

=== Building Clustered Cox Proportional Hazards Models ===

```

if (nrow(merged_data_complete) > 0 && sum(merged_data_complete$event) > 0) {

# 7.1 Full Clustered Cox Model (includes gdpir)
cat("\n--- Full Clustered Cox Model (with gdpir) ---\n")
tryCatch({
  cox_full_cluster <- coxph(
    Surv(survival_time, event) ~
      Reform + WTO + Distance + Union + gdpir + log_pop +
      cluster(country),
    data = merged_data_complete,
    robust = TRUE
  )

  cat("Full clustered Cox model results (with gdpir):\n")
  full_summary <- summary(cox_full_cluster)
  print(full_summary)

  full_rsqa <- full_summary$rsq
  cat("\nR2 for full clustered model:\n")
  cat("  R2 (explained variation):", round(full_rsqa[1], 4), "\n")
  cat("  Adjusted R2:", round(full_rsqa[2], 4), "\n")

  full_success <- TRUE
  final_full_model <- cox_full_cluster

  cat("\nFull model variables:\n")
  print(names(full_summary$coefficients[, 1]))

}, error = function(e) {
  cat("Full clustered model (with gdpir) failed:", e$message, "\n")
  full_success <- FALSE
})

# 7.2 Reduced Clustered Cox Model (without gdpir, as sensitivity analysis)
cat("\n--- Reduced Clustered Cox Model (without gdpir) ---\n")
tryCatch({
  cox_reduced_cluster <- coxph(
    Surv(survival_time, event) ~
      Reform + WTO + Distance + Union + log_pop +
      cluster(country),
    data = merged_data_complete,
    robust = TRUE
  )

  cat("Reduced clustered Cox model results (without gdpir):\n")
  reduced_summary <- summary(cox_reduced_cluster)
  print(reduced_summary)

  reduced_rsqa <- reduced_summary$rsq
  cat("\nR2 for reduced clustered model:\n")
  cat("  R2 (explained variation):", round(reduced_rsqa[1], 4), "\n")
  cat("  Adjusted R2:", round(reduced_rsqa[2], 4), "\n")

```

```

reduced_success <- TRUE
final_reduced_model <- cox_reduced_cluster

cat("\nReduced model variables:\n")
print(names(reduced_summary$coefficients[, 1]))

}, error = function(e) {
  cat("Reduced clustered model (without gdpir) failed:", e$message, "\n")
  reduced_success <- FALSE
})

cat("\n=== Proportional Hazards Assumption Test for Standard Cox Models ===\n")

# 7.3 Proportional Hazards Test for Full Model
if (exists("full_success") && full_success) {
  cat("\n--- Proportional Hazards Test for Full Cox Model (with gdpir) ---\n")
  tryCatch({
    ph_test_full <- cox.zph(final_full_model)
    cat("Global test for proportional hazards assumption:\n")
    print(ph_test_full)

    cat("\n--- Interpretation of Proportional Hazards Test for Full Model ---\n")
    if (ph_test_full$table["GLOBAL", "p"] > 0.05) {
      cat("Global test p-value =", round(ph_test_full$table["GLOBAL", "p"], 4),
        "> 0.05, indicating that the proportional hazards assumption holds overall.\n")
    } else {
      cat("Global test p-value =", round(ph_test_full$table["GLOBAL", "p"], 4),
        "≤ 0.05, indicating potential violation of proportional hazards assumption.\n")
    }
  })

  cat("\n--- Variable-specific Tests for Full Model ---\n")
  violations_full <- which(ph_test_full$table[-nrow(ph_test_full$table), "p"] < 0.05)
  if (length(violations_full) > 0) {
    cat("The following variables may violate the proportional hazards assumption (p < 0.05\n")
    for (var in names(violations_full)) {
      p_value <- ph_test_full$table[var, "p"]
      cat(sprintf(" %15s: p = %.4f\n", var, p_value))
    }
  } else {
    cat("All individual variables satisfy proportional hazards assumption (p ≥ 0.05).\n")
  }

  cat("\n--- Plotting Schoenfeld Residuals for Full Model ---\n")
  par(mfrow = c(3, 3), mar = c(4, 4, 2, 1))
  plot(ph_test_full, resid = FALSE, main = "Full Model Schoenfeld Residuals")
  par(mfrow = c(1, 1))

}, error = function(e) {
  cat("Proportional hazards test for full model failed:", e$message, "\n")
})
}

```

```

# 7.4 Proportional Hazards Test for Reduced Model (optional)
if (exists("reduced_success") && reduced_success) {
  cat("\n--- Proportional Hazards Test for Reduced Cox Model (without gdmr) ---\n")
  tryCatch({
    ph_test_reduced <- cox.zph(final_reduced_model)
    cat("Global test for proportional hazards assumption:\n")
    print(ph_test_reduced)

    cat("\n--- Interpretation of Proportional Hazards Test for Reduced Model ---\n")
    if (ph_test_reduced$table["GLOBAL", "p"] > 0.05) {
      cat("Global test p-value =", round(ph_test_reduced$table["GLOBAL", "p"], 4),
        "> 0.05, indicating that the proportional hazards assumption holds overall.\n")
    } else {
      cat("Global test p-value =", round(ph_test_reduced$table["GLOBAL", "p"], 4),
        "≤ 0.05, indicating potential violation of proportional hazards assumption.\n")
    }

    cat("\n--- Variable-specific Tests for Reduced Model ---\n")
    violations_reduced <- which(ph_test_reduced$table[-nrow(ph_test_reduced$table), "p"] < 0)
    if (length(violations_reduced) > 0) {
      cat("The following variables may violate the proportional hazards assumption (p < 0.05\n")
      for (var in names(violations_reduced)) {
        p_value <- ph_test_reduced$table[var, "p"]
        cat(sprintf(" %15s: p = %.4f\n", var, p_value))
      }
    } else {
      cat("All individual variables satisfy proportional hazards assumption (p ≥ 0.05).\n")
    }

    cat("\n--- Plotting Schoenfeld Residuals for Reduced Model ---\n")
    par(mfrow = c(3, 3), mar = c(4, 4, 2, 1))
    plot(ph_test_reduced, resid = FALSE, main = "Reduced Model Schoenfeld Residuals")
    par(mfrow = c(1, 1))

  }, error = function(e) {
    cat("Proportional hazards test for reduced model failed:", e$message, "\n")
  })
}

# 7.5 Time-dependent Covariates Full Cox Model with Linear Time Interaction (x * t)
cat("\n--- Time-dependent Covariates Full Cox Model (Linear Time Interaction) ---\n")

tryCatch({

  vars_with_tt <- c("Reform", "WTO", "Distance", "Union", "log_pop")
  vars_fixed <- c("gdmr")

  main_terms <- paste(c(vars_with_tt, vars_fixed), collapse = " + ")
  tt_terms <- paste0("tt(", vars_with_tt, ")", collapse = " + ")

  full_time_dep_formula <- as.formula(
    paste("Surv(survival_time, event) ~", main_terms, "+", tt_terms, "+ cluster(country)")
  )

```

```

)

cox_full_time_dep <- coxph(
  full_time_dep_formula,
  data = merged_data_complete,
  robust = TRUE,
  tt = function(x, t, ...) {

    if (is.factor(x) || is.character(x)) {
      x_num <- as.numeric(as.factor(x))
      return(x_num * t)
    } else {
      return(x * t)
    }
  }
)

cat("Time-dependent covariates full Cox model with LINEAR time interaction (x * t):\n")
full_time_dep_summary <- summary(cox_full_time_dep)
print(full_time_dep_summary)

full_time_dep_rsqr <- full_time_dep_summary$rsqr
cat("\nR2 for time-dependent covariates full model (linear time):\n")
cat("  R2 (explained variation):", round(full_time_dep_rsqr[1], 4), "\n")
cat("  Adjusted R2:", round(full_time_dep_rsqr[2], 4), "\n")

cat("\nSignificance of linear time interaction terms:\n")
time_terms <- grep("^tt\\(", rownames(full_time_dep_summary$coefficients), value = TRUE)
for (term in time_terms) {
  coef_idx <- which(rownames(full_time_dep_summary$coefficients) == term)
  p_val <- full_time_dep_summary$coefficients[coef_idx, "Pr(>|z|)"]
  if (p_val < 0.05) {
    cat(sprintf(" %20s: p = %.4f **\n", term, p_val))
  } else {
    cat(sprintf(" %20s: p = %.4f\n", term, p_val))
  }
}

full_time_dep_success <- TRUE
final_full_time_dep_model <- cox_full_time_dep

}, error = function(e) {
  cat("Time-dependent covariates full model (linear time) failed:", e$message, "\n")
  full_time_dep_success <- FALSE
})

# 8. Model Comparison and Selection
cat("\n=== Model Comparison and Selection ===\n")

models_list <- list()
models_names <- c()

```

```

models_aic <- c()

if (exists("full_success") && full_success) {
  models_list[["Full Model (with gdpir)"]] <- final_full_model
  models_names <- c(models_names, "Full Model (with gdpir)")
  models_aic <- c(models_aic, AIC(final_full_model))
}

if (exists("reduced_success") && reduced_success) {
  models_list[["Reduced Model (without gdpir)"]] <- final_reduced_model
  models_names <- c(models_names, "Reduced Model (without gdpir)")
  models_aic <- c(models_aic, AIC(final_reduced_model))
}

if (exists("full_time_dep_success") && full_time_dep_success) {
  models_list[["Full Time-Dep Model"]] <- final_full_time_dep_model
  models_names <- c(models_names, "Full Time-Dep Model")
  models_aic <- c(models_aic, AIC(final_full_time_dep_model))
}

if (length(models_list) > 0) {
  comparison_table <- data.frame(
    Model = models_names,
    AIC = round(models_aic, 2),
    Selected = rep("No", length(models_names)),
    stringsAsFactors = FALSE
  )

  min_aic_idx <- which.min(models_aic)
  comparison_table$Selected[min_aic_idx] <- "Yes"

  cat("\nModel comparison based on AIC:\n")
  print(comparison_table)

  final_model_name <- models_names[min_aic_idx]
  final_model <- models_list[[final_model_name]]

  cat("\nSelected model:", final_model_name, "\n")
  cat("AIC:", round(models_aic[min_aic_idx], 2), "\n")

  if (grepl("Time-Dep", final_model_name)) {
    model_type <- "full_time_dep"
  } else if (grepl("Reduced", final_model_name)) {
    model_type <- "reduced"
  } else {
    model_type <- "full"
  }

  cat("\n=== Final Model Details ===\n")
  final_summary <- summary(final_model)
  print(final_summary)
} else {
  cat("No models were successfully fitted\n")
}

```

```

final_model <- NULL
}

if (!is.null(final_model)) {

# 9. Model Diagnostics and Validation
cat("=== Model Diagnostics ===\n")

cat("\n--- Proportional Hazards Assumption Test for Final Model ---\n")

tryCatch({
  ph_test <- cox.zph(final_model)

  cat("Global test for proportional hazards assumption:\n")
  print(ph_test)

  cat("\n--- Interpretation of Proportional Hazards Test ---\n")
  if (ph_test$table["GLOBAL", "p"] > 0.05) {
    cat("Global test p-value =", round(ph_test$table["GLOBAL", "p"], 4),
        "> 0.05, indicating that the proportional hazards assumption holds overall.\n")
  } else {
    cat("Global test p-value =", round(ph_test$table["GLOBAL", "p"], 4),
        "≤ 0.05, indicating potential violation of proportional hazards assumption.\n")
  }

  cat("\n--- Variable-specific Tests ---\n")
  violations <- which(ph_test$table[-nrow(ph_test$table), "p"] < 0.05)
  if (length(violations) > 0) {
    cat("The following variables may violate the proportional hazards assumption (p < 0.05\n")
    for (var in names(violations)) {
      p_value <- ph_test$table[var, "p"]
      cat(sprintf(" %-15s: p = %.4f\n", var, p_value))
    }

    cat("\nRecommendations:\n")
    cat("1. Consider using stratified Cox model for variables violating PH assumption\n")
    cat("2. Consider time-dependent covariates (using tt() function in coxph)\n")
    cat("3. Consider alternative models (parametric survival models)\n")
  } else {
    cat("All individual variables satisfy proportional hazards assumption (p ≥ 0.05).\n")
  }

  cat("\n--- Plotting Schoenfeld Residuals ---\n")
  par(mfrow = c(3, 3), mar = c(4, 4, 2, 1))
  plot(ph_test, resid = FALSE, main = "Final Model Schoenfeld Residuals")
  par(mfrow = c(1, 1))

  cat("\n--- Time-varying Effect Analysis ---\n")
  cat("Plotting survival curves with time-dependent effects...\n")

  if (length(violations) > 0) {
    if ("Reform" %in% names(violations)) {
      cat("Reform variable shows time-varying effects. Plotting stratified curves...\n")
    }
  }
}

```



```

merged_data_complete <- merged_data_complete %>%
  mutate(
    time_period = cut(survival_time,
                      breaks = quantile(survival_time[event == 1],
                                         probs = c(0, 0.33, 0.67, 1)),
                      labels = c("Early", "Middle", "Late"),
                      include.lowest = TRUE)
  )

stratified_plot <- ggsurvplot(
  survfit(Surv(survival_time, event) ~ Reform,
           data = merged_data_complete),
  data = merged_data_complete,
  fun = "cumhaz",
  title = "Cumulative Hazard by Reform Status",
  xlab = "Time (years)",
  ylab = "Cumulative Hazard",
  palette = c("#FF0000", "#0000FF"),
  risk.table = TRUE,
  tables.theme = theme_cleantable(),
  ggtheme = theme_minimal()
)
print(stratified_plot)
}
}

}, error = function(e) {
  cat("Proportional hazards test failed:", e$message, "\n")
})

# 10. Organize Model Results
cat("\n=== Final Model Results ===")
if (exists("model_type")) {
  if (model_type == "full") {
    cat(" (Full Model with gdpir) ===\n")
  } else if (model_type == "reduced") {
    cat(" (Reduced Model without gdpir) ===\n")
  } else if (model_type == "full_time_dep") {
    cat(" (Full Time-Dependent Model) ===\n")
  }
} else {
  cat(" ===\n")
}

results_table <- tidy(final_model, conf.int = TRUE, exponentiate = TRUE) %>%
  mutate(
    significance = case_when(
      p.value < 0.001 ~ "****",
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      TRUE ~ ""
    ),
    HR_CI = sprintf("%.3f (%.3f-%.3f)%s",
                    estimate, conf.low, conf.high, significance),

```

```

    p_value_formatted = ifelse(p.value < 0.001, "<0.001",
                               sprintf("%.3f", p.value))
  ) %>%
  dplyr::select(term, estimate, std.error, conf.low, conf.high,
               p_value = p.value, significance, HR_CI, p_value_formatted)

print(results_table)

cat("\n=== Model Goodness-of-Fit Metrics ===\n")
model_summary <- summary(final_model)
cat("R² (explained variation):", round(model_summary$rsq[1], 4), "\n")
cat("Adjusted R²:", round(model_summary$rsq[2], 4), "\n")
cat("Concordance (C-index):", round(model_summary$concordance[1], 4), "\n")
cat("Likelihood ratio test p-value:", round(model_summary$logtest[3], 4), "\n")
cat("Wald test p-value:", round(model_summary$waldtest[3], 4), "\n")
cat("Score (logrank) test p-value:", round(model_summary$sctest[3], 4), "\n")

# 11. Visualize Results
tryCatch({
  fit_km <- survfit(Surv(survival_time, event) ~ Reform,
                   data = survival_plot_data)

  survival_plot <- ggsurvplot(
    fit_km,
    data = survival_plot_data,
    pval = TRUE,
    conf.int = TRUE,
    risk.table = TRUE,
    legend.labs = c("No Reform", "Reform"),
    palette = c("#FF0000", "#0000FF"),
    title = "Socialist System Survival Curve - Reform vs No Reform",
    xlab = "Time (years)",
    ylab = "Socialist System Survival Probability",
    risk.table.height = 0.25,
    ggtheme = theme_minimal()
  )

  cat("\nDisplaying survival curve plot...\n")
  print(survival_plot)

}, error = function(e) {
  cat("Survival curve plotting failed:", e$message, "\n")
})

# 12. Create Improved Forest Plots
cat("\n=== Creating Improved Forest Plots ===\n")

create_improved_forest_plot <- function(cox_model, model_name, max_hr = 50) {

  forest_data <- tidy(cox_model, conf.int = TRUE, exponentiate = TRUE) %>%
    filter(term != "(Intercept)") %>%

    mutate(
      variable = case_when(

```

```

term == "Reform" ~ "Reform (Yes vs No)",
term == "WTO" ~ "WTO Membership (Yes vs No)",
term == "Distance" ~ "Distance to Moscow (per 100 km increase)",
term == "Union" ~ "Union Membership (Yes vs No)",
term == "gdpir" ~ "GDP Increasing Rate (per 1% increase)",
term == "log_pop" ~ "Log Population (per 1000 people increase)",
grepl("^tt\\(", term) ~ paste("Time interaction:", gsub("^tt\\((.*)\\)", "\\1", term),
TRUE ~ term
),

estimate = ifelse(term == "Distance", estimate^100, estimate),
conf.low = ifelse(term == "Distance", conf.low^100, conf.low),
conf.high = ifelse(term == "Distance", conf.high^100, conf.high),

conf_high_truncated = ifelse(conf.high > max_hr, max_hr, conf.high),
conf_low_truncated = ifelse(conf.low < 1/max_hr, 1/max_hr, conf.low),
has_truncated = conf.high > max_hr | conf.low < 1/max_hr,

variable_label = ifelse(has_truncated,
                        paste0(variable, "*"),
                        variable),

variable_label = fct_reorder(variable_label, estimate)
)

extreme_values <- forest_data %>%
  filter(conf.high > max_hr | conf.low < 1/max_hr) %>%
  dplyr::select(variable, estimate, conf.low, conf.high)

if (nrow(extreme_values) > 0) {
  cat(sprintf("\n=== Extreme Value Details for %s ===\n", model_name))
  cat("The following variables have large confidence intervals that were truncated in the model")
  for (i in 1:nrow(extreme_values)) {
    row <- extreme_values[i, ]
    cat(sprintf("%s: HR=%.2e, 95% CI=(%.2e, %.2e)\n",
                row$variable, row$estimate, row$conf.low, row$conf.high))
  }
}

n_vars <- nrow(forest_data)
base_height <- 7
var_height <- 0.45
plot_height <- max(base_height, n_vars * var_height)

longest_label <- max(nchar(as.character(forest_data$variable_label)))
left_margin <- max(30, longest_label * 1.5)

main_plot <- ggplot(forest_data, aes(x = estimate, y = variable_label)) +
  geom_point(size = 3, shape = 18) +
  geom_errorbarh(aes(xmin = conf_low_truncated, xmax = conf_high_truncated),
                height = 0.2, linewidth = 0.8) +
  geom_vline(xintercept = 1, linetype = "dashed",
            color = "red", linewidth = 1) +
  geom_point(data = filter(forest_data, has_truncated),

```

```

        aes(color = "Truncated"),
        size = 4, shape = 17, alpha = 0.7) +
scale_color_manual(name = "", values = c("Truncated" = "orange")) +
scale_x_log10(
  breaks = c(0.01, 0.1, 0.5, 1, 2, 10, 50),
  labels = function(x) {
    ifelse(x >= 1000, sprintf("%.0e", x),
           ifelse(x >= 100, sprintf("%.0f", x),
                  ifelse(x >= 10, sprintf("%.1f", x),
                         ifelse(x >= 1, sprintf("%.2f", x),
                                ifelse(x >= 0.1, sprintf("%.3f", x),
                                       sprintf("%.4f", x))))))
  },
  limits = c(1/max_hr, max_hr)
) +
labs(
  title = paste("Cox Model Forest Plot -", model_name),
  subtitle = "Hazard Ratios with 95% Confidence Intervals (* indicates truncated confidence interval)",
  x = "Hazard Ratio (HR, log scale)",
  y = NULL
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16,
                             margin = margin(b = 10)),
  plot.subtitle = element_text(hjust = 0.5, size = 12,
                                margin = margin(b = 15)),
  axis.title.x = element_text(face = "bold", size = 12,
                                margin = margin(t = 10)),
  axis.text.x = element_text(size = 10, angle = 45, hjust = 1,
                              margin = margin(t = 5)),
  axis.text.y = element_text(size = 10, hjust = 1,
                              margin = margin(r = 5)),
  legend.position = "top",
  legend.margin = margin(b = 10),
  panel.grid.major = element_line(color = "grey90"),
  panel.grid.minor = element_blank(),
  plot.background = element_rect(fill = "white", color = NA),
  plot.margin = margin(30, 20, 20, left_margin),
  panel.spacing = unit(1, "lines")
)

print(main_plot)

return(list(plot = main_plot, data = forest_data, extreme_values = extreme_values))
}

create_comprehensive_forest_report <- function(cox_model, model_name) {
  cat(sprintf("\nCreating comprehensive forest plot report for '%s'...\n", model_name))

  result <- create_improved_forest_plot(cox_model, model_name, max_hr = 50)

  cat("\n=== Complete Statistical Results ===\n")
  model_summary <- summary(cox_model)

```

```

coefstable <- tidy(cox_model, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    Variable = case_when(
      term == "Reform" ~ "Reform (Yes vs No)",
      term == "WTO" ~ "WTO Membership (Yes vs No)",
      term == "Distance" ~ "Distance to Moscow (per 100 km increase)",
      term == "Union" ~ "Union Membership (Yes vs No)",
      term == "gdpir" ~ "GDP Increasing Rate (per 1% increase)",
      term == "log_pop" ~ "Log Population (per 1000 people increase)",
      grepl("^tt\\(", term) ~ paste("Time interaction:", gsub("^tt\\((.*)\\)", "\\1", term),
      TRUE ~ term
    ),

    estimate = ifelse(term == "Distance", estimate^100, estimate),
    conf.low = ifelse(term == "Distance", conf.low^100, conf.low),
    conf.high = ifelse(term == "Distance", conf.high^100, conf.high),

    HR = round(estimate, 3),
    `95% CI` = sprintf("(%.3f-%.3f)", conf.low, conf.high),
    p_value = ifelse(p.value < 0.001, "<0.001",
      sprintf("%.3f", p.value)),
    Significance = case_when(
      p.value < 0.001 ~ "****",
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "*",
      TRUE ~ ""
    )
  ) %>%
  dplyr::select(Variable, HR, `95% CI`, p_value, Significance)

print(coefstable)

cat("\n=== Model Fit Metrics ===\n")
cat(sprintf("R²: %.4f\n", model_summary$rsq[1]))
cat(sprintf("Adjusted R²: %.4f\n", model_summary$rsq[2]))
cat(sprintf("C-index: %.4f\n", model_summary$concordance[1]))
cat(sprintf("Likelihood ratio test p-value: %.4f\n", model_summary$logtest[3]))

return(result)
}

if (exists("final_full_model") && full_success) {
  cat("\nCreating improved forest plot for full model (with gdpir)...\n")
  full_forest_result <- create_comprehensive_forest_report(final_full_model, "Full Model (")
}

if (exists("final_reduced_model") && reduced_success) {
  cat("\nCreating improved forest plot for reduced model (without gdpir)...\n")
  reduced_forest_result <- create_comprehensive_forest_report(final_reduced_model, "Reduced")
}

if (exists("final_full_time_dep_model") && full_time_dep_success) {

```

```

    cat("\nCreating improved forest plot for full time-dependent model...\n")
    full_time_dep_result <- create_comprehensive_forest_report(final_full_time_dep_model, "F
}

# 13. Model Comparison Summary
cat("\n=== Complete Model Comparison Summary ===\n")

comparison_data <- data.frame()

if (exists("full_success") && full_success) {
  comparison_data <- bind_rows(comparison_data, data.frame(
    Model = "Full Model (with gdpir)",
    Variables = paste(attr(terms(final_full_model), "term.labels"), collapse = ", "),
    R2 = round(summary(final_full_model)$rsq[1], 4),
    Adj_R2 = round(summary(final_full_model)$rsq[2], 4),
    C_index = round(summary(final_full_model)$concordance[1], 4),
    AIC = round(AIC(final_full_model), 2),
    Type = "Regular",
    Selected = ifelse(exists("final_model_name") && final_model_name == "Full Model (with
  ))
}

if (exists("reduced_success") && reduced_success) {
  comparison_data <- bind_rows(comparison_data, data.frame(
    Model = "Reduced Model (without gdpir)",
    Variables = paste(attr(terms(final_reduced_model), "term.labels"), collapse = ", "),
    R2 = round(summary(final_reduced_model)$rsq[1], 4),
    Adj_R2 = round(summary(final_reduced_model)$rsq[2], 4),
    C_index = round(summary(final_reduced_model)$concordance[1], 4),
    AIC = round(AIC(final_reduced_model), 2),
    Type = "Regular (Sensitivity)",
    Selected = ifelse(exists("final_model_name") && final_model_name == "Reduced Model (wi
  ))
}

if (exists("full_time_dep_success") && full_time_dep_success) {
  comparison_data <- bind_rows(comparison_data, data.frame(
    Model = "Full Time-Dep Model",
    Variables = paste(attr(terms(final_full_time_dep_model), "term.labels"), collapse = ",
    R2 = round(summary(final_full_time_dep_model)$rsq[1], 4),
    Adj_R2 = round(summary(final_full_time_dep_model)$rsq[2], 4),
    C_index = round(summary(final_full_time_dep_model)$concordance[1], 4),
    AIC = round(AIC(final_full_time_dep_model), 2),
    Type = "Time-Dependent",
    Selected = ifelse(exists("final_model_name") && final_model_name == "Full Time-Dep Mod
  ))
}

if (nrow(comparison_data) > 0) {
  print(comparison_data)
}

} else {
  cat("All models failed, cannot proceed with further analysis\n")
}

```

```

}

} else {
  cat("Insufficient data or no events, cannot build Cox model\n")
  cat("Total observations:", nrow(merged_data_complete), "\n")
  cat("Number of events:", sum(merged_data_complete$event), "\n")
}

```

--- Full Clustered Cox Model (with gdpir) ---

Full clustered Cox model results (with gdpir):

Call:

```
coxph(formula = Surv(survival_time, event) ~ Reform + WTO + Distance +
      Union + gdpir + log_pop, data = merged_data_complete, robust = TRUE,
      cluster = country)
```

n= 431, number of events= 241

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)	
Reform	-3.757e+00	2.336e-02	3.486e-01	7.647e-01	-4.913	8.99e-07	***
WTO	5.387e+00	2.186e+02	4.048e-01	1.196e+00	4.503	6.69e-06	***
Distance	-4.414e-04	9.996e-01	5.077e-05	1.322e-04	-3.339	0.000842	***
Union	7.803e-01	2.182e+00	2.268e-01	3.702e-01	2.108	0.035058	*
gdpir	1.336e-02	1.013e+00	6.781e-01	2.537e-01	0.053	0.958008	
log_pop	-2.497e-01	7.790e-01	6.666e-02	1.085e-01	-2.301	0.021381	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	0.02336	42.813600	0.005218	0.1046
WTO	218.57968	0.004575	20.957999	2279.6582
Distance	0.99956	1.000441	0.999300	0.9998
Union	2.18213	0.458268	1.056214	4.5083
gdpir	1.01345	0.986733	0.616429	1.6662
log_pop	0.77901	1.283675	0.629756	0.9636

Concordance= 0.943 (se = 0.029)

Likelihood ratio test= 742.1 on 6 df, p=<2e-16

Wald test = 31.25 on 6 df, p=2e-05

Score (logrank) test = 628.1 on 6 df, p=<2e-16, Robust = 30.36 p=3e-05

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

R² for full clustered model:

R² (explained variation): 0.8212

Adjusted R²: 0.9983

Full model variables:

```
[1] "Reform" "WTO" "Distance" "Union" "gdpir" "log_pop"
```

--- Reduced Clustered Cox Model (without gdpir) ---

Reduced clustered Cox model results (without gdpir):

Call:

```
coxph(formula = Surv(survival_time, event) ~ Reform + WTO + Distance +  
      Union + log_pop, data = merged_data_complete, robust = TRUE,  
      cluster = country)
```

n= 431, number of events= 241

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)	
Reform	-3.757e+00	2.336e-02	3.486e-01	7.647e-01	-4.913	8.99e-07	***
WTO	5.388e+00	2.187e+02	4.044e-01	1.194e+00	4.512	6.43e-06	***
Distance	-4.414e-04	9.996e-01	5.076e-05	1.322e-04	-3.339	0.000841	***
Union	7.799e-01	2.181e+00	2.259e-01	3.706e-01	2.104	0.035340	*
log_pop	-2.497e-01	7.791e-01	6.659e-02	1.085e-01	-2.302	0.021349	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	0.02336	42.813941	0.005218	0.1046
WTO	218.66205	0.004573	21.055135	2270.8518
Distance	0.99956	1.000441	0.999300	0.9998
Union	2.18129	0.458444	1.055002	4.5100
log_pop	0.77906	1.283598	0.629858	0.9636

Concordance= 0.943 (se = 0.029)

Likelihood ratio test= 742.1 on 5 df, p=<2e-16

Wald test = 29.88 on 5 df, p=2e-05

Score (logrank) test = 626.8 on 5 df, p=<2e-16, Robust = 29.88 p=2e-05

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

R² for reduced clustered model:

R² (explained variation): 0.8212

Adjusted R²: 0.9983

Reduced model variables:

[1] "Reform" "WTO" "Distance" "Union" "log_pop"

=== Proportional Hazards Assumption Test for Standard Cox Models ===

--- Proportional Hazards Test for Full Cox Model (with gdpir) ---

Global test for proportional hazards assumption:

	chisq	df	p
Reform	15.38	1	8.8e-05
WTO	167.80	1	< 2e-16
Distance	91.89	1	< 2e-16
Union	23.98	1	9.7e-07
gdpir	0.19	1	0.66
log_pop	27.64	1	1.5e-07
GLOBAL	289.64	6	< 2e-16

--- Interpretation of Proportional Hazards Test for Full Model ---

Global test p-value = 0 ≤ 0.05, indicating potential violation of proportional hazards assumption.

--- Variable-specific Tests for Full Model ---

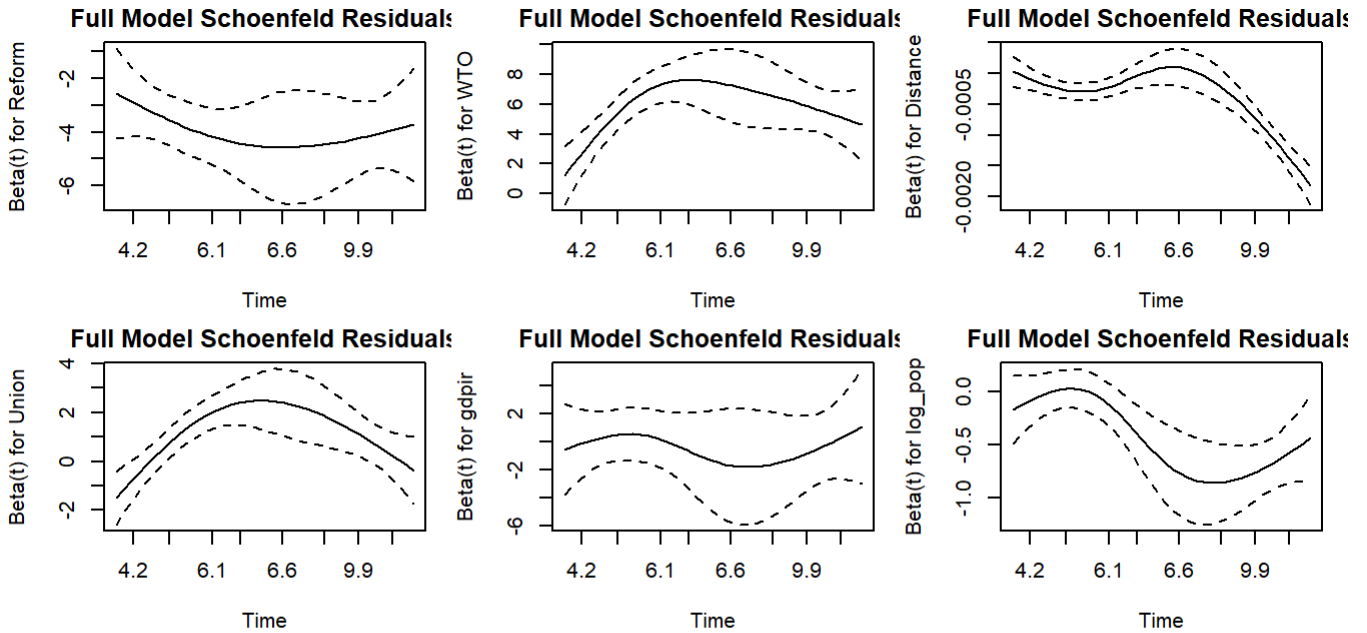
The following variables may violate the proportional hazards assumption ($p < 0.05$):

```

Reform      : p = 0.0001
WTO         : p = 0.0000
Distance    : p = 0.0000
Union       : p = 0.0000
log_pop     : p = 0.0000

```

--- Plotting Schoenfeld Residuals for Full Model ---



--- Proportional Hazards Test for Reduced Cox Model (without gdpir) ---

Global test for proportional hazards assumption:

	chisq	df	p
Reform	15.4	1	8.8e-05
WTO	167.8	1	< 2e-16
Distance	91.9	1	< 2e-16
Union	23.7	1	1.1e-06
log_pop	27.5	1	1.6e-07
GLOBAL	289.5	5	< 2e-16

--- Interpretation of Proportional Hazards Test for Reduced Model ---

Global test p-value = 0 ≤ 0.05, indicating potential violation of proportional hazards assumption.

--- Variable-specific Tests for Reduced Model ---

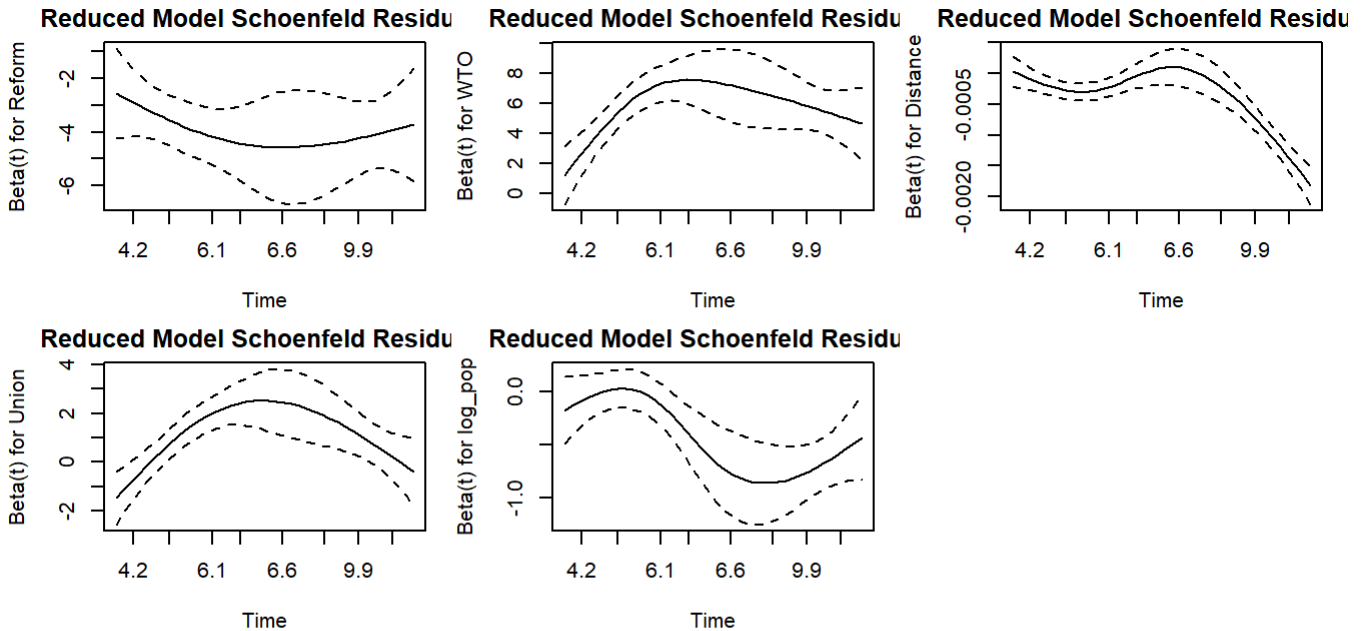
The following variables may violate the proportional hazards assumption ($p < 0.05$):

```

Reform      : p = 0.0001
WTO         : p = 0.0000
Distance    : p = 0.0000
Union       : p = 0.0000
log_pop     : p = 0.0000

```

--- Plotting Schoenfeld Residuals for Reduced Model ---



--- Time-dependent Covariates Full Cox Model (Linear Time Interaction) ---

Time-dependent covariates full Cox model with LINEAR time interaction ($x * t$):

Call:

```

coxph(formula = Surv(survival_time, event) ~ Reform + WTO + Distance +
      Union + log_pop + gdpir + tt(Reform) + tt(WTO) + tt(Distance) +
      tt(Union) + tt(log_pop), data = merged_data_complete, robust = TRUE,
      tt = function(x, t, ...) {
        if (is.factor(x) || is.character(x)) {
          x_num <- as.numeric(as.factor(x))
          return(x_num * t)
        }
        else {
          return(x * t)
        }
      }, cluster = country)

```

n= 431, number of events= 241

coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
------	-----------	----------	-----------	---	----------

Reform	8.722e-01	2.392e+00	1.303e+01	3.879e+00	0.225	0.82210	
WTO	-2.142e+01	4.979e-10	1.342e+01	7.962e+00	-2.690	0.00714	**
Distance	4.544e-04	1.000e+00	1.841e-04	2.425e-04	1.874	0.06100	.
Union	-1.288e-01	8.792e-01	9.900e-01	1.950e+00	-0.066	0.94733	
log_pop	1.488e+00	4.428e+00	4.094e-01	6.555e-01	2.270	0.02322	*
gdpir	3.595e-01	1.433e+00	6.705e-01	1.808e-01	1.988	0.04682	*
tt(Reform)	-2.312e+00	9.909e-02	3.272e+00	1.044e+00	-2.215	0.02675	*
tt(WTO)	6.292e+00	5.400e+02	3.364e+00	2.014e+00	3.124	0.00178	**
tt(Distance)	-1.126e-04	9.999e-01	3.223e-05	3.442e-05	-3.272	0.00107	**
tt(Union)	2.460e-01	1.279e+00	1.821e-01	4.249e-01	0.579	0.56270	
tt(log_pop)	-2.816e-01	7.546e-01	6.677e-02	1.106e-01	-2.545	0.01092	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	2.392e+00	4.180e-01	1.194e-03	4.793e+03
WTO	4.979e-10	2.009e+09	8.312e-17	2.982e-03
Distance	1.000e+00	9.995e-01	1.000e+00	1.001e+00
Union	8.792e-01	1.137e+00	1.925e-02	4.015e+01
log_pop	4.428e+00	2.259e-01	1.225e+00	1.600e+01
gdpir	1.433e+00	6.980e-01	1.005e+00	2.042e+00
tt(Reform)	9.909e-02	1.009e+01	1.281e-02	7.662e-01
tt(WTO)	5.400e+02	1.852e-03	1.043e+01	2.795e+04
tt(Distance)	9.999e-01	1.000e+00	9.998e-01	1.000e+00
tt(Union)	1.279e+00	7.820e-01	5.561e-01	2.941e+00
tt(log_pop)	7.546e-01	1.325e+00	6.075e-01	9.373e-01

Concordance= 0.978 (se = 0.013)

Likelihood ratio test= 965.8 on 11 df, p=<2e-16

Wald test = 120.6 on 11 df, p=<2e-16

Score (logrank) test = 1144 on 11 df, p=<2e-16, Robust = 35.76 p=2e-04

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

R² for time-dependent covariates full model (linear time):

R² (explained variation): 0.8936

Adjusted R²: 0.9983

Significance of linear time interaction terms:

tt(Reform)	: p = 0.0268 **
tt(WTO)	: p = 0.0018 **
tt(Distance)	: p = 0.0011 **
tt(Union)	: p = 0.5627
tt(log_pop)	: p = 0.0109 **

=== Model Comparison and Selection ===

Model comparison based on AIC:

	Model	AIC	Selected
1	Full Model (with gdpir)	2023.88	No
2	Reduced Model (without gdpir)	2021.88	No
3	Full Time-Dep Model	1810.14	Yes

Selected model: Full Time-Dep Model

AIC: 1810.14

=== Final Model Details ===

Call:

```
coxph(formula = Surv(survival_time, event) ~ Reform + WTO + Distance +
      Union + log_pop + gdpir + tt(Reform) + tt(WTO) + tt(Distance) +
      tt(Union) + tt(log_pop), data = merged_data_complete, robust = TRUE,
      tt = function(x, t, ...) {
        if (is.factor(x) || is.character(x)) {
          x_num <- as.numeric(as.factor(x))
          return(x_num * t)
        }
        else {
          return(x * t)
        }
      }, cluster = country)
```

n= 431, number of events= 241

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
Reform	8.722e-01	2.392e+00	1.303e+01	3.879e+00	0.225	0.82210
WTO	-2.142e+01	4.979e-10	1.342e+01	7.962e+00	-2.690	0.00714 **
Distance	4.544e-04	1.000e+00	1.841e-04	2.425e-04	1.874	0.06100 .
Union	-1.288e-01	8.792e-01	9.900e-01	1.950e+00	-0.066	0.94733
log_pop	1.488e+00	4.428e+00	4.094e-01	6.555e-01	2.270	0.02322 *
gdpir	3.595e-01	1.433e+00	6.705e-01	1.808e-01	1.988	0.04682 *
tt(Reform)	-2.312e+00	9.909e-02	3.272e+00	1.044e+00	-2.215	0.02675 *
tt(WTO)	6.292e+00	5.400e+02	3.364e+00	2.014e+00	3.124	0.00178 **
tt(Distance)	-1.126e-04	9.999e-01	3.223e-05	3.442e-05	-3.272	0.00107 **
tt(Union)	2.460e-01	1.279e+00	1.821e-01	4.249e-01	0.579	0.56270
tt(log_pop)	-2.816e-01	7.546e-01	6.677e-02	1.106e-01	-2.545	0.01092 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	2.392e+00	4.180e-01	1.194e-03	4.793e+03
WTO	4.979e-10	2.009e+09	8.312e-17	2.982e-03
Distance	1.000e+00	9.995e-01	1.000e+00	1.001e+00
Union	8.792e-01	1.137e+00	1.925e-02	4.015e+01
log_pop	4.428e+00	2.259e-01	1.225e+00	1.600e+01
gdpir	1.433e+00	6.980e-01	1.005e+00	2.042e+00
tt(Reform)	9.909e-02	1.009e+01	1.281e-02	7.662e-01
tt(WTO)	5.400e+02	1.852e-03	1.043e+01	2.795e+04
tt(Distance)	9.999e-01	1.000e+00	9.998e-01	1.000e+00
tt(Union)	1.279e+00	7.820e-01	5.561e-01	2.941e+00
tt(log_pop)	7.546e-01	1.325e+00	6.075e-01	9.373e-01

Concordance= 0.978 (se = 0.013)

Likelihood ratio test= 965.8 on 11 df, p=<2e-16

Wald test = 120.6 on 11 df, p=<2e-16

Score (logrank) test = 1144 on 11 df, p=<2e-16, Robust = 35.76 p=2e-04

(Note: the likelihood ratio and score tests assume independence of

observations within a cluster, the Wald and robust score tests do not).
=== Model Diagnostics ===

--- Proportional Hazards Assumption Test for Final Model ---
Proportional hazards test failed: function not defined for models with tt() terms

=== Final Model Results === (Full Time-Dependent Model) ===

A tibble: 11 × 9

	term	estimate	std.error	conf.low	conf.high	p_value	significance	HR_CI
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	Reform	2.39 e+ 0	1.30e+1	1.19 e- 3	4.79 e+3	0.822	" "	2.39...
2	WTO	4.98 e-10	1.34e+1	8.31 e-17	2.98 e-3	0.00714	"**"	0.00...
3	Distance	1.00 e+ 0	1.84e-4	1.000e+ 0	1.00 e+0	0.0610	" "	1.00...
4	Union	8.79 e- 1	9.90e-1	1.93 e- 2	4.01 e+1	0.947	" "	0.87...
5	log_pop	4.43 e+ 0	4.09e-1	1.23 e+ 0	1.60 e+1	0.0232	"**"	4.42...
6	gdpir	1.43 e+ 0	6.71e-1	1.01 e+ 0	2.04 e+0	0.0468	"**"	1.43...
7	tt(Reform)	9.91 e- 2	3.27e+0	1.28 e- 2	7.66 e-1	0.0268	"**"	0.09...
8	tt(WTO)	5.40 e+ 2	3.36e+0	1.04 e+ 1	2.80 e+4	0.00178	"**"	539...
9	tt(Distan...	1.000e+ 0	3.22e-5	1.000e+ 0	1.000e+0	0.00107	"**"	1.00...
10	tt(Union)	1.28 e+ 0	1.82e-1	5.56 e- 1	2.94 e+0	0.563	" "	1.27...
11	tt(log_po...	7.55 e- 1	6.68e-2	6.08 e- 1	9.37 e-1	0.0109	"**"	0.75...

i 1 more variable: p_value_formatted <chr>

=== Model Goodness-of-Fit Metrics ===

R² (explained variation): 0.8936

Adjusted R²: 0.9983

Concordance (C-index): 0.9781

Likelihood ratio test p-value: 0

Wald test p-value: 0

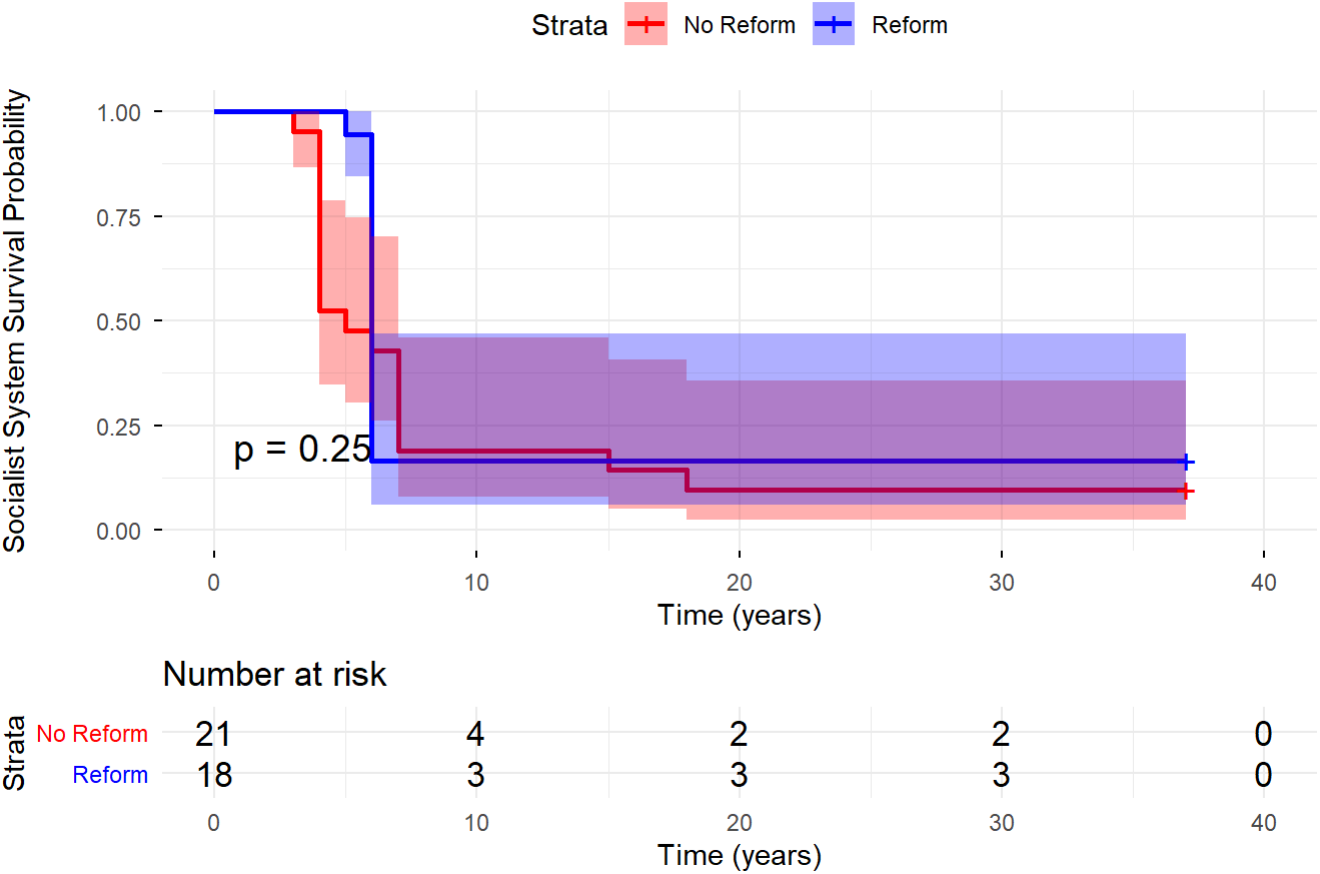
Score (logrank) test p-value: 0

Displaying survival curve plot...

Ignoring unknown labels:

- colour : "Strata"

Socialist System Survival Curve - Reform vs No Reform



=== Creating Improved Forest Plots ===

Creating improved forest plot for full model (with gdpir)...

Creating comprehensive forest plot report for 'Full Model (with gdpir)'...

=== Extreme Value Details for Full Model (with gdpir) ===

The following variables have large confidence intervals that were truncated in the plot:

Reform (Yes vs No): HR=2.34e-02, 95% CI=(5.22e-03, 1.05e-01)

WTO Membership (Yes vs No): HR=2.19e+02, 95% CI=(2.10e+01, 2.28e+03)

Warning: `geom\_errorbarh()` was deprecated in ggplot2 4.0.0.
i Please use the `orientation` argument of `geom\_errorbar()` instead.

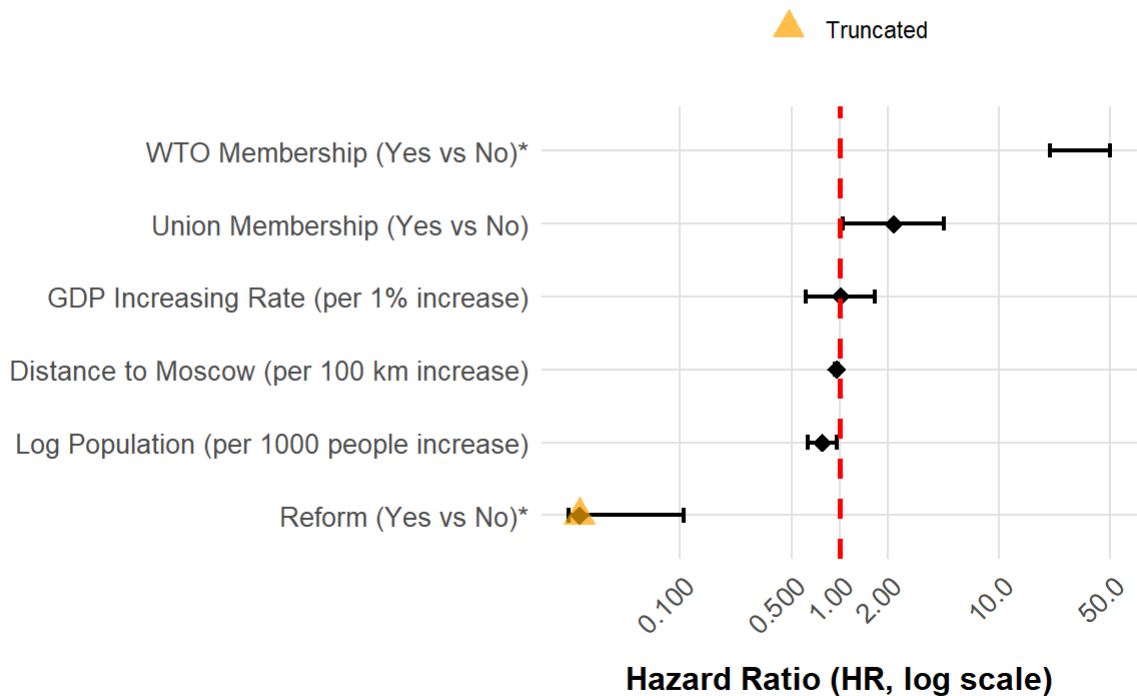
`height` was translated to `width`.

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Cox Model Forest Plot - Full Model (with

Hazard Ratios with 95% Confidence Intervals (* indicates truncated c



=== Complete Statistical Results ===

A tibble: 6 × 5

Variable	HR	95% CI	p_value	Significance
<chr>	<dbl>	<chr>	<chr>	<chr>
1 Reform (Yes vs No)	0.023	(0.005-...	<0.001	****
2 WTO Membership (Yes vs No)	219.	(20.958...	<0.001	****
3 Distance to Moscow (per 100 km increase)	0.957	(0.932-...	<0.001	****
4 Union Membership (Yes vs No)	2.18	(1.056-...	0.035	**
5 GDP Increasing Rate (per 1% increase)	1.01	(0.616-...	0.958	"
6 Log Population (per 1000 people increas...	0.779	(0.630-...	0.021	**

=== Model Fit Metrics ===

R²: 0.8212

Adjusted R²: 0.9983

C-index: 0.9430

Likelihood ratio test p-value: 0.0000

Creating improved forest plot for reduced model (without gdpir)...

Creating comprehensive forest plot report for 'Reduced Model (without gdpir)'...

=== Extreme Value Details for Reduced Model (without gdpir) ===

The following variables have large confidence intervals that were truncated in the plot:

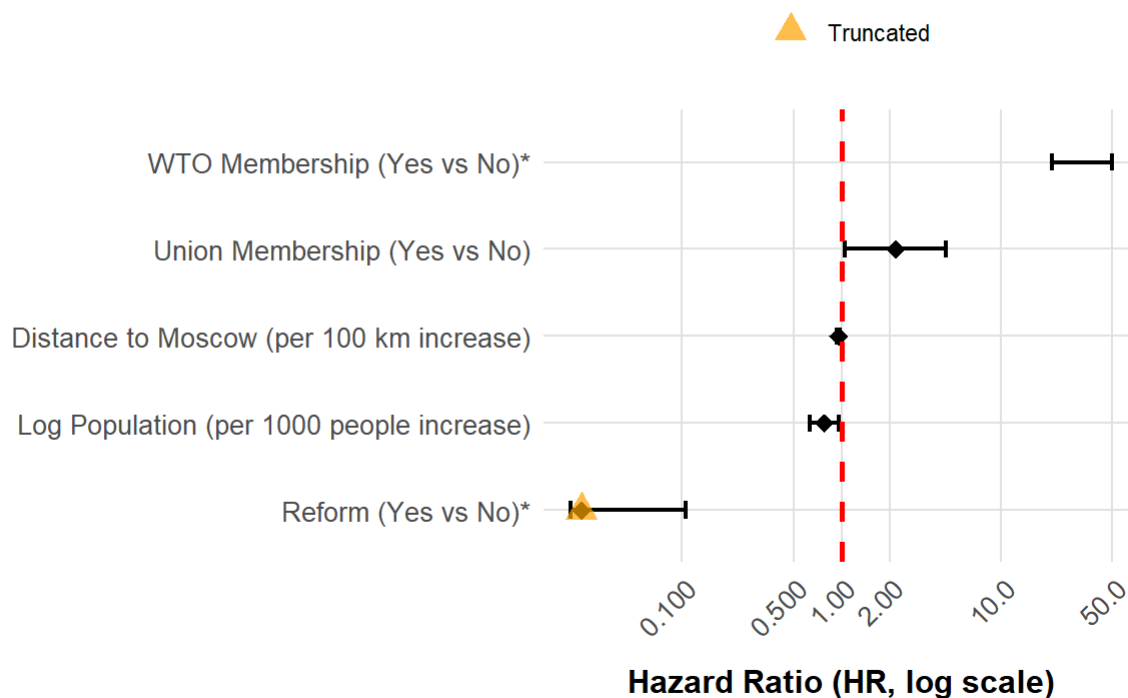
Reform (Yes vs No): HR=2.34e-02, 95% CI=(5.22e-03, 1.05e-01)

WTO Membership (Yes vs No): HR=2.19e+02, 95% CI=(2.11e+01, 2.27e+03)

`height` was translated to `width`.

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).
Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Cox Model Forest Plot - Reduced Model (with Hazard Ratios with 95% Confidence Intervals (* indicates truncated c



=== Complete Statistical Results ===

A tibble: 5 × 5

Variable	HR	95% CI	p_value	Significance
<chr>	<dbl>	<chr>	<chr>	<chr>
1 Reform (Yes vs No)	0.023	(0.005-...	<0.001	***
2 WTO Membership (Yes vs No)	219.	(21.055-...	<0.001	***
3 Distance to Moscow (per 100 km increase)	0.957	(0.932-...	<0.001	***
4 Union Membership (Yes vs No)	2.18	(1.055-...	0.035	*
5 Log Population (per 1000 people increas...	0.779	(0.630-...	0.021	*

=== Model Fit Metrics ===

R²: 0.8212

Adjusted R²: 0.9983

C-index: 0.9430

Likelihood ratio test p-value: 0.0000

Creating improved forest plot for full time-dependent model...

Creating comprehensive forest plot report for 'Full Time-Dependent Model'...

=== Extreme Value Details for Full Time-Dependent Model ===

The following variables have large confidence intervals that were truncated in the plot:

Reform (Yes vs No): HR=2.39e+00, 95% CI=(1.19e-03, 4.79e+03)
WTO Membership (Yes vs No): HR=4.98e-10, 95% CI=(8.31e-17, 2.98e-03)
Union Membership (Yes vs No): HR=8.79e-01, 95% CI=(1.93e-02, 4.01e+01)
Time interaction: Reform: HR=9.91e-02, 95% CI=(1.28e-02, 7.66e-01)
Time interaction: WTO: HR=5.40e+02, 95% CI=(1.04e+01, 2.80e+04)

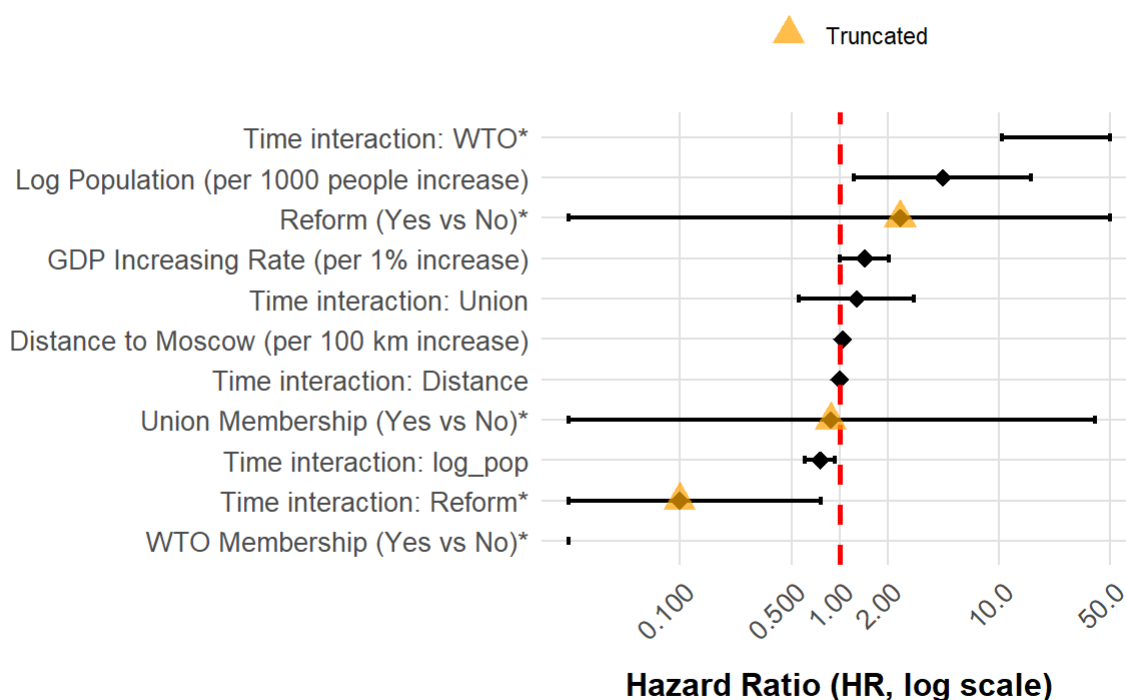
`height` was translated to `width`.

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom\_point()`).

Cox Model Forest Plot - Full Time-Depende

Hazard Ratios with 95% Confidence Intervals (* indicates truncated co



=== Complete Statistical Results ===

A tibble: 11 × 5

Variable	HR	95% CI	p_value	Significance
<chr>	<dbl>	<chr>	<chr>	<chr>
1 Reform (Yes vs No)	2.39	(0.001-...	0.822	" "
2 WTO Membership (Yes vs No)	0	(0.000-...	0.007	"**"
3 Distance to Moscow (per 100 km increas...	1.05	(0.998-...	0.061	" "
4 Union Membership (Yes vs No)	0.879	(0.019-...	0.947	" "
5 Log Population (per 1000 people increa...	4.43	(1.225-...	0.023	"**"
6 GDP Increasing Rate (per 1% increase)	1.43	(1.005-...	0.047	"**"
7 Time interaction: Reform	0.099	(0.013-...	0.027	"**"
8 Time interaction: WTO	540.	(10.432...	0.002	"**"
9 Time interaction: Distance	1	(1.000-...	0.001	"**"

```
10 Time interaction: Union                1.28 (0.556-... 0.563   ""
11 Time interaction: log_pop              0.755 (0.608-... 0.011   ""
```

```
=== Model Fit Metrics ===
```

```
R²: 0.8936
```

```
Adjusted R²: 0.9983
```

```
C-index: 0.9781
```

```
Likelihood ratio test p-value: 0.0000
```

```
=== Complete Model Comparison Summary ===
```

```
Model
rsq...1      Full Model (with gdpir)
rsq...2 Reduced Model (without gdpir)
rsq...3      Full Time-Dep Model
```

```
Variables
```

```
rsq...1      Reform, WTO, Distance,
```

```
Union, gdpir, log_pop
```

```
rsq...2      Reform, WTO,
```

```
Distance, Union, log_pop
```

```
rsq...3 Reform, WTO, Distance, Union, log_pop, gdpir, tt(Reform), tt(WTO), tt(Distance),
tt(Union), tt(log_pop)
```

	R2	Adj_R2	C_index	AIC	Type	Selected
rsq...1	0.8212	0.9983	0.9430	2023.88	Regular	No
rsq...2	0.8212	0.9983	0.9430	2021.88	Regular (Sensitivity)	No
rsq...3	0.8936	0.9983	0.9781	1810.14	Time-Dependent	Yes

```
cat("\n\n", paste(rep("=", 80), collapse = ""), "\n")
```

```
=====
```

```
cat(">>> Subgroup Analysis: Reform = 0 vs Reform = 1 <<<\n")
```

```
>>> Subgroup Analysis: Reform = 0 vs Reform = 1 <<<
```

```
cat(paste(rep("=", 80), collapse = ""), "\n\n")
```

```
=====
```

```
event_counts_sub <- merged_data_complete %>%
  group_by(Reform) %>%
  summarise(
    N_countries = n_distinct(country),
    N_obs = n(),
    N_events = sum(event),
    .groups = "drop"
  )
cat("\n--- Event Counts by Reform Status ---\n")
```

--- Event Counts by Reform Status ---

```
print(event_counts_sub)
```

A tibble: 2 × 4

	Reform	N_countries	N_obs	N_events
	<int>	<int>	<int>	<dbl>
1	0	21	213	137
2	1	18	218	104

```
for (reform_val in c(0, 1)) {

  cat("\n\n", paste(rep("-", 60), collapse = ""))
  cat(sprintf("\n>>> Analyzing Subset: Reform = %d <<<\n", reform_val))
  cat(paste(rep("-", 60), collapse = ""), "\n")

  sub_data <- merged_data_complete %>% filter(Reform == reform_val)

  if (sum(sub_data$event) < 5) {
    cat("WARNING: Too few events in this subgroup. Skipping detailed analysis.\n")
    next
  }

  subgroup_vars <- c("Distance", "gdpir", "log_pop")
  cat("Note: WTO and Union are excluded from all models due to perfect separation (all such co

  cat("\n--- Univariate Cox Analysis (Reform =", reform_val, ") ---\n")
  uni_res_sub <- data.frame()
  for (var in subgroup_vars) {
    tryCatch({
      fml <- as.formula(paste("Surv(survival_time, event) ~", var))
      fit <- coxph(fml, data = sub_data)
      s <- summary(fit)
      hr <- s$coefficients[1, 2]
      ci_low <- s$conf.int[1, 3]
      ci_high <- s$conf.int[1, 4]
      pval <- s$coefficients[1, 5]
      sig <- case_when(
        pval < 0.001 ~ "****",
        pval < 0.01 ~ "***",
        pval < 0.05 ~ "**",
        TRUE ~ ""
      )
      uni_res_sub <- bind_rows(uni_res_sub, data.frame(
        Variable = var,
        HR = round(hr, 3),
        CI_lower = round(ci_low, 3),
        CI_upper = round(ci_high, 3),
        p_value = round(pval, 4),
        Significance = sig,
        stringsAsFactors = FALSE
      ))
    })
  }
}
```

```

    }, error = function(e) {
      cat(" Univariate failed for", var, ":", e$message, "\n")
    })
  }
  if (nrow(uni_res_sub) > 0) {
    uni_res_sub <- uni_res_sub %>% mutate(
      HR_CI = sprintf("%.3f (%.3f-%.3f)%s", HR, CI_lower, CI_upper, Significance)
    )
    print(uni_res_sub %>% dplyr::select(Variable, HR_CI, p_value))
  } else {
    cat(" No univariate results.\n")
  }

  cat("\n--- Full Multivariate Clustered Model (Reform =", reform_val, ") ---\n")
  tryCatch({
    fml_full <- as.formula(paste("Surv(survival_time, event) ~",
                                paste(subgroup_vars, collapse = " + "),
                                "+ cluster(country)"))
    fit_full <- coxph(fml_full, data = sub_data, robust = TRUE)
    cat("Full model summary:\n")
    print(summary(fit_full))
    cat(sprintf("R² (explained variation): %.4f\n", summary(fit_full)$rsq[1]))
    cat(sprintf("Adjusted R²: %.4f\n", summary(fit_full)$rsq[2]))

    create_improved_forest_plot(fit_full, paste0("Reform=", reform_val, " Full Model"))
  }, error = function(e) {
    cat("Full clustered model failed:", e$message, "\n")
  })
}

```

```

-----
>>> Analyzing Subset: Reform = 0 <<<
-----

```

Note: WTO and Union are excluded from all models due to perfect separation (all such countries experienced the event).

```

--- Univariate Cox Analysis (Reform = 0 ) ---

```

	Variable	HR_CI	p_value
1	Distance	0.999 (0.999-0.999)***	0.0000
2	gdpir	0.517 (0.102-2.619)	0.4251
3	log_pop	0.575 (0.491-0.673)***	0.0000

```

--- Full Multivariate Clustered Model (Reform = 0 ) ---

```

Full model summary:

Call:

```

coxph(formula = Surv(survival_time, event) ~ Distance + gdpir +
      log_pop, data = sub_data, robust = TRUE, cluster = country)

```

n= 213, number of events= 137

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
--	------	-----------	----------	-----------	---	----------

```
Distance -6.496e-04  9.994e-01  7.323e-05  2.285e-04 -2.842  0.00448 **
gdpir      1.008e-01  1.106e+00  6.566e-01  2.518e-01  0.400  0.68881
log_pop    -3.446e-01  7.085e-01  8.385e-02  2.517e-01 -1.369  0.17100
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
Distance	0.9994	1.0006	0.9989	0.9998
gdpir	1.1061	0.9041	0.6753	1.8118
log_pop	0.7085	1.4114	0.4326	1.1604

```
Concordance= 0.826 (se = 0.082 )
Likelihood ratio test= 217 on 3 df,  p=<2e-16
Wald test               = 9.95 on 3 df,  p=0.02
Score (logrank) test = 179.6 on 3 df,  p=<2e-16,  Robust = 9.09 p=0.03
```

```
(Note: the likelihood ratio and score tests assume independence of
observations within a cluster, the Wald and robust score tests do not).
R² (explained variation): 0.6390
Adjusted R²: 0.9983
```

```
`height` was translated to `width`.
```

```
Warning: No shared levels found between `names(values)` of the manual scale and the
data's colour values.
```

Cox Model Forest Plot - Reform=0 Full M

Hazard Ratios with 95% Confidence Intervals (* indicates truncated co



```
>>> Analyzing Subset: Reform = 1 <<<
```

Note: WTO and Union are excluded from all models due to perfect separation (all such countries experienced the event).

```
--- Univariate Cox Analysis (Reform = 1 ) ---
```

	Variable	HR_CI	p_value
1	Distance	0.999 (0.999-0.999)***	0
2	gdpir	0.000 (0.000-0.000)***	0
3	log_pop	0.592 (0.510-0.687)***	0

```
--- Full Multivariate Clustered Model (Reform = 1 ) ---
```

Full model summary:

Call:

```
coxph(formula = Surv(survival_time, event) ~ Distance + gdpir +  
      log_pop, data = sub_data, robust = TRUE, cluster = country)
```

n= 218, number of events= 104

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
Distance	-6.641e-04	9.993e-01	6.585e-05	1.073e-04	-6.192	5.94e-10 ***
gdpir	-1.165e+00	3.121e-01	1.575e+00	6.569e-01	-1.773	0.07629 .
log_pop	-2.522e-01	7.771e-01	6.746e-02	8.675e-02	-2.907	0.00365 **

```
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Distance	0.9993	1.001	0.99913	0.9995
gdpir	0.3121	3.204	0.08611	1.1310
log_pop	0.7771	1.287	0.65561	0.9211

Concordance= 0.994 (se = 0.007)

Likelihood ratio test= 242.2 on 3 df, p=<2e-16

Wald test = 49.57 on 3 df, p=1e-10

Score (logrank) test = 268.2 on 3 df, p=<2e-16, Robust = 16.44 p=9e-04

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

R² (explained variation): 0.6707

Adjusted R²: 0.9923

`height` was translated to `width`.

Warning: No shared levels found between `names(values)` of the manual scale and the data's colour values.

Cox Model Forest Plot - Reform=1 Full Model

Hazard Ratios with 95% Confidence Intervals (* indicates truncated confidence interval)



```
cat("\n--- Reform Subgroup Analysis Completed ---\n")
```

```
--- Reform Subgroup Analysis Completed ---
```

```
cat("\n\n", paste(rep("=", 80), collapse = ""), "\n\n")
```

```
=====
```

```
cat(">>> Subgroup Analysis: WTO = 0 vs WTO = 1 <<<\n")
```

```
>>> Subgroup Analysis: WTO = 0 vs WTO = 1 <<<
```

```
cat(paste(rep("=", 80), collapse = ""), "\n\n")
```

```
=====
```

```
event_counts_wto <- merged_data_complete %>%  
  group_by(WTO) %>%  
  summarise(  
    N_countries = n_distinct(country),  
    N_obs = n(),  
    N_events = sum(event),
```

```

    .groups = "drop"
  )
  cat("\n--- Event Counts by WTO Status ---\n")

```

--- Event Counts by WTO Status ---

```
print(event_counts_wto)
```

A tibble: 2 × 4

	WTO	N_countries	N_obs	N_events
	<int>	<int>	<int>	<dbl>
1	0	17	292	102
2	1	22	139	139

```

for (wto_val in c(0, 1)) {

  cat("\n\n", paste(rep("-", 60), collapse = ""))
  cat(sprintf("\n>>> Analyzing Subset: WTO = %d <<<\n", wto_val))
  cat(paste(rep("-", 60), collapse = ""), "\n")

  sub_data <- merged_data_complete %>% filter(WTO == wto_val)

  if (sum(sub_data$event) < 5) {
    cat("WARNING: Too few events in this subgroup. Skipping detailed analysis.\n")
    next
  }

  subgroup_vars <- c("Reform", "Distance", "gdpin", "log_pop")

  cat("\n--- Univariate Cox Analysis (WTO =", wto_val, ") ---\n")
  uni_res_sub <- data.frame()
  for (var in subgroup_vars) {
    tryCatch({
      fml <- as.formula(paste("Surv(survival_time, event) ~", var))
      fit <- coxph(fml, data = sub_data)
      s <- summary(fit)
      hr <- s$coefficients[1, 2]
      ci_low <- s$conf.int[1, 3]
      ci_high <- s$conf.int[1, 4]
      pval <- s$coefficients[1, 5]
      sig <- case_when(
        pval < 0.001 ~ "****",
        pval < 0.01 ~ "***",
        pval < 0.05 ~ "**",
        TRUE ~ ""
      )
    })
    uni_res_sub <- bind_rows(uni_res_sub, data.frame(
      Variable = var,
      HR = round(hr, 3),
      CI_lower = round(ci_low, 3),
      CI_upper = round(ci_high, 3),
      p_value = round(pval, 4),

```



```

        Significance = sig,
        stringsAsFactors = FALSE
    ))
}, error = function(e) {
    cat("  Univariate failed for", var, ":", e$message, "\n")
})
}
if (nrow(uni_res_sub) > 0) {
    uni_res_sub <- uni_res_sub %>% mutate(
        HR_CI = sprintf("%.3f (%.3f-%.3f)%s", HR, CI_lower, CI_upper, Significance)
    )
    print(uni_res_sub %>% dplyr::select(Variable, HR_CI, p_value))
} else {
    cat("  No univariate results.\n")
}

cat("\n--- Full Multivariate Clustered Model (WTO =", wto_val, ") ---\n")
cat("Note: Union excluded from model.\n")
tryCatch({
    fml_full <- as.formula(paste("Surv(survival_time, event) ~",
                                paste(subgroup_vars, collapse = " + "),
                                "+ cluster(country)"))
    fit_full <- coxph(fml_full, data = sub_data, robust = TRUE)
    cat("Full model summary:\n")
    print(summary(fit_full))
    cat(sprintf("R² (explained variation): %.4f\n", summary(fit_full)$rsq[1]))
    cat(sprintf("Adjusted R²: %.4f\n", summary(fit_full)$rsq[2]))

    create_improved_forest_plot(fit_full, paste0("WTO=", wto_val, " Full Model"))
}, error = function(e) {
    cat("Full clustered model failed:", e$message, "\n")
})
}

```

```

-----
>>> Analyzing Subset: WTO = 0 <<<
-----

```

```

--- Univariate Cox Analysis (WTO = 0 ) ---

```

```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 1 ; coefficient may be infinite.

```

	Variable	HR_CI	p_value
1	Reform	0.000 (0.000-Inf)	0.9934
2	Distance	0.999 (0.999-0.999)***	0.0000
3	gdpir	0.028 (0.007-0.108)***	0.0000
4	log_pop	0.427 (0.360-0.508)***	0.0000

```

--- Full Multivariate Clustered Model (WTO = 0 ) ---
Note: Union excluded from model.

```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 1 ; coefficient may be infinite.

Full model summary:

Call:

```
coxph(formula = Surv(survival_time, event) ~ Reform + Distance +  
      gdpir + log_pop, data = sub_data, robust = TRUE, cluster = country)
```

n= 292, number of events= 102

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
Reform	-1.853e+01	8.966e-09	2.212e+03	1.122e+00	-16.519	< 2e-16 ***
Distance	-5.546e-04	9.994e-01	6.741e-05	1.935e-04	-2.866	0.00416 **
gdpir	6.242e-02	1.064e+00	7.567e-01	2.939e-01	0.212	0.83180
log_pop	-6.778e-01	5.077e-01	9.526e-02	2.765e-01	-2.452	0.01422 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	8.966e-09	1.115e+08	9.950e-10	8.080e-08
Distance	9.994e-01	1.001e+00	9.991e-01	9.998e-01
gdpir	1.064e+00	9.395e-01	5.984e-01	1.893e+00
log_pop	5.077e-01	1.970e+00	2.953e-01	8.729e-01

Concordance= 0.941 (se = 0.035)

Likelihood ratio test= 330.8 on 4 df, p=<2e-16

Wald test = 547.1 on 4 df, p=<2e-16

Score (logrank) test = 341.5 on 4 df, p=<2e-16, Robust = 11.22 p=0.02

(Note: the likelihood ratio and score tests assume independence of
observations within a cluster, the Wald and robust score tests do not).

R² (explained variation): 0.6779

Adjusted R²: 0.9782

=== Extreme Value Details for WT0=0 Full Model ===

The following variables have large confidence intervals that were truncated in the plot:

Reform (Yes vs No): HR=8.97e-09, 95% CI=(9.95e-10, 8.08e-08)

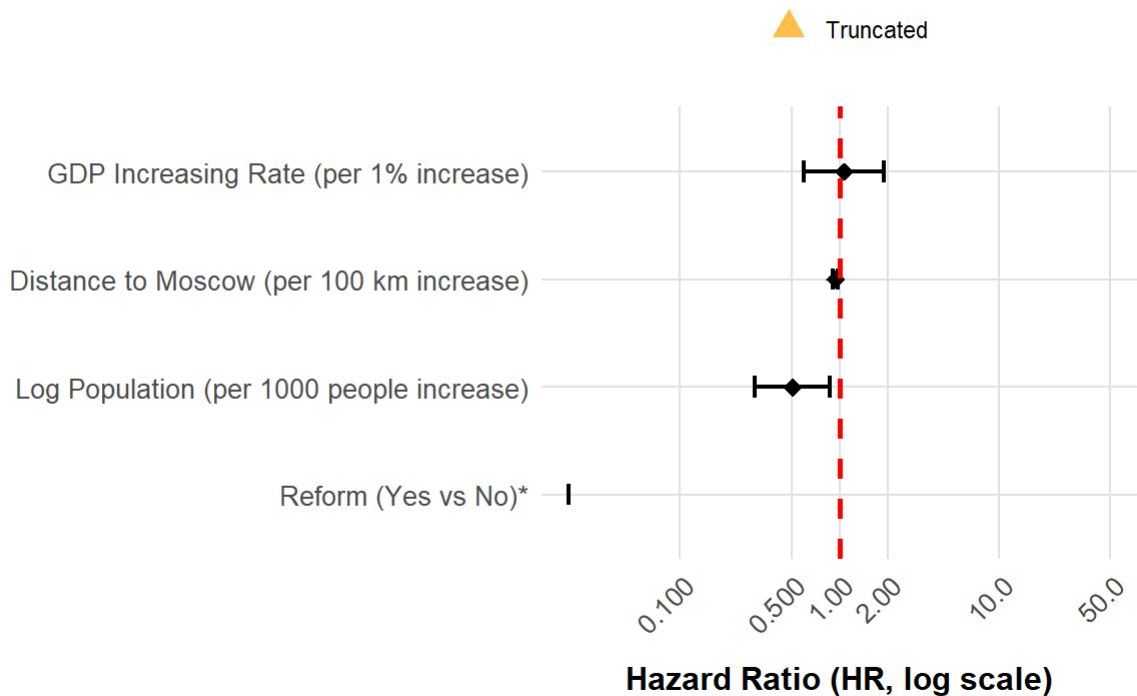
`height` was translated to `width`.

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Cox Model Forest Plot - WTO=0 Full Model

Hazard Ratios with 95% Confidence Intervals (* indicates truncated coefficient)



```
>>> Analyzing Subset: WTO = 1 <<<
```

```
--- Univariate Cox Analysis (WTO = 1 ) ---
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :  
Loglik converged before variable 1 ; coefficient may be infinite.
```

	Variable	HR	CI	p_value
1	Reform	0.000	(0.000-Inf)	0.9920
2	Distance	1.000	(1.000-1.000)	0.3343
3	gdpir	3.280	(0.184-58.408)	0.4189
4	log_pop	1.064	(0.928-1.219)	0.3771

```
--- Full Multivariate Clustered Model (WTO = 1 ) ---
```

```
Note: Union excluded from model.
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :  
Loglik converged before variable 1 ; coefficient may be infinite.
```

```
Full model summary:
```

```
Call:
```

```
coxph(formula = Surv(survival_time, event) ~ Reform + Distance +  
      gdpir + log_pop, data = sub_data, robust = TRUE, cluster = country)
```

n= 139, number of events= 139

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
Reform	-2.142e+01	4.991e-10	2.146e+03	2.591e-01	-82.675	<2e-16 ***
Distance	-6.515e-05	9.999e-01	1.012e-04	6.457e-05	-1.009	0.313
gdpir	2.573e-01	1.293e+00	1.588e+00	2.426e-01	1.061	0.289
log_pop	-4.221e-02	9.587e-01	8.153e-02	4.402e-02	-0.959	0.338

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	4.991e-10	2.004e+09	3.004e-10	8.293e-10
Distance	9.999e-01	1.000e+00	9.998e-01	1.000e+00
gdpir	1.293e+00	7.731e-01	8.040e-01	2.081e+00
log_pop	9.587e-01	1.043e+00	8.794e-01	1.045e+00

Concordance= 0.984 (se = 0.018)

Likelihood ratio test= 152.3 on 4 df, p=<2e-16

Wald test = 6910 on 4 df, p=<2e-16

Score (logrank) test = 221.9 on 4 df, p=<2e-16, Robust = 9.76 p=0.04

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

R² (explained variation): 0.6658

Adjusted R²: 0.9996

=== Extreme Value Details for WT0=1 Full Model ===

The following variables have large confidence intervals that were truncated in the plot:

Reform (Yes vs No): HR=4.99e-10, 95% CI=(3.00e-10, 8.29e-10)

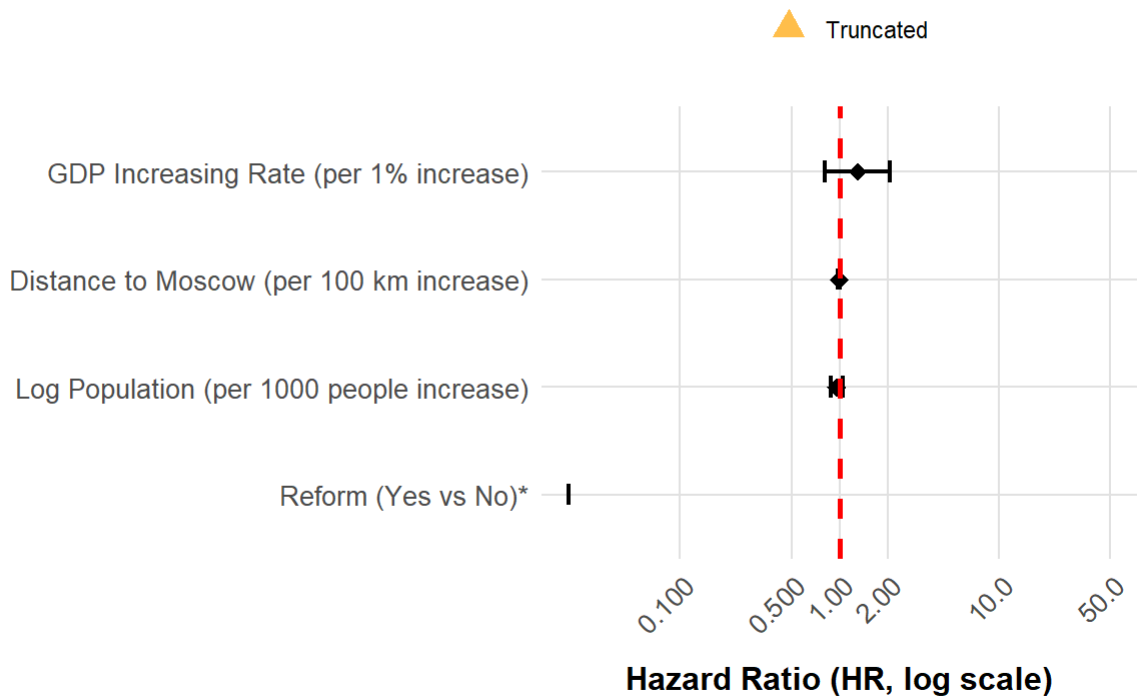
`height` was translated to `width`.

Warning: Removed 1 row containing missing values or values outside the scale range (`geom_point()`).

Removed 1 row containing missing values or values outside the scale range (`geom_point()`).

Cox Model Forest Plot - WTO=1 Full Model

Hazard Ratios with 95% Confidence Intervals (* indicates truncated confidence interval)



```
cat("\n--- WTO Subgroup Analysis Completed ---\n")
```

--- WTO Subgroup Analysis Completed ---

```
cat("\n\n", paste(rep("=", 80), collapse = ""), "\n")
```

=====

```
cat(">>> Subgroup Analysis: Union = 0 vs Union = 1 <<<\n")
```

>>> Subgroup Analysis: Union = 0 vs Union = 1 <<<

```
cat(paste(rep("=", 80), collapse = ""), "\n\n")
```

=====

```
event_counts_union <- merged_data_complete %>%
  group_by(Union) %>%
  summarise(
    N_countries = n_distinct(country),
    N_obs = n(),
    N_events = sum(event),
```

```

    .groups = "drop"
  )
  cat("\n--- Event Counts by Union Status ---\n")

```

--- Event Counts by Union Status ---

```
print(event_counts_union)
```

A tibble: 2 × 4

	Union	N_countries	N_obs	N_events
	<int>	<int>	<int>	<dbl>
1	0	16	275	85
2	1	23	156	156

```

for (union_val in c(0, 1)) {

  cat("\n\n", paste(rep("-", 60), collapse = ""))
  cat(sprintf("\n>>> Analyzing Subset: Union = %d <<<\n", union_val))
  cat(paste(rep("-", 60), collapse = ""), "\n")

  sub_data <- merged_data_complete %>% filter(Union == union_val)

  if (sum(sub_data$event) < 5) {
    cat("WARNING: Too few events in this subgroup. Skipping detailed analysis.\n")
    next
  }

  subgroup_vars <- c("Reform", "Distance", "gdpir", "log_pop")

  cat("\n--- Univariate Cox Analysis (Union =", union_val, ") ---\n")
  uni_res_sub <- data.frame()
  for (var in subgroup_vars) {
    tryCatch({
      fml <- as.formula(paste("Surv(survival_time, event) ~", var))
      fit <- coxph(fml, data = sub_data)
      s <- summary(fit)
      hr <- s$coefficients[1, 2]
      ci_low <- s$conf.int[1, 3]
      ci_high <- s$conf.int[1, 4]
      pval <- s$coefficients[1, 5]
      sig <- case_when(
        pval < 0.001 ~ "****",
        pval < 0.01 ~ "***",
        pval < 0.05 ~ "**",
        TRUE ~ ""
      )
    })
    uni_res_sub <- bind_rows(uni_res_sub, data.frame(
      Variable = var,
      HR = round(hr, 3),
      CI_lower = round(ci_low, 3),
      CI_upper = round(ci_high, 3),
      p_value = round(pval, 4),

```

```

    Significance = sig,
    stringsAsFactors = FALSE
  ))
}, error = function(e) {
  cat("  Univariate failed for", var, ":", e$message, "\n")
})
}
if (nrow(uni_res_sub) > 0) {
  uni_res_sub <- uni_res_sub %>% mutate(
    HR_CI = sprintf("%.3f (%.3f-%.3f)%s", HR, CI_lower, CI_upper, Significance)
  )
  print(uni_res_sub %>% dplyr::select(Variable, HR_CI, p_value))
} else {
  cat("  No univariate results.\n")
}

cat("\n--- Full Multivariate Clustered Model (Union =", union_val, ") ---\n")
cat("Note: WTO excluded from model.\n")
tryCatch({
  fml_full <- as.formula(paste("Surv(survival_time, event) ~",
                              paste(subgroup_vars, collapse = " + "),
                              "+ cluster(country)"))
  fit_full <- coxph(fml_full, data = sub_data, robust = TRUE)
  cat("Full model summary:\n")
  print(summary(fit_full))
  cat(sprintf("R² (explained variation): %.4f\n", summary(fit_full)$rsq[1]))
  cat(sprintf("Adjusted R²: %.4f\n", summary(fit_full)$rsq[2]))

  create_improved_forest_plot(fit_full, paste0("Union=", union_val, " Full Model"))
}, error = function(e) {
  cat("Full clustered model failed:", e$message, "\n")
})
}

```

```

-----
>>> Analyzing Subset: Union = 0 <<<
-----

```

```

--- Univariate Cox Analysis (Union = 0 ) ---

```

```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 1 ; coefficient may be infinite.

```

	Variable	HR_CI	p_value
1	Reform	0.000 (0.000-Inf)	0.9939
2	Distance	0.999 (0.999-0.999)***	0.0000
3	gdpir	0.057 (0.010-0.328)**	0.0013
4	log_pop	0.592 (0.486-0.721)***	0.0000

```

--- Full Multivariate Clustered Model (Union = 0 ) ---
Note: WTO excluded from model.

```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 1 ; coefficient may be infinite.

Full model summary:

Call:

```
coxph(formula = Surv(survival_time, event) ~ Reform + Distance +  
      gdpir + log_pop, data = sub_data, robust = TRUE, cluster = country)
```

n= 275, number of events= 85

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
Reform	-1.895e+01	5.902e-09	2.940e+03	1.156e+00	-16.397	< 2e-16 ***
Distance	-6.173e-04	9.994e-01	7.158e-05	2.074e-04	-2.976	0.00292 **
gdpir	2.594e-02	1.026e+00	7.517e-01	2.418e-01	0.107	0.91456
log_pop	-6.812e-01	5.060e-01	1.629e-01	4.828e-01	-1.411	0.15827

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	5.902e-09	1.694e+08	6.129e-10	5.684e-08
Distance	9.994e-01	1.001e+00	9.990e-01	9.998e-01
gdpir	1.026e+00	9.744e-01	6.389e-01	1.649e+00
log_pop	5.060e-01	1.976e+00	1.964e-01	1.304e+00

Concordance= 0.929 (se = 0.038)

Likelihood ratio test= 273 on 4 df, p=<2e-16

Wald test = 587.1 on 4 df, p=<2e-16

Score (logrank) test = 319.1 on 4 df, p=<2e-16, Robust = 9.05 p=0.06

(Note: the likelihood ratio and score tests assume independence of
observations within a cluster, the Wald and robust score tests do not).

R² (explained variation): 0.6294

Adjusted R²: 0.9655

=== Extreme Value Details for Union=0 Full Model ===

The following variables have large confidence intervals that were truncated in the plot:

Reform (Yes vs No): HR=5.90e-09, 95% CI=(6.13e-10, 5.68e-08)

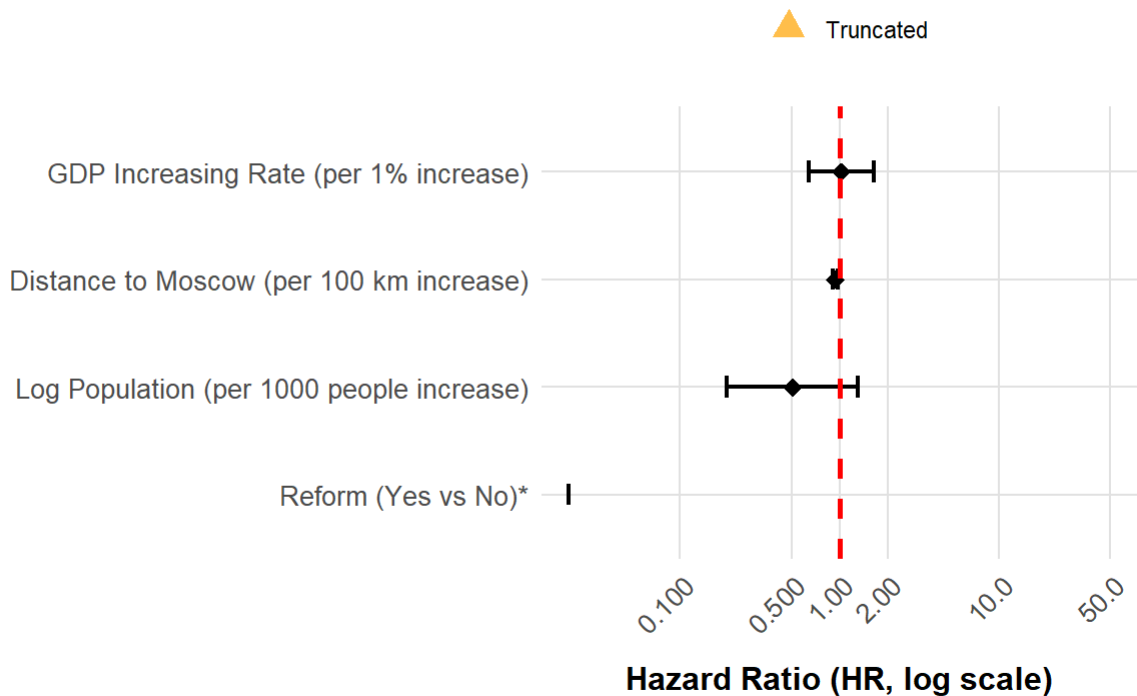
`height` was translated to `width`.

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Removed 1 row containing missing values or values outside the scale range
(`geom\_point()`).

Cox Model Forest Plot - Union=0 Full M

Hazard Ratios with 95% Confidence Intervals (* indicates truncated CI)



```
>>> Analyzing Subset: Union = 1 <<<
```

```
--- Univariate Cox Analysis (Union = 1 ) ---
```

Variable	HR	CI	p_value
1 Reform	2.753	(1.754-4.321)***	0.0000
2 Distance	1.000	(1.000-1.000)	0.1805
3 gdpir	26.522	(2.398-293.349)**	0.0075
4 log_pop	1.142	(1.008-1.293)*	0.0364

```
--- Full Multivariate Clustered Model (Union = 1 ) ---
```

Note: WTO excluded from model.

Full model summary:

Call:

```
coxph(formula = Surv(survival_time, event) ~ Reform + Distance +  
      gdpir + log_pop, data = sub_data, robust = TRUE, cluster = country)
```

n= 156, number of events= 156

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
Reform	8.992e-01	2.458e+00	2.464e-01	7.456e-01	1.206	0.22783
Distance	-8.035e-05	9.999e-01	1.006e-04	6.187e-05	-1.299	0.19402
gdpir	2.185e+00	8.893e+00	1.243e+00	8.160e-01	2.678	0.00741 **
log_pop	1.020e-02	1.010e+00	7.013e-02	9.026e-02	0.113	0.91007

```
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Reform	2.4576	0.4069	0.5699	10.597
Distance	0.9999	1.0001	0.9998	1.000
gdpir	8.8930	0.1124	1.7966	44.020
log_pop	1.0102	0.9899	0.8464	1.206

Concordance= 0.586 (se = 0.139)

Likelihood ratio test= 26.64 on 4 df, p=2e-05

Wald test = 44.39 on 4 df, p=5e-09

Score (logrank) test = 24.33 on 4 df, p=7e-05, Robust = 11.31 p=0.02

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

R² (explained variation): 0.1570

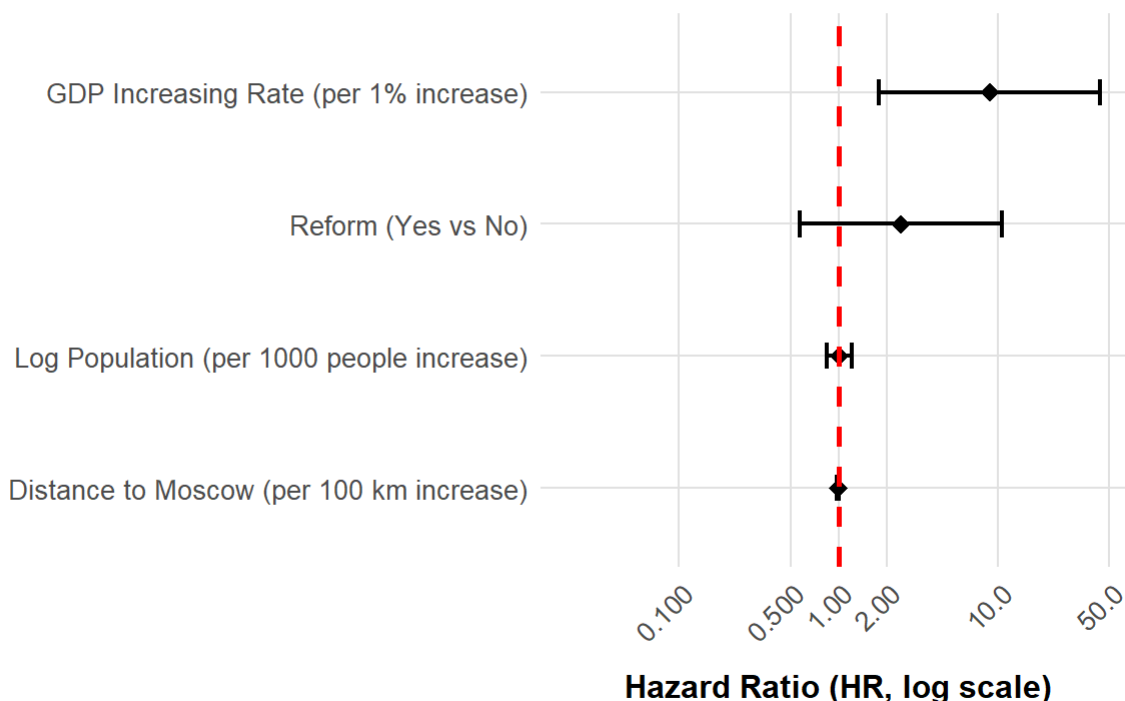
Adjusted R²: 0.9997

`height` was translated to `width`.

Warning: No shared levels found between `names(values)` of the manual scale and the data's colour values.

Cox Model Forest Plot - Union=1 Full M

Hazard Ratios with 95% Confidence Intervals (* indicates truncated c



```
cat("\n--- Union Subgroup Analysis Completed ---\n")
```

--- Union Subgroup Analysis Completed ---

```
cat("=== All Subgroup Analyses Completed ===\n")
```

=== All Subgroup Analyses Completed ===

```
# 14. Generate Analysis Report
```

```
cat("\n=== Analysis Summary ===\n")
```

=== Analysis Summary ===

```
cat("Number of countries analyzed:", n_distinct(merged_data_complete$country), "\n")
```

Number of countries analyzed: 39

```
cat("Total observation years:", nrow(merged_data_complete), "\n")
```

Total observation years: 431

```
cat("Institutional transformation events:", sum(merged_data_complete$event), "\n")
```

Institutional transformation events: 241

```
cat("Censored observations:", nrow(merged_data_complete) - sum(merged_data_complete$event), "\n")
```

Censored observations: 190

```
cat("\n=== ICC Analysis Summary ===\n")
```

=== ICC Analysis Summary ===

```
icc_values <- list()
if (exists("icc_gdpir")) icc_values$gdpir <- icc_gdpir
if (exists("icc_log_pop")) icc_values$log_pop <- icc_log_pop

if (length(icc_values) > 0) {
  for (var_name in names(icc_values)) {
    cat(var_name, "ICC:", round(icc_values[[var_name]], 4), "\n")
  }
  avg_icc <- mean(unlist(icc_values), na.rm = TRUE)
  cat("Average ICC across key variables:", round(avg_icc, 4), "\n")

  if (avg_icc >= 0.05) {
    cat("Conclusion: ICC ≥ 0.05 indicates non-negligible within-country correlation.\n")
    cat("Methodological decision: Clustered Cox models were used.\n")
  } else {
    cat("Conclusion: ICC < 0.05 indicates negligible within-country correlation.\n")
  }
}
```

```
}
}
```

gdpir ICC: 0.1198

log_pop ICC: 0.9947

Average ICC across key variables: 0.5573

Conclusion: ICC ≥ 0.05 indicates non-negligible within-country correlation.

Methodological decision: Clustered Cox models were used.

```
if (exists("final_model") && !is.null(final_model)) {
  cat("\nModel fitting successful\n")
  model_summary <- summary(final_model)
  if (exists("model_type")) {
    cat("Final model type:",
      switch(model_type,
        "full" = "Full model (with gdpir)",
        "reduced" = "Reduced model (without gdpir)",
        "full_time_dep" = "Full time-dependent covariates model",
        "Unknown"), "\n")
  }
  cat("Model likelihood ratio test p-value:", round(model_summary$logtest["pvalue"], 4), "\n")
  cat("R2 (explained variation):", round(model_summary$rsq[1], 4), "\n")
  cat("Adjusted R2:", round(model_summary$rsq[2], 4), "\n")
  cat("Concordance (C-index):", round(model_summary$concordance[1], 4), "\n")

  final_vars <- names(model_summary$coefficients[, 1])
  if ("gdpir" %in% final_vars) {
    cat("\nGDP growth rate (gdpir) in final model: YES\n")
    gdpir_index <- which(final_vars == "gdpir")
    gdpir_hr <- round(model_summary$coefficients[gdpir_index, 2], 3)
    gdpir_p <- round(model_summary$coefficients[gdpir_index, 5], 4)
    cat(sprintf("gdpir HR: %.3f, p-value: %.4f\n", gdpir_hr, gdpir_p))
  } else {
    cat("\nGDP growth rate (gdpir) in final model: NO\n")
    cat("Possible reasons:\n")
    cat("1. Not selected in stepwise regression (if any)\n")
    cat("2. Highly correlated with other variables in the model\n")
    cat("3. Not statistically significant for event risk\n")
  }

  time_interaction_vars <- grep("^tt\\(", final_vars, value = TRUE)
  if (length(time_interaction_vars) > 0) {
    cat("\nTime interaction effects in final model:\n")
    for (var in time_interaction_vars) {
      var_index <- which(final_vars == var)
      var_hr <- round(model_summary$coefficients[var_index, 2], 3)
      var_p <- round(model_summary$coefficients[var_index, 5], 4)
      var_name_clean <- gsub("^tt\\((.*)\\)$", "\\1", var)
      cat(sprintf(" %20s: HR = %.3f, p-value = %.4f\n",
        paste(var_name_clean, "(time interaction)", var_hr, var_p))
    }
  }
} else {
```

```
cat("\nModel fitting failed\n")
}
```

Model fitting successful

Final model type: Full time-dependent covariates model

Model likelihood ratio test p-value: 0

R² (explained variation): 0.8936

Adjusted R²: 0.9983

Concordance (C-index): 0.9781

GDP growth rate (gdpir) in final model: YES

gdpir HR: 1.433, p-value: 1.9879

Time interaction effects in final model:

Reform (time interaction): HR = 0.099, p-value = -2.2151

WTO (time interaction): HR = 539.997, p-value = 3.1244

Distance (time interaction): HR = 1.000, p-value = -3.2716

Union (time interaction): HR = 1.279, p-value = 0.5788

log_pop (time interaction): HR = 0.755, p-value = -2.5452

GDP Growth Rate Comparison Analysis Before and After Transition

```
# 15. Load necessary packages
library(tidyverse)
library(broom)

# 16. Data Import
dataset1_path <- "D:/paper/dataset1.csv"
dataset1 <- read.csv(dataset1_path, stringsAsFactors = FALSE, fileEncoding = "UTF-8-BOM")

cat("Dataset1 dimensions:", dim(dataset1), "\n")
```

Dataset1 dimensions: 39 219

```
cat("Dataset1 year range:", range(dataset1$Year, na.rm = TRUE), "\n")
```

Dataset1 year range: 1985 2022

```
if (!exists("static_data")) {
  stop("static_data does not exist, please ensure static data is loaded")
}

# 17. Identify countries that underwent transition and their transition years (excluding East Germany)
transform_countries <- static_data %>%
  filter(event == 1 & !is.na(Endpoint) & country != "Eg") %>%
  dplyr::select(country, transform_year = Endpoint)

cat("Number of countries that underwent transition (excluding East Germany):", nrow(transform_countries))
```

Number of countries that underwent transition (excluding East Germany): 33

```
print(transform_countries)
```

	country	transform_year
1	Mongolia	1992
2	Kyrgyzstan	1991
3	Kazakhstan	1991
4	Tajikistan	1991
5	Turkmenistan	1991
6	Uzbekistan	1991
7	Cambodia	1989
8	Myanmar	1988
9	Armenia	1991
10	Azerbaijan	1991
11	Georgia	1991
12	Iraq	2003
13	Syria	2000
14	Bulgaria	1989
15	Belarus	1991
16	Czech	1989
17	Hungary	1989
18	Moldova	1991
19	Poland	1989
20	Romania	1989
21	Russia	1991
22	Slovakia	1989
23	Ukraine	1991
24	Estonia	1991
25	Lithuania	1990
26	Latvia	1991
27	Albania	1992
28	BAH	1992
29	Serbia	1992
30	Croatia	1990
31	Montenegro	1992
32	Macedonia	1991
33	Slovenia	1989

```
# 18. Extract GDP growth rate data for transition countries
transform_gdpir_data <- data.frame()

for(i in 1:nrow(transform_countries)) {
  country <- transform_countries$country[i]
  transform_year <- transform_countries$transform_year[i]

  gdpir_col <- paste0(country, "gdpir")

  if(gdpir_col %in% names(dataset1)) {
    years <- c((transform_year-3):(transform_year-1), (transform_year+1):(transform_year+3))
    country_data <- dataset1 %>%
      filter(Year %in% years) %>%
      dplyr::select(Year, gdpir = all_of(gdpir_col)) %>%
      mutate(
        country = country,
```

```

      transform_year = transform_year,
      period = ifelse(Year < transform_year, "pre", "post"),
      years_from_transform = Year - transform_year
    )

    transform_gdpir_data <- bind_rows(transform_gdpir_data, country_data)
  } else {
    cat("Warning: GDP growth rate column", gdpir_col, "for", country, "not found in dataset1\n")
  }
}

# 19. Data cleaning and quality check
transform_gdpir_clean <- transform_gdpir_data %>%
  filter(!is.na(gdpir) & gdpir != "") %>%
  mutate(
    gdpir_clean = gsub("%", "", gdpir),
    gdpir_clean = gsub(",", ".", gdpir_clean),
    gdpir_numeric = as.numeric(gdpir_clean) / 100
  ) %>%
  filter(!is.na(gdpir_numeric) & is.finite(gdpir_numeric)) %>%
  filter(gdpir_numeric >= -0.5 & gdpir_numeric <= 0.5)

cat("Number of observations after cleaning:", nrow(transform_gdpir_clean), "\n")

```

Number of observations after cleaning: 197

```
cat("Countries involved:", paste(unique(transform_gdpir_clean$country), collapse = ", "), "\n")
```

Countries involved: Mongolia, Kyrgyzstan, Kazakhstan, Tajikistan, Turkmenistan, Uzbekistan, Cambodia, Myanmar, Armenia, Azerbaijan, Georgia, Iraq, Syria, Bulgaria, Belarus, Czech, Hungary, Moldova, Poland, Romania, Russia, Slovakia, Ukraine, Estonia, Lithuania, Latvia, Albania, BAH, Serbia, Croatia, Montenegro, Macedonia, Slovenia

```

country_data_summary <- transform_gdpir_clean %>%
  group_by(country, period) %>%
  summarise(
    n_years = n(),
    years_list = paste(sort(unique(years_from_transform)), collapse = ", "),
    .groups = 'drop'
  )

cat("\n=== Data Completeness Check by Country ===\n")

```

=== Data Completeness Check by Country ===

```
print(country_data_summary)
```

```

# A tibble: 66 × 4
  country    period n_years years_list

```

```

  <chr>      <chr>      <int> <chr>
1 Albania   post          3 1, 2, 3
2 Albania   pre          3 -3, -2, -1
3 Armenia   post          3 1, 2, 3
4 Armenia   pre          3 -3, -2, -1
5 Azerbaijan post          3 1, 2, 3
6 Azerbaijan pre          3 -3, -2, -1
7 BAH       post          3 1, 2, 3
8 BAH       pre          3 -3, -2, -1
9 Belarus   post          3 1, 2, 3
10 Belarus  pre          3 -3, -2, -1
# i 56 more rows

```

```
# 20. Descriptive statistics
```

```
cat("\n=== Descriptive Statistics of GDP Growth Rate Before and After Transition (3-year window) ===")
```

```
=== Descriptive Statistics of GDP Growth Rate Before and After Transition (3-year window) ===
```

```

desc_stats <- transform_gdpir_clean %>%
  group_by(period) %>%
  summarise(
    n_countries = n_distinct(country),
    n_observations = n(),
    mean_gdpir = mean(gdpir_numeric, na.rm = TRUE),
    sd_gdpir = sd(gdpir_numeric, na.rm = TRUE),
    median_gdpir = median(gdpir_numeric, na.rm = TRUE),
    min_gdpir = min(gdpir_numeric, na.rm = TRUE),
    max_gdpir = max(gdpir_numeric, na.rm = TRUE)
  )
print(desc_stats)

```

```
# A tibble: 2 × 8
```

```

  period n_countries n_observations mean_gdpir sd_gdpir median_gdpir min_gdpir
  <chr>      <int>          <int>      <dbl>    <dbl>        <dbl>    <dbl>
1 post          33             98   -0.0865   0.123       -0.0806   -0.452
2 pre           33             99    0.00156  0.0668       0.0079   -0.276
# i 1 more variable: max_gdpir <dbl>

```

```
# 21. Calculate average GDP growth rate for each country 3 years before and after transition
```

```

country_means <- transform_gdpir_clean %>%
  filter(years_from_transform >= -3 & years_from_transform <= 3 & years_from_transform != 0) %
  group_by(country, period) %>%
  summarise(
    mean_gdpir = mean(gdpir_numeric, na.rm = TRUE),
    n_years = n(),
    years_used = paste(sort(unique(years_from_transform)), collapse = ", "),
    .groups = 'drop'
  ) %>%
  pivot_wider(names_from = period, values_from = c(mean_gdpir, n_years, years_used)) %>%
  filter(!is.na(mean_gdpir_pre) & !is.na(mean_gdpir_post) &

```



```
n_years_pre >= 2 & n_years_post >= 2) # Allow missing 1 year of data

cat("\nNumber of countries available for paired analysis:", nrow(country_means), "\n")
```

Number of countries available for paired analysis: 33

```
if(nrow(country_means) > 0) {
  print(country_means %>% dplyr::select(country, mean_gdpir_pre, mean_gdpir_post, n_years_pre,
}
```

A tibble: 33 × 5

	country	mean_gdpir_pre	mean_gdpir_post	n_years_pre	n_years_post
2	Armenia	-0.0112	-0.193	3	3
3	Azerbaijan	-0.0461	-0.210	3	3
4	BAH	-0.0769	-0.0849	3	3
5	Belarus	0.0093	-0.107	3	3
6	Bulgaria	0.00733	-0.0862	3	3
7	Cambodia	0.0210	0.0328	3	3
8	Croatia	-0.00847	-0.123	3	3
9	Czech	0.0238	-0.0298	3	3
10	Estonia	0.0106	-0.0940	3	3

i 23 more rows

22. Statistical testing

```
if(nrow(country_means) >= 3) {
  paired_data <- country_means %>%
    dplyr::select(country, pre = mean_gdpir_pre, post = mean_gdpir_post)

  differences <- paired_data$post - paired_data$pre

  cat("\n=== GDP Growth Rate Differences Before and After Transition ===\n")
  diff_summary <- data.frame(
    Country = paired_data$country,
    Pre = round(paired_data$pre, 4),
    Post = round(paired_data$post, 4),
    Difference = round(differences, 4),
    Percent_Change = round((differences / paired_data$pre) * 100, 1)
  )
  print(diff_summary)

# 22.1 Normality test
shapiro_test <- shapiro.test(differences)
cat("\n=== Normality Test ===\n")
cat("Shapiro-Wilk normality test p-value:", round(shapiro_test$p.value, 4), "\n")

# 22.2 Select test method based on normality
if(shapiro_test$p.value > 0.05) {
  cat("Differences satisfy normal distribution, using paired t-test\n")
}
```

```

paired_t_test <- t.test(paired_data$post, paired_data$pre,
                        paired = TRUE, conf.level = 0.95)

cat("\n=== Paired t-test Results ===\n")
print(paired_t_test)

mean_diff <- mean(differences)
sd_diff <- sd(differences)
cohens_d <- mean_diff / sd_diff
cat("\nEffect size analysis:\n")
cat("Mean difference:", round(mean_diff, 4), "(", round(mean_diff * 100, 2), "%)\n")
cat("Standard deviation:", round(sd_diff, 4), "\n")
cat("Cohen's d (effect size):", round(cohens_d, 3), "\n")

effect_size_interpretation <- case_when(
  abs(cohens_d) < 0.2 ~ "negligible",
  abs(cohens_d) < 0.5 ~ "small",
  abs(cohens_d) < 0.8 ~ "medium",
  TRUE ~ "large"
)
cat("Effect size magnitude:", effect_size_interpretation, "\n")

test_method <- "Paired t-test"
p_value <- paired_t_test$p.value
ci_lower <- paired_t_test$conf.int[1]
ci_upper <- paired_t_test$conf.int[2]

} else {
  cat("Differences do not satisfy normal distribution, using Wilcoxon signed-rank test\n")

  wilcoxon_test <- wilcox.test(paired_data$post, paired_data$pre,
                              paired = TRUE, exact = FALSE, conf.int = TRUE)

  cat("\n=== Wilcoxon Signed-Rank Test Results ===\n")
  print(wilcoxon_test)

  n_pairs <- nrow(paired_data)
  z_value <- qnorm(wilcoxon_test$p.value / 2)
  r_effect <- abs(z_value) / sqrt(n_pairs * 2) # Modified effect size calculation

  cat("\nEffect size analysis:\n")
  cat("Median difference:", round(median(differences), 4), "(", round(median(differences) *
  cat("Effect size r:", round(r_effect, 3), "\n")

  effect_size_interpretation <- case_when(
    r_effect < 0.1 ~ "negligible",
    r_effect < 0.3 ~ "small",
    r_effect < 0.5 ~ "medium",
    TRUE ~ "large"
  )
  cat("Effect size magnitude:", effect_size_interpretation, "\n")

  test_method <- "Wilcoxon signed-rank test"

```

```

p_value <- wilcoxon_test$p.value
ci_lower <- wilcoxon_test$conf.int[1]
ci_upper <- wilcoxon_test$conf.int[2]
}

# 23. Detailed country-level analysis
cat("\n=== Country-level Detailed Analysis ===\n")
for(i in 1:nrow(paired_data)) {
  country <- paired_data$country[i]
  pre_mean <- paired_data$pre[i]
  post_mean <- paired_data$post[i]
  change <- post_mean - pre_mean
  change_pct <- ifelse(pre_mean != 0, (change / pre_mean) * 100, NA)

  cat(sprintf("%-12s: Before=%.2f%%, After=%.2f%%, Change=%.2f%%",
              country, pre_mean*100, post_mean*100, change*100))
  if(!is.na(change_pct)) {
    cat(sprintf(" (%.1f%%)\n", change_pct))
  } else {
    cat(" (N/A)\n")
  }
}

# 24. Visualization
# 24.1 Paired comparison plot
comparison_data <- paired_data %>%
  pivot_longer(cols = c(pre, post),
               names_to = "period",
               values_to = "gdpir") %>%
  mutate(period = factor(period, levels = c("pre", "post"),
                        labels = c("3-year average before transition", "3-year average after

paired_plot <- ggplot(comparison_data,
                      aes(x = period, y = gdpir, group = country)) +
  geom_line(alpha = 0.6, color = "gray", size = 1) +
  geom_point(size = 3, alpha = 0.8, aes(color = country)) +
  stat_summary(aes(group = 1), fun = mean, geom = "line",
               size = 2, color = "red", group = 1) +
  stat_summary(aes(group = 1), fun = mean, geom = "point",
               size = 4, color = "red", group = 1, shape = 18) +
  labs(
    title = "Paired Comparison of GDP Growth Rates Before and After Socialist Transition",
    subtitle = "Red line shows overall mean change trend",
    x = "Period",
    y = "GDP Growth Rate",
    color = "Country"
  ) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.1)) +
  theme_minimal() +
  theme(legend.position = "bottom")

print(paired_plot)

# 24.2 Difference distribution plot

```

```

diff_plot <- ggplot(data.frame(difference = differences),
                      aes(x = difference)) +
  geom_histogram(binwidth = 0.01, fill = "lightblue", color = "black", alpha = 0.7) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red", size = 1) +
  geom_vline(xintercept = mean(differences), linetype = "solid", color = "blue", size = 1) +
  labs(
    title = "Distribution of GDP Growth Rate Differences Before and After Transition",
    x = "After - Before GDP Growth Rate Difference",
    y = "Frequency"
  ) +
  scale_x_continuous(labels = scales::percent_format(accuracy = 0.1)) +
  theme_minimal()

print(diff_plot)

# 24.3 Trend plot (annual data for all countries)
trend_plot <- ggplot(transform_gdpir_clean,
                      aes(x = years_from_transform, y = gdpir_numeric,
                          group = country, color = country)) +
  geom_line(size = 1, alpha = 0.8) +
  geom_point(size = 2) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red", size = 1) +
  geom_smooth(aes(group = 1), method = "lm", se = TRUE, color = "black", linetype = "dashed")
  labs(
    title = "Annual Trend of GDP Growth Rates Before and After Socialist Transition",
    subtitle = "Dashed line shows transition year, black dashed line shows overall trend",
    x = "Years from Transition",
    y = "GDP Growth Rate",
    color = "Country"
  ) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.1)) +
  theme_minimal() +
  theme(legend.position = "bottom")

print(trend_plot)

} else {
  cat("\nInsufficient number of countries for meaningful statistical testing\n")
  cat("Current number of available countries:", nrow(country_means), ", need at least 3\n")
  test_method <- "Cannot perform"
  p_value <- NA
  ci_lower <- NA
  ci_upper <- NA
}

```

=== GDP Growth Rate Differences Before and After Transition ===

	Country	Pre	Post	Difference	Percent_Change
1	Albania	-0.0745	0.0987	0.1733	-232.5
2	Armenia	-0.0112	-0.1934	-0.1822	1621.7
3	Azerbaijan	-0.0461	-0.2095	-0.1634	354.5
4	BAH	-0.0769	-0.0849	-0.0080	10.4
5	Belarus	0.0093	-0.1068	-0.1161	-1248.0

6	Bulgaria	0.0073	-0.0862	-0.0935	-1275.5
7	Cambodia	0.0210	0.0328	0.0118	56.1
8	Croatia	-0.0085	-0.1229	-0.1144	1351.6
9	Czech	0.0238	-0.0298	-0.0536	-224.9
10	Estonia	0.0106	-0.0940	-0.1046	-987.1
11	Georgia	-0.0241	-0.2853	-0.2613	1085.6
12	Hungary	0.0169	-0.0608	-0.0777	-460.5
13	Iraq	0.1719	0.1041	-0.0678	-39.4
14	Kazakhstan	0.0213	-0.0855	-0.1068	-500.6
15	Kyrgyzstan	0.0483	-0.1638	-0.2121	-439.2
16	Latvia	0.0346	-0.1447	-0.1793	-518.1
17	Lithuania	0.0308	-0.1529	-0.1837	-595.9
18	Macedonia	-0.0384	-0.0486	-0.0102	26.7
19	Moldova	0.0143	-0.2178	-0.2321	-1626.6
20	Mongolia	-0.0105	0.0685	0.0790	-750.3
21	Montenegro	-0.0500	-0.0524	-0.0024	4.8
22	Myanmar	-0.0073	0.0143	0.0216	-296.3
23	Poland	0.0135	-0.0417	-0.0552	-408.6
24	Romania	-0.0021	-0.0849	-0.0827	3878.1
25	Russia	-0.0006	-0.1001	-0.0995	17564.7
26	Serbia	-0.0654	-0.0519	0.0134	-20.6
27	Slovakia	0.0285	-0.0793	-0.1078	-378.1
28	Slovenia	-0.0010	-0.0760	-0.0750	7503.3
29	Syria	-0.0327	-0.0067	0.0260	-79.4
30	Tajikistan	0.0142	-0.2189	-0.2330	-1644.9
31	Turkmenistan	0.0038	-0.0923	-0.0961	-2551.3
32	Ukraine	0.0018	-0.1425	-0.1443	-8016.7
33	Uzbekistan	0.0287	-0.0767	-0.1054	-367.4

=== Normality Test ===

Shapiro-Wilk normality test p-value: 0.5442

Differences satisfy normal distribution, using paired t-test

=== Paired t-test Results ===

Paired t-test

data: paired_data\$post and paired_data\$pre

t = -5.2033, df = 32, p-value = 1.1e-05

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

-0.11988569 -0.05242946

sample estimates:

mean difference

-0.08615758

Effect size analysis:

Mean difference: -0.0862 (-8.62 %)

Standard deviation: 0.0951

Cohen's d (effect size): -0.906

Effect size magnitude: large

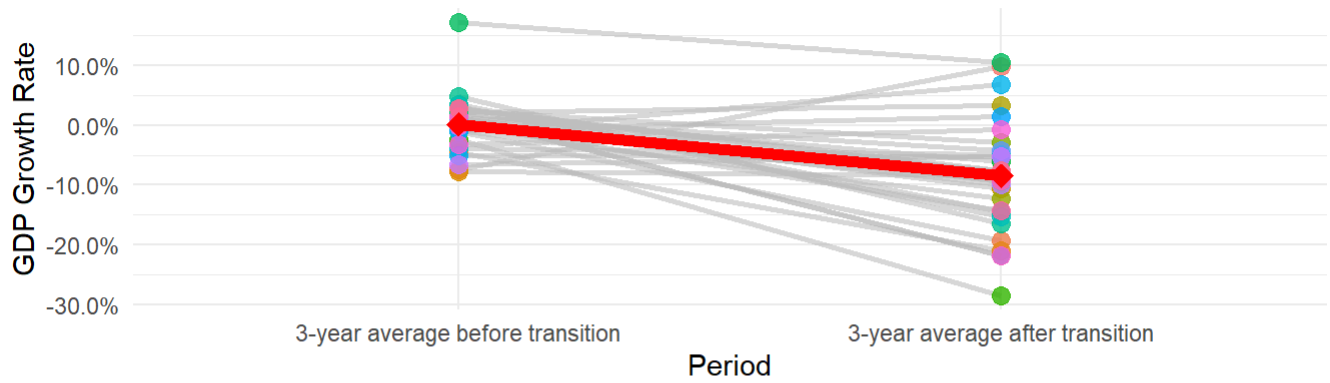
=== Country-level Detailed Analysis ===

Albania	: Before=-7.45%, After=9.87%, Change=17.33% (-232.5%)
Armenia	: Before=-1.12%, After=-19.34%, Change=-18.22% (1621.7%)
Azerbaijan	: Before=-4.61%, After=-20.95%, Change=-16.34% (354.5%)
BAH	: Before=-7.69%, After=-8.49%, Change=-0.80% (10.4%)
Belarus	: Before=0.93%, After=-10.68%, Change=-11.61% (-1248.0%)
Bulgaria	: Before=0.73%, After=-8.62%, Change=-9.35% (-1275.5%)
Cambodia	: Before=2.10%, After=3.28%, Change=1.18% (56.1%)
Croatia	: Before=-0.85%, After=-12.29%, Change=-11.44% (1351.6%)
Czech	: Before=2.38%, After=-2.98%, Change=-5.36% (-224.9%)
Estonia	: Before=1.06%, After=-9.40%, Change=-10.46% (-987.1%)
Georgia	: Before=-2.41%, After=-28.53%, Change=-26.13% (1085.6%)
Hungary	: Before=1.69%, After=-6.08%, Change=-7.77% (-460.5%)
Iraq	: Before=17.19%, After=10.41%, Change=-6.78% (-39.4%)
Kazakhstan	: Before=2.13%, After=-8.55%, Change=-10.68% (-500.6%)
Kyrgyzstan	: Before=4.83%, After=-16.38%, Change=-21.21% (-439.2%)
Latvia	: Before=3.46%, After=-14.47%, Change=-17.93% (-518.1%)
Lithuania	: Before=3.08%, After=-15.29%, Change=-18.37% (-595.9%)
Macedonia	: Before=-3.84%, After=-4.86%, Change=-1.02% (26.7%)
Moldova	: Before=1.43%, After=-21.78%, Change=-23.21% (-1626.6%)
Mongolia	: Before=-1.05%, After=6.85%, Change=7.90% (-750.3%)
Montenegro	: Before=-5.00%, After=-5.24%, Change=-0.24% (4.8%)
Myanmar	: Before=-0.73%, After=1.43%, Change=2.16% (-296.3%)
Poland	: Before=1.35%, After=-4.17%, Change=-5.52% (-408.6%)
Romania	: Before=-0.21%, After=-8.49%, Change=-8.27% (3878.1%)
Russia	: Before=-0.06%, After=-10.01%, Change=-9.95% (17564.7%)
Serbia	: Before=-6.54%, After=-5.19%, Change=1.34% (-20.6%)
Slovakia	: Before=2.85%, After=-7.93%, Change=-10.78% (-378.1%)
Slovenia	: Before=-0.10%, After=-7.60%, Change=-7.50% (7503.3%)
Syria	: Before=-3.27%, After=-0.67%, Change=2.60% (-79.4%)
Tajikistan	: Before=1.42%, After=-21.89%, Change=-23.30% (-1644.9%)
Turkmenistan	: Before=0.38%, After=-9.23%, Change=-9.61% (-2551.3%)
Ukraine	: Before=0.18%, After=-14.25%, Change=-14.43% (-8016.7%)
Uzbekistan	: Before=2.87%, After=-7.67%, Change=-10.54% (-367.4%)

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

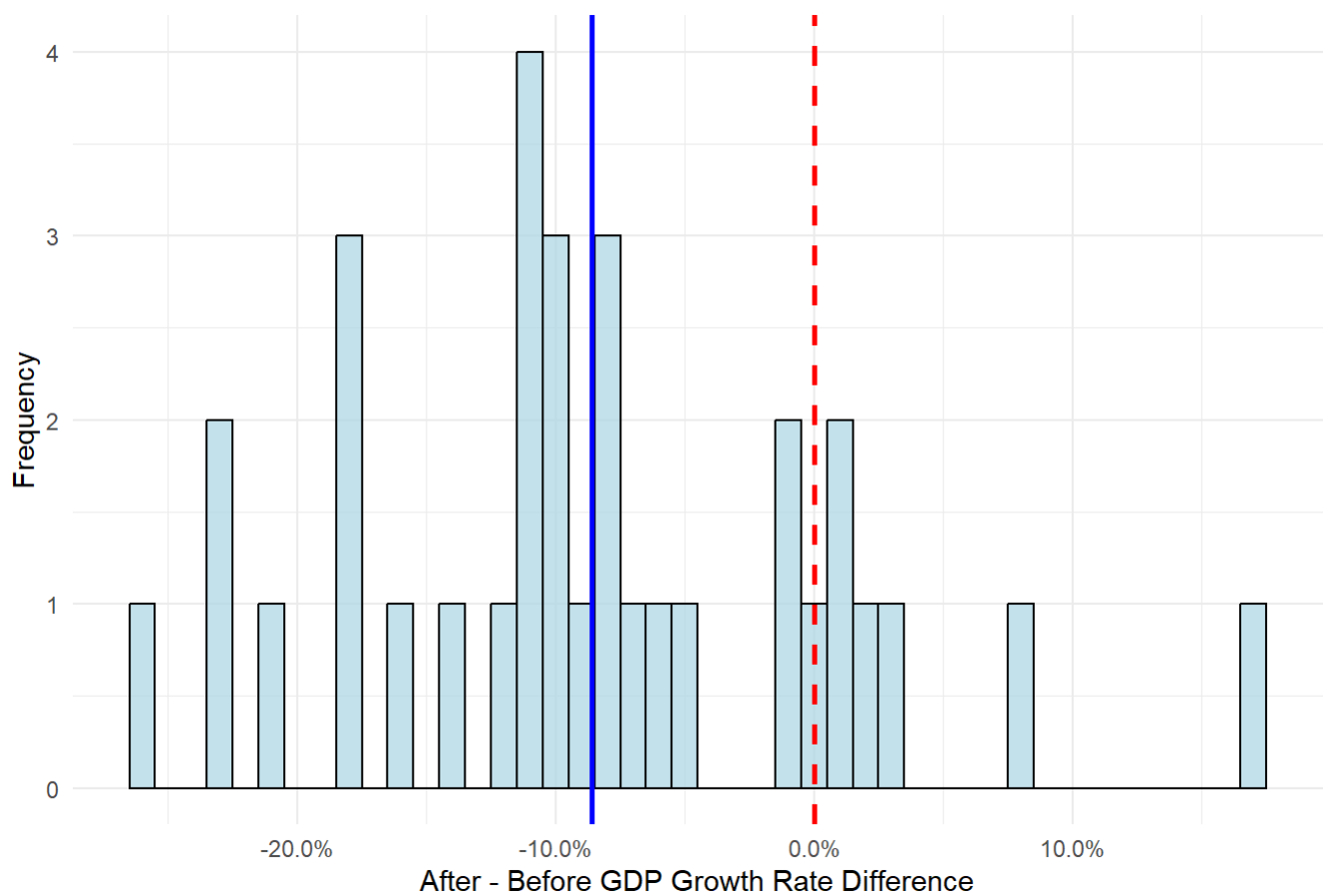
Paired Comparison of GDP Growth Rates Before and After Socialist Transition

Red line shows overall mean change trend



Albania	Croatia	Kyrgyzstan	Myanmar	Syria
Armenia	Czech	Latvia	Poland	Tajikistan
Azerbaijan	Estonia	Lithuania	Romania	Turkmenistan
BAH	Georgia	Macedonia	Russia	Ukraine
Belarus	Hungary	Moldova	Serbia	Uzbekistan
Bulgaria	Iraq	Mongolia	Slovakia	
Cambodia	Kazakhstan	Montenegro	Slovenia	

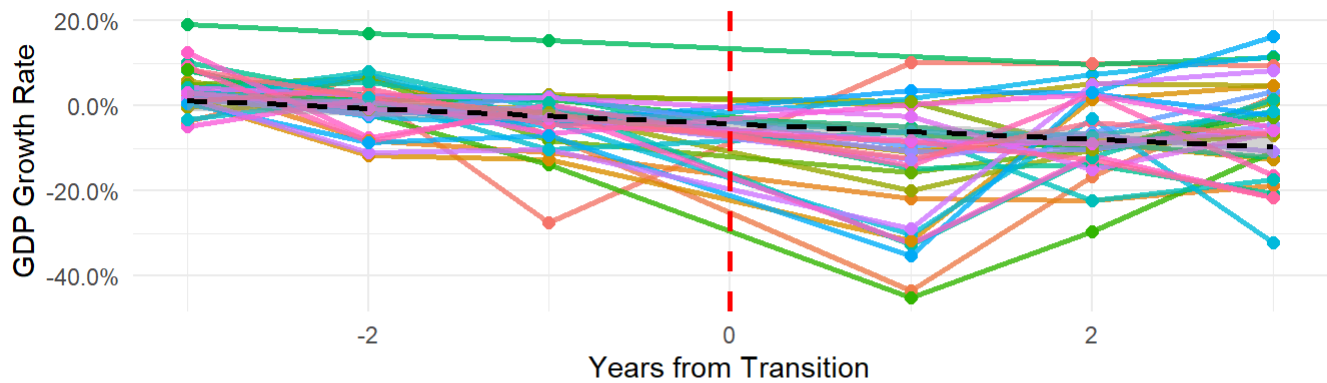
Distribution of GDP Growth Rate Differences Before and After Transition



`geom\_smooth()` using formula = 'y ~ x'

Annual Trend of GDP Growth Rates Before and After Socialist Transition

Dashed line shows transition year, black dashed line shows overall trend



Country

Albania	Croatia	Kyrgyzstan	Myanmar	Syria
Armenia	Czech	Latvia	Poland	Tajikistan
Azerbaijan	Estonia	Lithuania	Romania	Turkmenistan
BAH	Georgia	Macedonia	Russia	Ukraine
Belarus	Hungary	Moldova	Serbia	Uzbekistan
Bulgaria	Iraq	Mongolia	Slovakia	
Cambodia	Kazakhstan	Montenegro	Slovenia	

```
# 25. Results summary
```

```
cat("\n", strrep("=", 60), "\n", sep = "")
```

```
cat("Summary of GDP Growth Rate Comparison Analysis Before and After Transition\n")
```

Summary of GDP Growth Rate Comparison Analysis Before and After Transition

```
cat(strrep("=", 60), "\n", sep = "")
```

```
if(exists("paired_data") && nrow(paired_data) > 0) {
  analysis_countries <- paste(unique(paired_data$country), collapse = ", ")
} else {
  analysis_countries <- paste(unique(country_means$country), collapse = ", ")
}
cat("Analysis countries:", analysis_countries, "\n")
```

Analysis countries: Albania, Armenia, Azerbaijan, BAH, Belarus, Bulgaria, Cambodia, Croatia, Czech, Estonia, Georgia, Hungary, Iraq, Kazakhstan, Kyrgyzstan, Latvia, Lithuania, Macedonia, Moldova, Mongolia, Montenegro, Myanmar, Poland, Romania, Russia, Serbia, Slovakia, Slovenia, Syria, Tajikistan, Turkmenistan, Ukraine, Uzbekistan


```
cat("Time window: 3 years before and after transition (excluding transition year itself)\n")
```

Time window: 3 years before and after transition (excluding transition year itself)

```
cat("Data source: dataset1.csv (contains complete post-transition data)\n")
```

Data source: dataset1.csv (contains complete post-transition data)

```
if(exists("paired_data") && nrow(paired_data) > 0) {  
  cat("Average GDP growth rate before transition:", round(mean(paired_data$pre, na.rm = TRUE)  
  cat("Average GDP growth rate after transition:", round(mean(paired_data$post, na.rm = TRUE)  
  cat("Average difference:", round(mean(differences, na.rm = TRUE) * 100, 2), "%\n")  
  
  if(nrow(paired_data) >= 3) {  
    cat("Statistical test:", test_method, "\n")  
    cat("p-value:", round(p_value, 4), "\n")  
    if(!is.na(ci_lower) && !is.na(ci_upper)) {  
      cat("95% confidence interval: [",  
        round(ci_lower * 100, 2), "%, ",  
        round(ci_upper * 100, 2), "%]\n", sep = "")  
    }  
  
    if(p_value < 0.05) {  
      significance <- "statistically significant"  
      direction <- ifelse(mean(differences) > 0, "increased", "decreased")  
      cat("Conclusion: There is a", significance, "difference in GDP growth rates before and a  
    } else {  
      cat("Conclusion: No statistically significant difference in GDP growth rates before and  
    }  
  }  
} else {  
  cat("Insufficient data for statistical analysis\n")  
}
```

Average GDP growth rate before transition: 0.16 %

Average GDP growth rate after transition: -8.46 %

Average difference: -8.62 %

Statistical test: Paired t-test

p-value: 0

95% confidence interval: [-11.99%, -5.24%]

Conclusion: There is a statistically significant difference in GDP growth rates before and after transition, GDP growth rate decreased after transition

```
cat("Analysis completed!\n")
```

Analysis completed!

```
# 26. Analysis results table  
if(exists("paired_data") && nrow(paired_data) > 0) {  
  results_df <- data.frame(  

```

```

Country = paired_data$country,
Avg_GDP_Growth_Pre = round(paired_data$pre * 100, 2),
Avg_GDP_Growth_Post = round(paired_data$post * 100, 2),
Difference = round(differences * 100, 2),
Percent_Change = round((differences / paired_data$pre) * 100, 1)
)

overall_stats <- data.frame(
  Country = "Overall",
  Avg_GDP_Growth_Pre = round(mean(paired_data$pre) * 100, 2),
  Avg_GDP_Growth_Post = round(mean(paired_data$post) * 100, 2),
  Difference = round(mean(differences) * 100, 2),
  Percent_Change = round((mean(differences) / mean(paired_data$pre)) * 100, 1)
)

results_df <- bind_rows(results_df, overall_stats)

cat("\n=== Analysis Results Table ===\n")
print(results_df)
}

```

=== Analysis Results Table ===

	Country	Avg_GDP_Growth_Pre	Avg_GDP_Growth_Post	Difference
1	Albania	-7.45	9.87	17.33
2	Armenia	-1.12	-19.34	-18.22
3	Azerbaijan	-4.61	-20.95	-16.34
4	BAH	-7.69	-8.49	-0.80
5	Belarus	0.93	-10.68	-11.61
6	Bulgaria	0.73	-8.62	-9.35
7	Cambodia	2.10	3.28	1.18
8	Croatia	-0.85	-12.29	-11.44
9	Czech	2.38	-2.98	-5.36
10	Estonia	1.06	-9.40	-10.46
11	Georgia	-2.41	-28.53	-26.13
12	Hungary	1.69	-6.08	-7.77
13	Iraq	17.19	10.41	-6.78
14	Kazakhstan	2.13	-8.55	-10.68
15	Kyrgyzstan	4.83	-16.38	-21.21
16	Latvia	3.46	-14.47	-17.93
17	Lithuania	3.08	-15.29	-18.37
18	Macedonia	-3.84	-4.86	-1.02
19	Moldova	1.43	-21.78	-23.21
20	Mongolia	-1.05	6.85	7.90
21	Montenegro	-5.00	-5.24	-0.24
22	Myanmar	-0.73	1.43	2.16
23	Poland	1.35	-4.17	-5.52
24	Romania	-0.21	-8.49	-8.27
25	Russia	-0.06	-10.01	-9.95
26	Serbia	-6.54	-5.19	1.34
27	Slovakia	2.85	-7.93	-10.78
28	Slovenia	-0.10	-7.60	-7.50
29	Syria	-3.27	-0.67	2.60
30	Tajikistan	1.42	-21.89	-23.30

31	Turkmenistan	0.38	-9.23	-9.61
32	Ukraine	0.18	-14.25	-14.43
33	Uzbekistan	2.87	-7.67	-10.54
34	Overall	0.16	-8.46	-8.62

Percent_Change

1	-232.5
2	1621.7
3	354.5
4	10.4
5	-1248.0
6	-1275.5
7	56.1
8	1351.6
9	-224.9
10	-987.1
11	1085.6
12	-460.5
13	-39.4
14	-500.6
15	-439.2
16	-518.1
17	-595.9
18	26.7
19	-1626.6
20	-750.3
21	4.8
22	-296.3
23	-408.6
24	3878.1
25	17564.7
26	-20.6
27	-378.1
28	7503.3
29	-79.4
30	-1644.9
31	-2551.3
32	-8016.7
33	-367.4
34	-5538.7

Specialized Analysis of Belarus Before and After Transitions

```
# 27. Load required packages
library(tidyverse)
library(broom)
library(patchwork)
```

Attaching package: 'patchwork'

The following object is masked from 'package:MASS':

area

```
# 28. Data Import
dataset1_path <- "D:/paper/dataset1.csv"
dataset1 <- read.csv(dataset1_path, stringsAsFactors = FALSE, fileEncoding = "UTF-8-BOM")

cat("Specialized Analysis of Belarus\n")
```

Specialized Analysis of Belarus

```
cat("dataset1 data dimensions:", dim(dataset1), "\n")
```

dataset1 data dimensions: 39 219

```
cat("dataset1 year range:", range(dataset1$Year, na.rm = TRUE), "\n")
```

dataset1 year range: 1985 2022

```
# 29. Extract Belarus Data
belarus_data <- dataset1 %>%
  dplyr::select(Year,
               gdpir = Belarusgdpir, # GDP growth rate
               gdppc = Belarusgdppc) %>% # GDP per capita
  filter(Year >= 1985 & Year <= 2004) %>% # 1985-2004
  mutate(
    gdpir_clean = gsub("%", "", gdpir),
    gdpir_clean = gsub(",", ".", gdpir_clean),
    gdpir_numeric = as.numeric(gdpir_clean) / 100,
    gdppc_numeric = as.numeric(gdppc)
  ) %>%
  filter(!is.na(gdpir_numeric) & !is.na(gdppc_numeric)) # Remove missing values

cat("Belarus data year range:", range(belarus_data$Year), "\n")
```

Belarus data year range: 1985 2004

```
cat("Belarus data observations count:", nrow(belarus_data), "\n")
```

Belarus data observations count: 20

```
# 30. Transition Point Definitions

transform_1991 <- 1991 # Became a capitalist country
transform_1994 <- 1994 # Became a socialist country again

cat("\n=== Belarus Transition Points ===\n")
```

=== Belarus Transition Points ===

```
cat("1991: Became a capitalist country\n")
```

1991: Became a capitalist country

```
cat("1994: Became a socialist country again\n")
```

1994: Became a socialist country again

```
# 31. 1991 Transition Comparison
```

```
cat("\n", strrep("=", 50), "\n", sep = "")
```

=====

```
cat("1991 Transition Comparison Analysis\n")
```

1991 Transition Comparison Analysis

```
cat(strrep("=", 50), "\n", sep = "")
```

=====

```
belarus_1991_data <- belarus_data %>%  
  filter(Year >= (transform_1991 - 3) & Year <= (transform_1991 + 3) & Year != transform_1991)  
  mutate(  
    period_1991 = ifelse(Year < transform_1991, "pre_1991", "post_1991")  
  )
```

```
belarus_1991_means <- belarus_1991_data %>%  
  group_by(period_1991) %>%  
  summarise(  
    mean_gdpir = mean(gdpir_numeric, na.rm = TRUE),  
    n_years = n(),  
    years = paste(sort(unique(Year)), collapse = ", ")  
  )
```

```
cat("1991 transition data:\n")
```



1991 transition data:

```
print(belarus_1991_means)
```

A tibble: 2 × 4

	period_1991	mean_gdpir	n_years	years
	<chr>	<dbl>	<int>	<chr>
1	post_1991	-0.107	3	1992, 1993, 1994
2	pre_1991	0.0093	3	1988, 1989, 1990

```
if(nrow(belarus_1991_means) == 2) {  
  pre_1991 <- belarus_1991_means$mean_gdpir[belarus_1991_means$period_1991 == "pre_1991"]
```

```

post_1991 <- belarus_1991_means$mean_gdpir[belarus_1991_means$period_1991 == "post_1991"]
diff_1991 <- post_1991 - pre_1991

cat("\n1991 transition GDP growth rate comparison:\n")
cat("Pre-transition average:", round(pre_1991 * 100, 2), "% (",
    belarus_1991_means$years[belarus_1991_means$period_1991 == "pre_1991"], ")\n")
cat("Post-transition average:", round(post_1991 * 100, 2), "% (",
    belarus_1991_means$years[belarus_1991_means$period_1991 == "post_1991"], ")\n")
cat("Difference:", round(diff_1991 * 100, 2), "%\n")

change_pct_1991 <- (diff_1991 / pre_1991) * 100
cat("Percentage change:", round(change_pct_1991, 1), "%\n")
} else {
  cat("1991 transition data incomplete, cannot compare\n")
}

```

1991 transition GDP growth rate comparison:
 Pre-transition average: 0.93 % (1988, 1989, 1990)
 Post-transition average: -10.68 % (1992, 1993, 1994)
 Difference: -11.61 %
 Percentage change: -1248 %

32. 1994 Transition Comparison

```

cat("\n", strrep("=", 50), "\n", sep = "")

```

=====

```

cat("1994 Transition Comparison Analysis\n")

```

1994 Transition Comparison Analysis

```

cat(strrep("=", 50), "\n", sep = "")

```

=====

```

belarus_1994_data <- belarus_data %>%
  filter(Year >= (transform_1994 - 3) & Year <= (transform_1994 + 3) & Year != transform_1994)
mutate(
  period_1994 = ifelse(Year < transform_1994, "pre_1994", "post_1994")
)

belarus_1994_means <- belarus_1994_data %>%
  group_by(period_1994) %>%
  summarise(
    mean_gdpir = mean(gdpir_numeric, na.rm = TRUE),
    n_years = n(),
    years = paste(sort(unique(Year)), collapse = ", ")
  )

```

```
cat("1994 transition data:\n")
```

1994 transition data:

```
print(belarus_1994_means)
```

```
# A tibble: 2 × 4
  period_1994 mean_gdpir n_years years
  <chr>         <dbl>   <int> <chr>
1 post_1994    -0.00127     3 1995, 1996, 1997
2 pre_1994     -0.0728     3 1991, 1992, 1993
```

```
if(nrow(belarus_1994_means) == 2) {
  pre_1994 <- belarus_1994_means$mean_gdpir[belarus_1994_means$period_1994 == "pre_1994"]
  post_1994 <- belarus_1994_means$mean_gdpir[belarus_1994_means$period_1994 == "post_1994"]
  diff_1994 <- post_1994 - pre_1994

  cat("\n1994 transition GDP growth rate comparison:\n")
  cat("Pre-transition average:", round(pre_1994 * 100, 2), "% (",
      belarus_1994_means$years[belarus_1994_means$period_1994 == "pre_1994"], ")\n")
  cat("Post-transition average:", round(post_1994 * 100, 2), "% (",
      belarus_1994_means$years[belarus_1994_means$period_1994 == "post_1994"], ")\n")
  cat("Difference:", round(diff_1994 * 100, 2), "%\n")

  change_pct_1994 <- (diff_1994 / pre_1994) * 100
  cat("Percentage change:", round(change_pct_1994, 1), "%\n")
} else {
  cat("1994 transition data incomplete, cannot compare\n")
}
```

1994 transition GDP growth rate comparison:
Pre-transition average: -7.28 % (1991, 1992, 1993)
Post-transition average: -0.13 % (1995, 1996, 1997)
Difference: 7.16 %
Percentage change: -98.3 %

```
# 33. Plot Belarus 1985-2004 Trend Graphs
```

```
cat("\n", strrep("=", 50), "\n", sep = "")
```

=====

```
cat("Plotting Belarus 1985-2004 Trend Graphs\n")
```

Plotting Belarus 1985-2004 Trend Graphs

```
cat(strrep("=", 50), "\n", sep = "")
```

```

=====

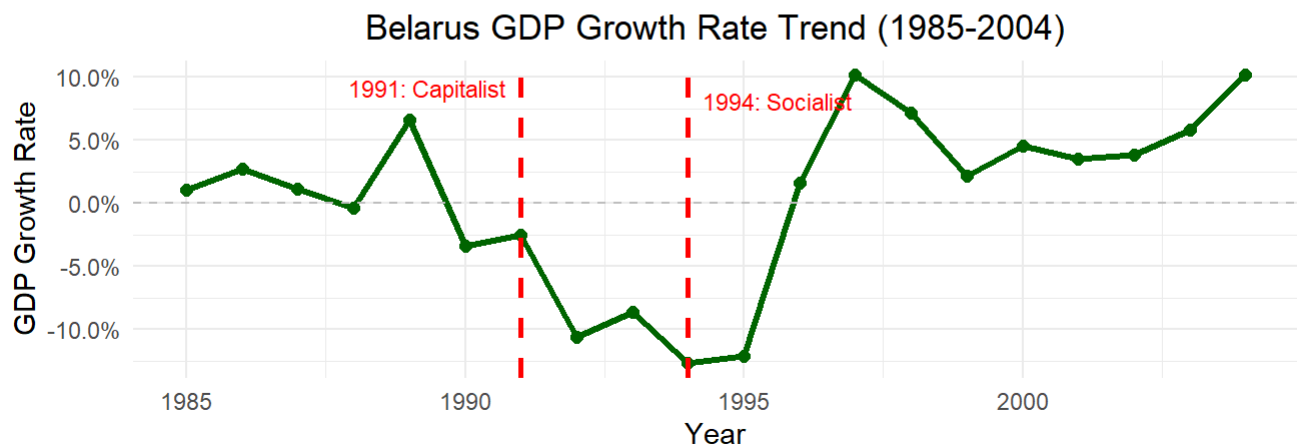
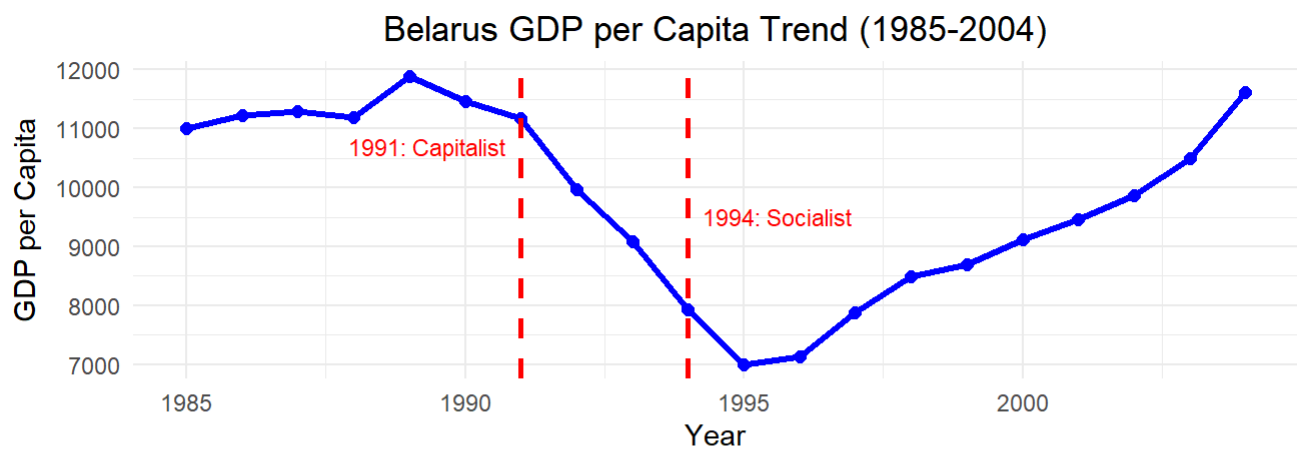
# 33.1 GDP per capita trend plot
gdppc_plot <- ggplot(belarus_data, aes(x = Year, y = gdppc_numeric)) +
  geom_line(color = "blue", size = 1.2) +
  geom_point(color = "blue", size = 2) +
  geom_vline(xintercept = c(transform_1991, transform_1994),
             linetype = "dashed", color = "red", size = 1) +
  annotate("text", x = transform_1991, y = max(belarus_data$gdppc_numeric) * 0.9,
           label = "1991: Capitalist", hjust = 1.1, color = "red", size = 3) +
  annotate("text", x = transform_1994, y = max(belarus_data$gdppc_numeric) * 0.8,
           label = "1994: Socialist", hjust = -0.1, color = "red", size = 3) +
  labs(
    title = "Belarus GDP per Capita Trend (1985-2004)",
    x = "Year",
    y = "GDP per Capita"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

# 33.2 GDP growth rate trend plot
gdpir_plot <- ggplot(belarus_data, aes(x = Year, y = gdpir_numeric)) +
  geom_line(color = "darkgreen", size = 1.2) +
  geom_point(color = "darkgreen", size = 2) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", size = 0.5) +
  geom_vline(xintercept = c(transform_1991, transform_1994),
             linetype = "dashed", color = "red", size = 1) +
  annotate("text", x = transform_1991, y = max(belarus_data$gdpir_numeric) * 0.9,
           label = "1991: Capitalist", hjust = 1.1, color = "red", size = 3) +
  annotate("text", x = transform_1994, y = max(belarus_data$gdpir_numeric) * 0.8,
           label = "1994: Socialist", hjust = -0.1, color = "red", size = 3) +
  labs(
    title = "Belarus GDP Growth Rate Trend (1985-2004)",
    x = "Year",
    y = "GDP Growth Rate"
  ) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.1)) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

# 33.3 Combine plots
combined_plot <- gdppc_plot / gdpir_plot # Using patchwork layout

print(combined_plot)

```

```
# 34. Transition Period Detailed Analysis
```

```
cat("\n", strrep("=", 50), "\n", sep = "")
```

```
cat("Belarus Transition Period Detailed Analysis\n")
```

Belarus Transition Period Detailed Analysis

```
cat(strrep("=", 50), "\n", sep = "")
```

```
belarus_detailed <- belarus_data %>%
  mutate(
    period = case_when(
      Year < transform_1991 ~ "Socialist Period (1985-1990)",
      Year >= transform_1991 & Year < transform_1994 ~ "Capitalist Period (1991-1993)",
      Year >= transform_1994 ~ "Socialist Period (1994-2004)"
    )
  )

period_stats <- belarus_detailed %>%
```

```

group_by(period) %>%
summarise(
  n_years = n(),
  mean_gdpir = mean(gdpir_numeric) * 100,
  sd_gdpir = sd(gdpir_numeric) * 100,
  mean_gdppc = mean(gdppc_numeric),
  sd_gdppc = sd(gdppc_numeric),
  years = paste(min(Year), "-", max(Year))
)

cat("Statistical indicators by period:\n")

```

Statistical indicators by period:

```
print(period_stats)
```

A tibble: 3 × 7

	period <chr>	n_years <int>	mean_gdpir <dbl>	sd_gdpir <dbl>	mean_gdppc <dbl>	sd_gdppc <dbl>	years <chr>
1	Capitalist Period (1991...	3	-7.28	4.22	10070.	1051.	1991...
2	Socialist Period (1985-...	6	1.29	3.31	11348.	307.	1985...
3	Socialist Period (1994-...	11	2.19	7.76	8878.	1420.	1994...

35. Transition Period Visualization

```

period_colors <- c("Socialist Period (1985-1990)" = "red",
                  "Capitalist Period (1991-1993)" = "blue",
                  "Socialist Period (1994-2004)" = "red")

```

35.1 GDP per capita trend colored by period

```

gdppc_period_plot <- ggplot(belarus_detailed, aes(x = Year, y = gdppc_numeric, color = period))
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  scale_color_manual(values = period_colors) +
  labs(
    title = "Belarus GDP per Capita Trend - By Political System Period",
    x = "Year",
    y = "GDP per Capita",
    color = "Political System Period"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom")

```

35.2 GDP growth rate trend colored by period

```

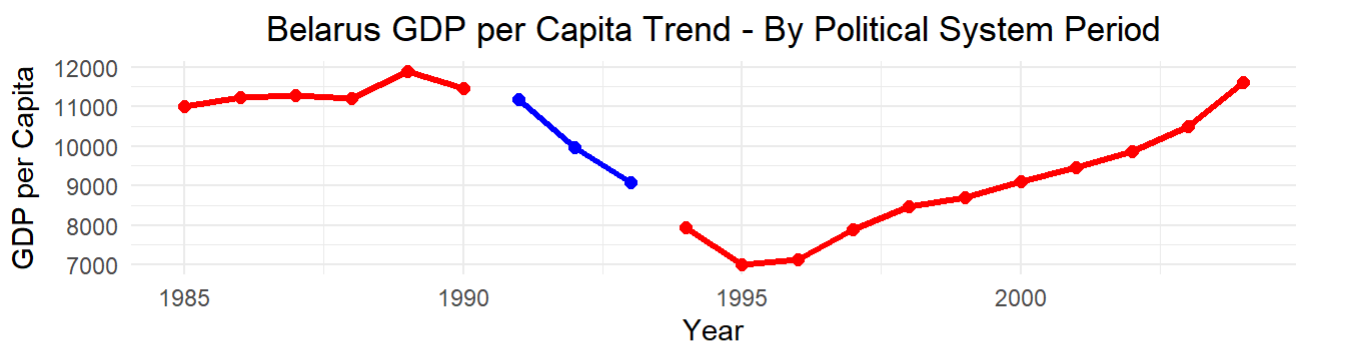
gdpir_period_plot <- ggplot(belarus_detailed, aes(x = Year, y = gdpir_numeric, color = period))
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", size = 0.5) +
  scale_color_manual(values = period_colors) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.1)) +
  labs(
    title = "Belarus GDP Growth Rate Trend - By Political System Period",

```

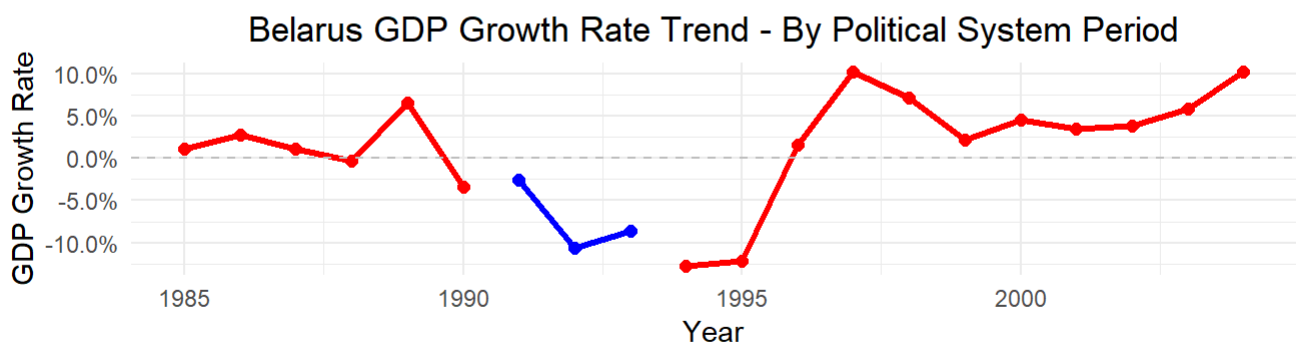
```
x = "Year",
y = "GDP Growth Rate",
color = "Political System Period"
) +
theme_minimal() +
theme(plot.title = element_text(hjust = 0.5),
      legend.position = "bottom")
```

```
combined_period_plot <- gdppc_period_plot / gdpgr_period_plot
```

```
print(combined_period_plot)
```



Political System Period ● Capitalist Period (1991-1993) ● Socialist Period (1985-1990) ● Socialist Period (1994-2005)



Political System Period ● Capitalist Period (1991-1993) ● Socialist Period (1985-1990) ● Socialist Period (1994-2005)

```
# 36. Results Summary
```

```
cat("\n", strrep("=", 60), "\n", sep = "")
```

```
cat("Belarus Transition Analysis Summary\n")
```

Belarus Transition Analysis Summary

```
cat(strrep("=", 60), "\n", sep = "")
```

```
cat("Analysis period: 1985-2004\n")
```

Analysis period: 1985-2004

```
cat("Transition point 1: 1991 - Became a capitalist country\n")
```

Transition point 1: 1991 - Became a capitalist country

```
cat("Transition point 2: 1994 - Became a socialist country again\n\n")
```

Transition point 2: 1994 - Became a socialist country again

```
if(exists("diff_1991")) {  
  cat("=== 1991 Transition Comparison ===\n")  
  cat("Pre-transition GDP growth rate (1988-1990):", round(pre_1991 * 100, 2), "%\n")  
  cat("Post-transition GDP growth rate (1992-1994):", round(post_1991 * 100, 2), "%\n")  
  cat("Change:", round(diff_1991 * 100, 2), "%\n")  
  cat("Percentage change:", round(change_pct_1991, 1), "%\n\n")  
}
```

```
=== 1991 Transition Comparison ===  
Pre-transition GDP growth rate (1988-1990): 0.93 %  
Post-transition GDP growth rate (1992-1994): -10.68 %  
Change: -11.61 %  
Percentage change: -1248 %
```

```
if(exists("diff_1994")) {  
  cat("=== 1994 Transition Comparison ===\n")  
  cat("Pre-transition GDP growth rate (1991-1993):", round(pre_1994 * 100, 2), "%\n")  
  cat("Post-transition GDP growth rate (1995-1997):", round(post_1994 * 100, 2), "%\n")  
  cat("Change:", round(diff_1994 * 100, 2), "%\n")  
  cat("Percentage change:", round(change_pct_1994, 1), "%\n\n")  
}
```

```
=== 1994 Transition Comparison ===  
Pre-transition GDP growth rate (1991-1993): -7.28 %  
Post-transition GDP growth rate (1995-1997): -0.13 %  
Change: 7.16 %  
Percentage change: -98.3 %
```

```
cat("=== Average Performance by Period ===\n")
```

```
=== Average Performance by Period ===
```

```
for(i in 1:nrow(period_stats)) {  
  cat(period_stats$period[i], ":\n")  
  cat(" GDP growth rate:", round(period_stats$mean_gdpir[i], 2), "%\n")  
  cat(" GDP per capita:", round(period_stats$mean_gdppc[i], 2), "%\n")  
}
```

```
cat("  Years:", period_stats$years[i], "\n\n")
}
```

Capitalist Period (1991-1993) :

GDP growth rate: -7.28 %

GDP per capita: 10069.67

Years: 1991 - 1993

Socialist Period (1985-1990) :

GDP growth rate: 1.29 %

GDP per capita: 11347.5

Years: 1985 - 1990

Socialist Period (1994-2004) :

GDP growth rate: 2.19 %

GDP per capita: 8878.45

Years: 1994 - 2004

```
cat("Analysis complete!\n")
```

Analysis complete!

```

import geopandas as gpd import matplotlib.pyplot as plt

red_countries = ['CHN', 'PRK', 'LAO', 'VNM', 'CUB']

violet_countries = [ 'MNG', 'KGZ', 'KAZ', 'TJK', 'TKM', 'UZB', 'KHM', 'MMR', 'ARM', 'AZE', 'GEO', 'IRQ',
'SYR', 'BGR', 'CZE', 'HUN', 'MDA', 'POL', 'ROU', 'RUS', 'SVK', 'UKR', 'EST', 'LTU', 'LVA', 'ALB', 'BIH', 'SRB',
'HRV', 'MNE', 'MKD', 'SVN']

pink_countries = ['BLR']

east_germany_states = [ 'Berlin', 'Brandenburg', 'Mecklenburg-Vorpommern', 'Sachsen',
'Sachsen-Anhalt', 'Thüringen']

world_url = "https://naciscdn.org/naturalearth/10m/cultural/ne_10m_admin_0_countries.zip"
world = gpd.read_file(world_url)

world_no_germany = world[world['ADM0_A3'] != 'DEU']

states_url =
"https://naciscdn.org/naturalearth/10m/cultural/ne_10m_admin_1_states_provinces.zip" states =
gpd.read_file(states_url) german_states = states[states['adm0_a3'] == 'DEU'].copy()

world_red = world_no_germany[world_no_germany['ADM0_A3'].isin(red_countries)] world_violet
= world_no_germany[world_no_germany['ADM0_A3'].isin(violet_countries)] world_pink =
world_no_germany[world_no_germany['ADM0_A3'].isin(pink_countries)] world_default =
world_no_germany[~world_no_germany['ADM0_A3'].isin( red_countries + violet_countries +
pink_countries )]

german_states_violet = german_states[german_states['name'].isin(east_germany_states)]
german_states_default = german_states[~german_states['name'].isin(east_germany_states)]

fig, ax = plt.subplots(1, 1, figsize=(15, 10))

if not world_default.empty: world_default.plot(ax=ax, color='lightgray', edgecolor='black',
linewidth=0.1) if not german_states_default.empty: german_states_default.plot(ax=ax,
color='lightgray', edgecolor='black', linewidth=0.1)

if not world_red.empty: world_red.plot(ax=ax, color='red', edgecolor='black', linewidth=0.1)

if not world_violet.empty: world_violet.plot(ax=ax, color='violet', edgecolor='black',
linewidth=0.1)

if not german_states_violet.empty: german_states_violet.plot(ax=ax, color='violet',
edgecolor='black', linewidth=0.1)

if not world_pink.empty: world_pink.plot(ax=ax, color='pink', edgecolor='black', linewidth=0.1)

ax.set_title('World Map with Highlighted Countries', fontsize=16) ax.axis('off')
plt.tight_layout()

fig

```


Countries	Region	Endpoint	Survival	tin Region	fmt WTO	Distance	Capital	Reform
China	East Asia		37+		5	0	5796.23 Beijing	1
North Kor	East Asia		37+		5	0	6423.18 Pyongyang	0
Mongolia	East Asia	1992		7	5	0	4264.51 Ulan Bator	0
Kyrgyzstan	Central As	1991		6	4	1	3002.76 Bishkek	1
Kazakhstan	Central As	1991		6	4	1	3169.42 Alma-Ata	1
Tajikistan	Central As	1991		6	4	1	3026.41 Dushanbe	1
Turkmenistan	Central As	1991		6	4	1	2513.88 Ashgabat	1
Uzbekistan	Central As	1991		6	4	1	2845.67 Tashkent	1
Cambodia	Southeast	1989		4	6	0	7531.69 Phnom Penh	0
Laos	Southeast Asia		37+		6	0	6524.9 Vientiane	1
Myanmar	Southeast	1988		3	6	0	6033.52 Nay Pyi Taw	0
Vietnam	Southeast Asia		37+		6	0	6514.62 Hanoi	1
Armenia	Western A	1991		6	3	1	1800.07 Yerevan	1
Azerbaijan	Western A	1991		6	3	1	2003.42 Baku	1
Georgia	Western A	1991		6	3	1	1693.27 Tbilisi	1
Iraq	Western A	2003		18	3	0	2001.34 Baghdad	0
Syria	Western A	2000		15	3	0	2008.88 Damascus	0
Bulgaria	Eastern Eu	1989		4	1	1	1765.31 Sofia	0
Belarus	Eastern Eu	1991		6	1	1	674.89 Minsk	1
Czech Rep	Eastern Eu	1989		4	1	1	1560.53 Prague	0
Hungary	Eastern Eu	1989		4	1	1	1491.45 Budapest	0
Moldova	Eastern Eu	1991		6	1	1	1117.99 Chisinau	1
Poland	Eastern Eu	1989		4	1	1	1155.93 Warsaw	0
Romania	Eastern Eu	1989		4	1	1	1487.48 Bucharest	0
Russia	Eastern Eu	1991		6	1	1	0 Moscow	1
Slovakia	Eastern Eu	1989		4	1	1	1343.9 Bratislava	0
Ukraine	Eastern Eu	1991		6	1	1	756.32 Kiev	1
Estonia	Eastern Eu	1991		6	1	1	853.52 Tallinn	1
Lithuania	Eastern Eu	1990		5	1	1	797.89 Vilnius	1
Latvia	Eastern Eu	1991		6	1	1	853.33 Riga	1
Albania	Eastern Eu	1992		7	1	0	1839.11 Tirana	0
Bosnia and Herz	Eastern Eu	1992		7	1	0	1868.66 Sarajevo	0
Serbia	Eastern Eu	1992		7	1	0	1803.93 Belgrade	0
Croatia	Eastern Eu	1990		5	1	0	1877.88 Zagreb	0
Montenegro	Eastern Eu	1992		7	1	0	1978.42 Podgorica	0
Macedonia	Eastern Eu	1991		6	1	0	1881.97 Skopje	0
Slovenia	Eastern Eu	1989		4	1	0	1935.08 Ljubljana	0
East Germany	Central Eu	1989		4	2	1	1607.56 Berlin	0
Cuba	North America		37+		7	0	9088.13 Havana	0

Capitalism	Socialism	Capitalism	Union
1912	1949	37	0
	1949	0	0
	1921	0	0
1861	1921	60	1
1861	1924	63	1
1861	1924	63	1
1861	1921	60	1
1861	1921	60	1
1953	1975	22	0
1953	1975	22	0
1948	1962	14	0
1945	1954	9	0
1861	1920	59	1
1861	1920	59	1
1861	1921	60	1
1932	1968	36	0
1946	1970	24	0
1878	1946	68	0
1861	1920	59	1
1867	1948	81	1
1867	1949	82	0
1861	1924	63	1
1861	1945	84	0
1877	1947	70	0
1861	1917	56	1
1867	1948	81	1
1861	1922	61	1
1861	1940	79	1
1861	1940	79	1
1861	1940	79	1
1912	1946	34	0
1878	1945	67	1
1878	1945	67	1
1867	1945	78	1
1878	1945	67	1
1878	1945	67	1
1867	1945	78	1
1871	1949	78	0
1902	1959	57	0

Year	Suvival	tim	Chinapop	NKpop	Mongolia	Kyrgyzstar	Kazakhsta	Tajikistan	Turkmenis
1985	0		1051040	18831	1859	4006	15790	4569	3240
1986	1		1066790	19137	1903	4089	15946	4716	3324
1987	2		1084035	19449	1947	4176	16109	4865	3411
1988	3		1101630	19769	1992	4244	16207	5006	3492
1989	4		1118650	20103	2070	4302	16265	5148	3572
1990	5		1135185	20451	2121	4382	16398	5272	3658
1991	6		1149979	20813	2170	4446	16522	5407	3745
1992	7		1163363	21182	2210				
1993	8		1176727	21563					
1994	9		1189962	21852					
1995	10		1202581	22107					
1996	11		1215171	22324					
1997	12		1227453	22443					
1998	13		1238722	22483					
1999	14		1248899	22594					
2000	15		1258401	22785					
2001	16		1267178	22977					
2002	17		1275379	23165					
2003	18		1283064	23330					
2004	19		1290620	23477					
2005	20		1298245	23621					
2006	21		1305116	23764					
2007	22		1311877	23908					
2008	23		1318559	24051					
2009	24		1324993	24191					
2010	25		1331357	24326					
2011	26		1339549	24457					
2012	27		1349537	24589					
2013	28		1357520	24720					
2014	29		1366654	24852					
2015	30		1373406	24983					
2016	31		1382401	25115					
2017	32		1390136	25248					
2018	33		1395398	25381					
2019	34		1400035	25497					
2020	35		1402060	25608					
2021	36		1402537	25711					
2022	37		1401693	25808					

Uzbekistan	Cambodia	Laos	Myanmar	Vietnam	Armenia	Azerbaijan	Georgia	Iraq
18215	7822	3657	36346	60093	3374	6682	5193	15694
18718	8079	3762	36998	61440	3422	6776	5219	16247
19219	8380	3869	37625	62826	3469	6874	5240	16543
19625	8692	3980	38196	64211	3516	6976	5320	17038
20033	9014	4094		65868	3326	7102	5398	17568
20530		4210		67283	3377	7200	5426	18135
21059		4331		68675	3434	7300	5426	17474
		4454		70032				17863
		4581		71357				18404
		4712		72654				18968
		4846		73904				19555
		4948		75096				20163
		5055		76234				20785
		5168		77323				21422
		5286		78394				22076
		5397		79472				22748
		5499		80546				23440
		5578		81637				24212
		5654		82720				24979
		5746		83821				
		5836		84942				
		5931		86036				
		6035		87136				
		6145		88253				
		6257		89373				
		6368		90481				
		6477		91591				
		6586		92720				
		6695		93883				
		6804		95055				
		6912		96222				
		7019		97380				
		7127		98507				
		7234		99604				
		7343		100695				
		7452		101771				
		7560		102815				
		7666		103574				

Syriapop	Bulgariapop	Belaruspop	Czechpop	Hungarypop	Moldovapop	Polandpop	Romaniapop	Russiapop
10481	8944	9982	10310	10649	4148	37226	22521	143938
10857	8959	10044	10309	10631	4183	37504	22600	144967
11243	8972	10097	10312	10613	4217	37741	22686	145959
11632	8982	10150	10314	10443	4290	37867	22769	146866
12018	8990	10184	10314	10398	4359	37970	22852	147352
12436		10201			4394			147973
12867		10206			4401			148398
13243								
13615								
13989								
14366								
14750								
15139								
15533								
15930								
16329								

Slovakia	Ukraine	Estonia	Lithuania	Latvia	Albania	BAH	Serbia	Croatia
5145	50944	1538	3587	2606	2957	4275	7720	4458
5172	51095	1550	3620	2627	3015	4308	7740	4472
5199	51218	1562	3653	2649	3075	4339	7756	4484
5223	51423	1569	3674	2664	3137	4370	7769	4494
5245	51501	1571	3681	2667	3196	4398	7780	4501
	51622	1569	3695	2664	3251	4424	7786	4508
	51730	1560		2651	3187	4437	7685	
					3078	4233	7646	

Monteneg	Macedoni	Sloveni	apEgpop	Cubapop	Chinagdp	NKgdppc	Mongolia	Kyrgyzstar
561	1845	1914	16660	10084	2619	2415	2004	5616
565	1851	1932	16630	10183	2748	2689	2142	5413
569	1856	1946	16590	10279	2931	2796	2168	5241
574	1859	1962	16560	10382	3045	2553	2227	5679
579	1863	1976	16430	10489	3022	2710	2232	5719
583	1861			10597	2982	2456	2123	5742
591	1868			10697	3051	2316	1976	5162
622				10780	3228	2121	1841	
				10850	3506	1996		
				10893	3730	1926		
				10921	4000	1816		
				10963	4220	1733		
				11014	4311	1605		
				11064	4310	1580		
				11108	4467	1665		
				11147	4730	1657		
				11179	4995	1704		
				11210	5343	1711		
				11241	5649	1730		
				11260	6068	1754		
				11268	6627	1808		
				11266	7271	1778		
				11261	7896	1747		
				11258	8190	1795		
				11260	8734	1769		
				11262	9658	1751		
				11264	10221	1757		
				11262	10947	1771		
				11250	11731	1781		
				11238	12515	1791		
				11214	13325	1763		
				11180	14139	1821		
				11147	15030	1749		
				11116	15977	1669		
				11105	16880	1667		
				11089	17226	1584		
				11046	18667	1577		
				11003	19238	1569		

Kazakhstan	Tajikistan	Turkmenistan	Uzbekistan	Cambodia	Laos	Myanmar	Vietnam	Armenia
11545	5364	6186	7181	1540	1382	1506	1471	10485
11576	5319	6204	6904	1578	1409	1465	1481	10394
11389	5040	6194	6680	1490	1350	1382	1500	9789
12256	5515	6601	7101	1473	1286	1207	1556	9301
12153	4962	5915	7093	1455	1428		1588	10525
11888	4803	5796	6790		1481		1634	9669
10557	4270	5458	6486		1512		1688	8142
					1564		1793	
					1623		1895	
					1718		2018	
					1802		2167	
					1900		2323	
					2002		2467	
					2051		2564	
					2168		2641	
					2262		2773	
					2365		2916	
					2487		3071	
					2625		3242	
					2785		3438	
					2949		3638	
					2896		3829	
					3389		4037	
					3614		4200	
					3844		4360	
					4114		4572	
					4401		4786	
					4670		4988	
					4961		5200	
					5251		5465	
					5559		5776	
					5860		6090	
					6168		6438	
					6455		6842	
					6706		7266	
					6824		7395	
					6961		7508	
					7168		8050	

Azerbaijan	Georgia	Iraq	Syria	Bulgaria	Belarus	Czech	Hungary	Moldova
8558	13868	6268	10026	9924	11000	12855	10452	9398
8663	13839	5992	9200	10170	11230	13211	10678	9937
8995	13707	6052	9055	10173	11297	13514	10861	9852
9328	14665	4635	9913	10098	11198	13791	11207	9731
8421	14161	4098	8735	9908	11893	14027	11003	10288
7394	12140	3918	9087		11467			9835
7319	9535	1519	9424		11171			7958
		2046	10002					
		2132	10072					
		2109	9969					
		1997	9856					
		2338	9517					
		2338	8827					
		2795	8743					
		3157	7960					
		3649	7668					
		4148						
		4632						
		3782						

[illegible]

[illegible]

Myanmar	Vietnam	Armenia	Azerbaijan	Georgia	Iraq	Syria	Bulgaria	Belarus
54736507	88396903	35371483	57181877	72015928	98367610	1.05E+08	88756018	1.1E+08
54201446	90992382	35568039	58704586	72225367	97354061	99884050	91110691	1.13E+08
51998274	94239737	33956641	61831432	71824269	1E+08	1.02E+08	91271729	1.14E+08
46103010	99912856	32698726	65070244	78021437	78971519	1.15E+08	90700488	1.14E+08
	1.05E+08	35006066	59803710	76442579	71995361	1.05E+08	89073465	1.21E+08
	1.1E+08	32650115	53235602	65874153	71051762	1.13E+08		1.17E+08
	1.16E+08	27959801	53426622	51732417	26538332	1.21E+08		1.14E+08
	1.26E+08				36550622	1.32E+08		
	1.35E+08				39244450	1.37E+08		
	1.47E+08				39993905	1.39E+08		
	1.6E+08				39049963	1.42E+08		
	1.74E+08				47130675	1.4E+08		
	1.88E+08				48594364	1.34E+08		
	1.98E+08				59869190	1.36E+08		
	2.07E+08				69697267	1.27E+08		
	2.2E+08				83019213	1.25E+08		
	2.35E+08				97219513			
	2.51E+08				1.12E+08			
	2.68E+08				94480598			
	2.88E+08							
	3.09E+08							
	3.29E+08							
	3.52E+08							
	3.71E+08							
	3.9E+08							
	4.14E+08							
	4.38E+08							
	4.62E+08							
	4.88E+08							
	5.2E+08							
	5.56E+08							
	5.93E+08							
	6.34E+08							
	6.82E+08							
	7.32E+08							
	7.53E+08							
	7.72E+08							
	8.34E+08							

Czechgdp	Hungaryg	Moldovag	Polandgdp	Romaniag	Russiagdp	Sloveniagdp	Sloveniagdp	Ukraine	Ukraine	Estoniagdp
1.33E+08	1.11E+08	38984520	3.36E+08	1.49E+08	1.8E+09	60783827	4.78E+08	25443763		
1.36E+08	1.14E+08	41569283	3.47E+08	1.52E+08	1.84E+09	63316652	4.81E+08	26143514		
1.39E+08	1.15E+08	41542751	3.42E+08	1.49E+08	1.84E+09	64851093	4.94E+08	26398461		
1.42E+08	1.17E+08	41742672	3.49E+08	1.48E+08	1.87E+09	66120767	4.99E+08	27713412		
1.45E+08	1.14E+08	44850495	3.44E+08	1.44E+08	1.88E+09	66785526	5.19E+08	29500419		
		43214990			1.83E+09		4.96E+08	27066097		
		35018365			1.78E+09		4.6E+08	23940083		

Lithuaniagdp	Latviagdp	Albaniagdp	BAHgdgdp	Serbiagdp	Croatiagdp	Monteneg	Macedoni	Sloveniagdp
47660894	37201219	11371457	29393066	83673828	62708914	4374626	18251858	39553528
49468978	38189519	11669745	30438102	86531389	64464777	4528320	19526514	40810429
50529332	38039137	11975300	29878196	85105491	64403660	4323439	19278486	40573335
54561831	39628012	12296425	29138453	83849587	63841655	4192553	18652403	39398327
54088606	41772726	12617018	29517139	85287251	62835001	4213824	19013849	39202217
51021990	42100704	12947849	26068546	75948516	58374077	3852049	17074264	
	36517763	9375857	22759464	68073340		3585538	16091740	
		8745674	16213892	50213316		2862750		

Eggdp	Cubagdp	Chinagdp	piNKgdp	pir	Mongolia	Kyrgyzstan	Kazakhstan	Tajikistan	Turkmenistan
2.72E+08	51630080	8.63%	-1.28%		5.74%	-3.67%	0.95%	1.13%	0.29%
2.81E+08	51678725	6.50%	13.16%		9.42%	-1.60%	1.26%	2.35%	2.87%
2.9E+08	50233473	8.38%	5.67%		3.55%	-1.14%	-0.61%	-2.25%	2.45%
3.02E+08	51847708	5.58%	-7.19%		5.10%	10.12%	8.26%	12.60%	9.12%
3.08E+08	52350599	0.78%	7.94%		4.15%	2.08%	-0.48%	-7.48%	-8.34%
	49943661	0.13%	-7.81%		-2.54%	2.29%	-1.38%	-0.87%	0.35%
	42598296	3.66%	-4.03%		-4.77%	-8.81%	-10.52%	-8.82%	-3.60%
	36651620	7.01%	-6.81%		-5.12%				
	28833506	9.88%	-4.17%						
	29226992	7.58%	-2.23%						
	31242330	8.36%	-4.61%						
	35625752	6.60%	-3.63%						
	37729137	3.19%	-6.90%						
	37789593	0.90%	-1.38%						
	40250897	4.49%	5.88%						
	44692112	6.71%	0.39%						
	47301969	6.33%	3.72%						
	48976942	7.66%	1.21%						
	50659543	6.37%	1.81%						
	51357048	8.03%	2.03%						
	53114043	9.86%	3.73%						
	61018247	10.31%	-1.04%						
	63463485	9.15%	-1.18%						
	67603095	4.26%	3.35%						
	70355907	7.16%	-0.86%						
	74083582	11.11%	-0.43%						
	78036992	6.48%	0.84%						
	80388156	7.90%	1.36%						
	83013750	7.80%	1.12%						
	85723464	7.40%	1.10%						
	88467246	7.00%	-1.09%						
	88635598	6.80%	3.88%						
	90020493	6.90%	-3.45%						
	91854652	6.70%	-4.10%						
	91703239	6.00%	0.35%						
	81663256	2.20%	-4.53%						
	82687389	8.40%	-0.06%						
	84154969	3.00%	-0.15%						

Uzbekistan	Cambodia	Laos	Myanmar	Vietnam	Armenia	Azerbaijan	Georgia	Iraq
2.79%	6.32%	5.13%	2.85%	5.94%	4.60%	3.97%	6.27%	-1.59%
-1.20%	5.83%	4.88%	-0.98%	2.94%	0.56%	2.66%	0.29%	-1.03%
-0.65%	-2.06%	-1.46%	-4.06%	3.57%	-4.53%	5.33%	-0.56%	2.84%
8.55%	2.54%	-2.01%	-11.34%	6.02%	-3.70%	5.24%	8.63%	-21.12%
1.96%	2.44%	14.22%		4.69%	7.06%	-8.09%	-2.02%	-8.83%
-1.90%		6.65%		5.11%	-6.73%	-10.98%	-13.83%	-1.31%
-2.02%		5.06%		5.45%	-14.37%	0.36%	-21.47%	-62.65%
		6.34%		8.29%				37.73%
		6.71%		7.70%				7.37%
		8.92%		8.45%				1.91%
		7.84%		9.21%				-2.36%
		7.68%		8.93%				20.69%
		7.67%		7.80%				3.11%
		4.72%		5.43%				23.20%
		8.11%		4.43%				16.42%
		6.55%		6.43%				19.11%
		6.52%		6.56%				17.10%
		6.66%		6.74%				15.35%
		7.00%		6.99%				-15.75%
		7.79%		7.44%				
		7.55%		7.23%				
		-0.18%		6.61%				
		19.08%		6.79%				
		8.59%		5.37%				
		8.28%		5.12%				
		8.93%		6.16%				
		8.81%		5.97%				
		7.90%		5.50%				
		7.99%		5.55%				
		7.57%		6.42%				
		7.55%		6.99%				
		7.04%		6.69%				
		6.88%		6.94%				
		6.22%		7.46%				
		5.46%		7.36%				
		3.28%		2.87%				
		3.48%		2.56%				
		4.42%		8.02%				

Syriagdp	Bulgariagdp	Belarusgdp	Czechgdp	Hungarygdp	Moldovagdp	Polandgdp	Romaniagdp	Russiangdp
13.03%	-3.00%	1.08%	2.25%	-2.46%	-9.80%	1.05%	-0.15%	1.49%
-4.94%	2.65%	2.73%	2.76%	1.99%	6.63%	3.18%	1.71%	2.48%
1.92%	0.18%	1.13%	2.32%	1.54%	-0.06%	-1.34%	-2.10%	-0.08%
13.27%	-0.63%	-0.36%	2.07%	1.53%	0.48%	2.21%	-0.25%	1.60%
-8.96%	-1.79%	6.57%	1.72%	-2.24%	7.45%	-1.55%	-3.19%	0.69%
7.64%		-3.42%			-3.65%			-2.46%
7.30%		-2.54%			-18.97%			-2.85%
9.24%								
3.53%								
1.70%								
1.53%								
-0.86%								
-4.81%								
1.63%								
-6.63%								
-1.25%								

Sloveniagdp	Sloveniagdp	Sloveniagdp	Sloveniagdp	Sloveniagdp	Sloveniagdp	Sloveniagdp	Sloveniagdp	Sloveniagdp
4.13%	-0.60%	0.07%	-2.82%	-2.11%	2.64%	1.27%	1.95%	0.10%
4.17%	0.52%	2.75%	3.79%	2.66%	2.62%	3.56%	3.42%	2.80%
2.42%	2.80%	0.98%	2.14%	-0.39%	2.62%	-1.84%	-1.65%	-0.09%
1.96%	0.98%	4.98%	7.98%	4.18%	2.68%	-2.48%	-1.48%	-0.87%
1.01%	4.05%	6.45%	-0.87%	5.41%	2.61%	1.30%	1.71%	-1.58%
	-4.49%	-8.25%	-5.67%	0.79%	2.62%	-11.68%	-10.95%	-7.10%
	-7.17%	-11.55%		-13.26%	-27.59%	-12.69%	-10.37%	
					-6.72%	-28.76%	-26.24%	

Montenegro	Macedonia	Slovenia	Croatia	Egypt	Cuba	China	Malta	Norway	Mongolia	Kyrgyzstan
0.23%	-0.70%	1.01%	3.16%	2.39%	50.89%	48.17%	49.44%	48.47%		
3.51%	6.98%	3.18%	3.28%	0.09%	50.90%	48.23%	49.42%	48.19%		
-4.52%	-1.27%	-0.58%	3.37%	-2.80%	50.91%	48.29%	49.39%	47.89%		
-3.03%	-3.25%	-2.90%	4.01%	3.21%	50.92%	48.35%	49.35%	47.58%		
0.51%	1.94%	-0.50%	2.11%	0.97%	50.93%	48.41%	49.32%	47.45%		
-8.59%	-10.20%			-4.60%	50.95%	48.47%	49.32%	47.52%		
-6.92%	-5.75%			-14.71%	50.97%	48.53%	49.30%	47.60%		
-20.16%				-13.96%	50.98%	48.58%	49.28%			
				-21.33%	51.00%	48.62%				
				1.36%	51.01%	48.66%				
				6.90%	51.02%	48.69%				
				14.03%	51.03%	48.72%				
				5.90%	51.04%	48.74%				
				0.16%	51.05%	48.76%				
				6.51%	51.07%	48.78%				
				11.03%	51.08%	48.80%				
				5.84%	51.09%	48.81%				
				3.54%	51.10%	48.83%				
				3.44%	51.11%	48.83%				
				1.38%	51.12%	48.84%				
				3.42%	51.13%	48.84%				
				14.88%	51.14%	48.85%				
				4.01%	51.14%	48.85%				
				6.52%	51.15%	48.86%				
				4.07%	51.16%	48.88%				
				5.30%	51.16%	48.90%				
				5.34%	51.17%	48.94%				
				3.01%	51.17%	48.98%				
				3.27%	51.17%	49.02%				
				3.26%	51.17%	49.07%				
				3.20%	51.16%	49.12%				
				0.19%	51.15%	49.17%				
				1.56%	51.14%	49.21%				
				2.04%	51.12%	49.25%				
				-0.16%	51.10%	49.28%				
				-10.95%	51.08%	49.30%				
				1.25%	51.05%	49.34%				
				1.77%	51.02%	49.38%				

Kazakhstan	Tajikistan	Turkmenistan	Uzbekistan	Cambodia	Laos	Myanmar	Vietnam	Armenia
47.72%	49.71%	49.22%	49.94%	46.73%	50.04%	50.16%	48.97%	47.77%
47.80%	49.34%	49.18%	49.98%	46.90%	50.08%	50.16%	48.96%	47.74%
47.89%	49.00%	49.15%	50.02%	47.15%	50.11%	50.16%	48.94%	47.72%
47.95%	48.69%	49.12%	50.06%	47.32%	50.15%	50.16%	48.94%	47.70%
47.99%	48.55%	49.13%	50.10%	47.49%	50.18%		48.94%	47.74%
48.03%	48.61%	49.19%	50.14%		50.20%		48.94%	47.82%
48.06%	48.70%	49.23%	50.17%		50.23%		48.95%	47.86%
					50.25%		48.96%	
					50.27%		48.96%	
					50.29%		48.98%	
					50.30%		48.99%	
					50.32%		48.99%	
					50.33%		49.01%	
					50.35%		49.03%	
					50.36%		49.04%	
					50.37%		49.03%	
					50.38%		48.99%	
					50.38%		48.96%	
					50.38%		48.90%	
					50.38%		48.85%	
					50.37%		48.81%	
					50.36%		48.75%	
					50.35%		48.69%	
					50.34%		48.64%	
					50.33%		48.62%	
					50.33%		48.65%	
					50.32%		48.67%	
					50.32%		48.70%	
					50.32%		48.73%	
					50.31%		48.77%	
					50.31%		48.80%	
					50.30%		48.85%	
					50.30%		48.90%	
					50.30%		48.93%	
					50.29%		48.95%	
					50.28%		48.96%	
					50.27%		48.97%	
					50.26%		48.98%	

Azerbaijan	Georgia	Iraq	Syria	Bulgaria	Belarus	Czechia	Hungary	Moldova
48.20%	46.53%	49.45%	50.79%	49.60%	46.65%	48.50%	48.22%	46.44%
48.10%	46.58%	49.40%	50.77%	49.51%	46.68%	48.51%	48.17%	46.42%
48.00%	46.63%	49.35%	50.75%	49.46%	46.72%	48.52%	48.13%	46.40%
47.98%	46.68%	49.29%	50.74%	49.42%	46.78%	48.52%	48.10%	46.37%
48.04%	46.74%	49.25%	50.72%	49.36%	46.85%	48.53%	48.05%	46.38%
48.08%	46.77%	49.23%	50.70%		46.89%			46.42%
48.12%	46.71%	49.24%	50.68%		46.92%			46.46%
		49.26%	50.66%					
		49.30%	50.64%					
		49.34%	50.62%					
		49.39%	50.60%					
		49.43%	50.57%					
		49.48%	50.55%					
		49.52%	50.53%					
		49.57%	50.51%					
		49.62%	50.50%					
		49.66%						
		49.70%						
		49.73%						

Poland	Romania	Russia	Malta	Slovakia	Ukraine	€ Estonia	Malta	Lithuania	Latvia	Malta	€ Albania	Malta
48.77%	48.53%	46.43%	49.00%	45.97%	46.53%	47.21%	46.21%	50.52%				
48.78%	48.52%	46.51%	48.98%	46.04%	46.57%	47.24%	46.27%	50.47%				
48.78%	48.52%	46.61%	48.95%	46.11%	46.63%	47.29%	46.34%	50.43%				
48.76%	48.51%	46.71%	48.91%	46.20%	46.69%	47.32%	46.42%	50.39%				
48.74%	48.50%	46.79%	48.87%	46.30%	46.74%	47.34%	46.48%	50.33%				
		46.84%		46.37%	46.80%	47.32%	46.51%	50.23%				
		46.87%		46.43%	46.83%		46.50%	50.13%				
								50.01%				

BAHmale	Serbiama	Croatiam	Monteneg	Macedoni	Sloveniam	Egma	Cubamale
48.66%	48.22%	48.00%	48.60%	48.99%	48.45%	47.76%	50.72%
48.68%	48.19%	48.00%	48.62%	48.92%	48.46%	47.83%	50.70%
48.70%	48.15%	48.00%	48.64%	48.84%	48.46%	47.95%	50.68%
48.72%	48.11%	48.01%	48.65%	48.77%	48.46%	48.07%	50.67%
48.73%	48.06%	48.02%	48.65%	48.71%	48.45%	48.17%	50.65%
48.75%	48.01%	48.02%	48.65%	48.63%			50.64%
48.76%	47.95%		48.62%	48.62%			50.63%
48.53%	47.89%		48.58%				50.62%
							50.61%
							50.57%
							50.52%
							50.46%
							50.41%
							50.35%
							50.29%
							50.22%
							50.16%
							50.10%
							50.05%
							50.00%
							49.96%
							49.91%
							49.86%
							49.81%
							49.76%
							49.72%
							49.67%
							49.65%
							49.64%
							49.63%
							49.61%
							49.59%
							49.57%
							49.54%
							49.51%
							49.48%
							49.43%
							49.38%

Year	Suvival	tim	Chinapop	NKpop	Mongolia	Kyrgyzstar	Kazakhsta	Tajikistan	Turkmenis
1985	0		1051040	18831	1859	4006	15790	4569	3240
1986	1		1066790	19137	1903	4089	15946	4716	3324
1987	2		1084035	19449	1947	4176	16109	4865	3411
1988	3		1101630	19769	1992	4244	16207	5006	3492
1989	4		1118650	20103	2070	4302	16265	5148	3572
1990	5		1135185	20451	2121	4382	16398	5272	3658
1991	6		1149979	20813	2170	4446	16522	5407	3745
1992	7		1163363	21182	2210	4489	16583	5500	3830
1993	8		1176727	21563	2242	4480	16523	5546	3917
1994	9		1189962	21852	2272	4472	16289	5604	4003
1995	10		1202581	22107	2306	4516	16022	5678	4078
1996	11		1215171	22324	2341	4580	15872	5764	4142
1997	12		1227453	22443	2372	4644	15692	5864	4200
1998	13		1238722	22483	2403	4708	15485	5983	4262
1999	14		1248899	22594	2433	4773	15357	6110	4324
2000	15		1258401	22785	2461	4826	15335	6229	4385
2001	16		1267178	22977	2489	4865	15383	6339	4444
2002	17		1275379	23165	2519	4901	15447	6452	4502
2003	18		1283064	23330	2547	4946	15526	6567	4556
2004	19		1290620	23477	2574	4999	15629	6688	4608
2005	20		1298245	23621	2602	5048	15758	6814	4664
2006	21		1305116	23764	2633	5096	15914	6943	4719
2007	22		1311877	23908	2666	5137	16086	7076	4773
2008	23		1318559	24051	2702	5180	16271	7211	4828
2009	24		1324993	24191	2741	5238	16480	7348	4884
2010	25		1331357	24326	2783	5289	16697	7485	4940
2011	26		1339549	24457	2827	5328	16911	7624	4996
2012	27		1349537	24589	2870	5373	17124	7765	5053
2013	28		1357520	24720	2912	5423	17333	7905	5110
2014	29		1366654	24852	2953	5478	17540	8045	5169
2015	30		1373406	24983	2993	5537	17743	8185	5227
2016	31		1382401	25115	3031	5598	17941	8322	5287
2017	32		1390136	25248	3068	5658	18133	8458	5346
2018	33		1395398	25381	3103	5717	18316	8593	5405
2019	34		1400035	25497	3171	5774	18491	8727	5464
2020	35		1402060	25608	3231	5830	18656	8860	5522
2021	36		1402537	25711	3284	5884	18813	8990	5579
2022	37		1401693	25808	3333	5976	19011	9177	5657

Uzbekistan	Cambodia	Laos	Myanmar	Vietnam	Armenia	Azerbaijan	Georgia	Iraq
18215	7822	3657	36346	60093	3374	6682	5193	15694
18718	8079	3762	36998	61440	3422	6776	5219	16247
19219	8380	3869	37625	62826	3469	6874	5240	16543
19625	8692	3980	38196	64211	3516	6976	5320	17038
20033	9014	4094	38721	65868	3326	7102	5398	17568
20530	9345	4210	39243	67283	3377	7200	5426	18135
21059	9683	4331	39885	68675	3434	7300	5426	17474
21609	10117	4454	40419	70032	3357	7412	5366	17863
22127	10596	4581	41003	71357	3185	7528	5264	18404
22604	10944	4712	41732	72654	3060	7637	5141	18968
23067	11204	4846	42463	73904	2995	7731	5037	19555
23524	11454	4948	43143	75096	2985	7815	4961	20163
23952	11690	5055	43782	76234	2978	7897	4912	20785
24342	11913	5168	44407	77323	2972	7979	4878	21422
24701	12122	5286	45018	78394	2970	8056	4848	22076
25042	12317	5397	45616	79472	2966	8127	4814	22748
25372	12504	5499	46171	80546	2965	8194	4789	23440
25696	12691	5578	46690	81637	2962	8257	4771	24212
25992	12881	5654	47201	82720	2955	8319	4746	24979
26268	13076	5746	47699	83821	2953	8391	4739	25726
26540	13276	5836	48196	84942	2951	8475	4786	26531
26812	13461	5931	48674	86036	2948	8565	4826	26706
27081	13630	6035	49151	87136	2946	8657	4820	26429
27347	13808	6145	49505	88253	2943	8750	4819	26969
27609	13990	6257	49896	89373	2941	8842	4851	28024
27869	14162	6368	50285	90481	2939	8933	4898	28956
28132	14322	6477	50813	91591	2937	9025	4934	29974
28398	14480	6586	51369	92720	2934	9117	4946	31321
28666	14637	6695	51878	93883	2932	9210	4938	32779
28934	14793	6804	52374	95055	2928	9302	4931	33857
29205	14994	6912	52900	96222	2924	9393	4927	34674
29480	15238	7019	53416	97380	2919	9481	4924	35437
29756	15480	7127	53914	98507	2913	9567	4922	36200
30031	15717	7234	54403	99604	2907	9649	4922	37015
30305	15944	7343	54884	100695	2899	9727	4923	37846
30576	16149	7452	55355	101771	2891	9802	4926	38673
30843	16338	7560	55815	102815	2881	9874	4929	39491
31337	16514	7666	56210	103574	2871	9918	4911	40364

Syriapop	Bulgariapop	Belaruspop	Czechpop	Hungarypop	Moldovapop	Polandpop	Romaniapop	Russiapop
10481	8944	9982	10310	10649	4148	37226	22521	143938
10857	8959	10044	10309	10631	4183	37504	22600	144967
11243	8972	10097	10312	10613	4217	37741	22686	145959
11632	8982	10150	10314	10443	4290	37867	22769	146866
12018	8990	10184	10314	10398	4359	37970	22852	147352
12436	8894	10201	10310	10372	4394	38119	22866	147973
12867	8772	10206	10256	10353	4401	38198	22826	148398
13243	8659	10228	10265	10331	4402	38262	22797	148542
13615	8442	10251	10278	10301	4401	38310	22766	148462
13989	8363	10239	10283	10268	4392	38341	22733	148412
14366	8284	10205	10278	10336	4375	38363	22687	148379
14750	8241	10171	10262	10318	4354	38378	22622	148164
15139	8191	10129	10251	10297	4324	38381	22558	147919
15533	8137	10092	10242	10273	4286	38372	22511	147674
15930	8091	10064	10230	10244	4244	38358	22414	147218
16329	8052	10034	10220	10218	4199	38344	22262	146600
16730	7798	10000	10172	10194	4154	38339	22106	145980
17132	7755	9955	10148	10165	4108	38320	21931	145310
17534	7710	9905	10149	10136	4059	38283	21828	144652
17942	7668	9856	10154	10114	4013	38268	21704	144071
18423	7628	9807	10182	10094	3966	38249	21571	143522
19252	7587	9764	10214	10078	3919	38220	21443	143053
20493	7549	9734	10270	10062	3873	38204	21129	142809
21221	7513	9713	10376	10045	3829	38204	20779	142746
21542	7475	9697	10438	10029	3788	38572	20607	142789
22050	7425	9681	10463	10007	3748	38606	20485	142853
22458	7242	9662	10443	9978	3709	38615	20385	142964
22258	7200	9643	10455	9927	3671	38623	20296	143205
20901	7160	9624	10457	9900	3633	38591	20224	143510
19318	7119	9605	10471	9873	3595	38573	20151	146094
18272	7074	9585	10489	9850	3558	38544	20055	146410
17669	7024	9564	10511	9821	3522	38516	19938	146678
17327	6973	9542	10535	9794	3485	38511	19823	146846
17601	6923	9519	10572	9782	3448	38502	19713	146834
18440	6874	9495	10615	9778	3412	38475	19622	146768
19241	6833	9469	10645	9757	3376	38443	19496	146463
20226	6778	9442	10642	9716	3339	38250	19351	145849
20985	6720	9400	10635	9690	3570	37914	19266	146693

Slovakia	Ukraine	Estonia	Lithuania	Latvia	Albania	BAH	Serbia	Croatia
5145	50944	1538	3587	2606	2957	4275	7720	4458
5172	51095	1550	3620	2627	3015	4308	7740	4472
5199	51218	1562	3653	2649	3075	4339	7756	4484
5223	51423	1569	3674	2664	3137	4370	7769	4494
5245	51501	1571	3681	2667	3196	4398	7780	4501
5263	51622	1569	3695	2664	3251	4424	7786	4508
5248	51730	1560	3676	2651	3187	4437	7685	4541
5272	51869	1529	3629	2613	3078	4233	7646	4491
5289	51887	1491	3565	2561	3072	3913	7640	4478
5311	51638	1468	3533	2517	3117	3727	7634	4499
5328	51248	1447	3520	2480	3164	3636	7625	4481
5338	50813	1424	3494	2453	3144	3649	7641	4420
5348	50371	1404	3468	2428	3125	3681	7627	4398
5355	49941	1392	3443	2406	3105	3713	7609	4396
5360	49491	1378	3419	2386	3085	3811	7611	4377
5365	49014	1400	3395	2363	3065	3901	7587	4334
5344	48520	1391	3367	2333	3047	3919	7574	4171
5343	48074	1382	3340	2306	3038	3923	7570	4176
5343	47692	1374	3313	2284	3026	3940	7551	4176
5346	47342	1365	3276	2259	3014	3975	7533	4179
5351	47003	1357	3223	2235	2998	3991	7510	4182
5355	46685	1349	3172	2215	2979	3993	7481	4183
5361	46407	1341	3134	2197	2957	3990	7451	4183
5370	46156	1337	3102	2174	2934	3987	7419	4181
5382	45951	1334	3068	2138	2915	3985	7389	4177
5394	45768	1332	3004	2093	2900	3983	7360	4167
5362	45604	1328	2937	2055	2892	3980	7304	4154
5370	45492	1324	2898	2030	2888	3977	7269	4141
5377	45389	1319	2869	2009	2882	3973	7234	4126
5382	44944	1314	2844	1991	2876	3968	7199	4109
5386	44429	1312	2818	1974	2868	3963	7162	4081
5395	44208	1314	2782	1956	2863	3957	7124	4047
5402	44029	1314	2744	1938	2861	3951	7087	4006
5410	43944	1318	2718	1923	2854	3944	7048	3968
5417	43951	1323	2710	1910	2842	3937	7010	3945
5424	43904	1327	2711	1897	2825	3929	6964	3925
5404	43746	1329	2724	1880	2799	3921	6898	3839
5464	39897	1330	2748	1883	2787	3876	6833	3790

Montenegro	Macedonia	Slovenia	Croatia	Cuba	China	India	Nigeria	Mongolia	Kyrgyzstan
561	1845	1914	16660	10084	2619	2415	2004	5616	
565	1851	1932	16630	10183	2748	2689	2142	5413	
569	1856	1946	16590	10279	2931	2796	2168	5241	
574	1859	1962	16560	10382	3045	2553	2227	5679	
579	1863	1976	16430	10489	3022	2710	2232	5719	
583	1861	1991		10597	2982	2456	2123	5742	
591	1868	1999		10697	3051	2316	1976	5162	
622	1889	1999		10780	3228	2121	1841	4362	
623	1918	2000		10850	3506	1996	1846	3765	
623	1935	2001		10893	3730	1926	1955	2994	
624	1954	2003		10921	4000	1816	2148	2776	
624	1971	2004		10963	4220	1733	2272	2902	
624	1985	2000		11014	4311	1605	2445	3115	
623	1995	1996		11064	4310	1580	2620	3106	
623	2006	1998		11108	4467	1665	2802	3144	
623	2017	2004		11147	4730	1657	2936	3246	
623	2026	2006		11179	4995	1704	3134	3358	
622	2011	2009		11210	5343	1711	3402	3299	
621	2018	2010		11241	5649	1730	3775	3465	
621	2024	2011		11260	6068	1754	4334	3632	
621	2028	2015		11268	6627	1808	4822	3554	
620	2031	2022		11266	7271	1778	5427	3595	
620	2035	2033		11261	7896	1747	6197	3831	
621	2038	2036		11258	8190	1795	6982	4047	
622	2042	2056		11260	8734	1769	7127	4076	
623	2046	2063		11262	9658	1751	7830	3978	
622	2049	2067		11264	10221	1757	9484	4142	
622	2052	2071		11262	10947	1771	10493	4104	
622	2055	2074		11250	11731	1781	11545	4510	
621	2058	2076		11238	12515	1791	12283	4644	
620	2061	2078		11214	13325	1763	12398	4773	
618	2063	2079		11180	14139	1821	12314	4925	
616	2065	2081		11147	15030	1749	12728	5104	
614	2067	2087		11116	15977	1669	13420	5226	
613	2068	2104		11105	16880	1667	13871	5413	
612	2065	2118		11089	17226	1584	12990	4899	
611	2057	2123		11046	18667	1577	12992	5036	
610	2048	2124		11003	19238	1569	13443	5305	

Kazakhstan	Tajikistan	Turkmenistan	Uzbekistan	Cambodia	Laos	Myanmar	Vietnam	Armenia
11545	5364	6186	7181	1540	1382	1506	1471	10485
11576	5319	6204	6904	1578	1409	1465	1481	10394
11389	5040	6194	6680	1490	1350	1382	1500	9789
12256	5515	6601	7101	1473	1286	1207	1556	9301
12153	4962	5915	7093	1455	1428	1234	1588	10525
11888	4803	5796	6790	1404	1481	1253	1634	9669
10557	4270	5458	6486	1425	1512	1205	1688	8142
10013	2833	4588	5530	1427	1564	1284	1793	4700
9174	2482	4604	5193	1384	1623	1322	1895	4130
8179	1928	3770	4744	1416	1718	1374	2018	4393
7674	1666	3474	4535	1439	1802	1424	2167	4703
7827	1560	3227	4453	1451	1900	1469	2323	4811
8094	1553	2856	4529	1466	2002	1506	2467	4836
8089	1589	3038	4576	1476	2051	1547	2564	4994
8421	1621	3527	4631	1586	2168	1668	2641	5000
9309	1723	4173	4667	1659	2262	1843	2773	5140
10589	1845	5014	4725	1726	2365	1996	2916	5458
11640	1983	5796	4777	1771	2487	2176	3071	6082
12726	2136	6785	4845	1850	2625	2411	3242	6743
13930	2301	7782	5068	1966	2785	2665	3438	7230
15234	2417	8792	5284	2145	2949	2947	3638	8007
16787	2525	9752	5535	2286	2896	3247	3829	8799
18181	2647	10829	5908	2423	3389	3543	4037	9713
18666	2802	12427	6278	2482	3614	3586	4200	10081
18751	2854	13188	6618	2433	3844	3673	4360	8399
19965	2976	14397	7003	2427	4114	3773	4572	8331
21302	3111	16518	7396	2496	4401	3884	4786	8465
22090	3284	17334	7847	2649	4670	4091	4988	9076
23110	3464	17080	8341	2814	4961	4371	5200	9395
23820	3632	17537	8832	2983	5251	4685	5465	9745
23783	3784	17857	9381	3150	5559	4985	5776	10076
23732	3979	17485	9845	3312	5860	5253	6090	10113
24397	4192	18104	10183	3489	6168	5504	6438	10895
25143	4440	18064	10682	3692	6455	5803	6842	11490
26026	4696	17264	11219	3897	6706	6141	7266	12399
25124	4828	16581	11342	3728	6824	6283	7395	11546
25936	5206	17169	12076	3796	6961	5113	7508	12242
26492	5508	17235	12559	3944	7168	5177	8050	13838

Azerbaijan	Georgia	Iraq	Syria	Bulgaria	Belarus	Czechia	Hungary	Moldova
8558	13868	6268	10026	9924	11000	12855	10452	9398
8663	13839	5992	9200	10170	11230	13211	10678	9937
8995	13707	6052	9055	10173	11297	13514	10861	9852
9328	14665	4635	9913	10098	11198	13791	11207	9731
8421	14161	4098	8735	9908	11893	14027	11003	10288
7394	12140	3918	9087	8922	11467	14178	10296	9835
7319	9535	1519	9424	8320	11171	12725	9256	7958
5639	5287	2046	10002	7846	9961	12810	9154	5546
4315	3793	2132	10072	7959	9077	12974	9294	5384
3453	3463	2109	9969	8212	7932	13518	9772	3662
2999	3611	1997	9856	8567	6994	14550	10135	3559
3074	4031	2338	9517	7901	7128	15386	10359	3305
3348	4481	2338	8827	7927	7885	15494	10942	3322
3550	4620	2795	8743	8322	8482	15657	11663	3078
3960	4763	3157	7960	7922	8690	16103	12317	2950
4215	4892	3649	7668	8411	9111	17056	13129	2990
4499	5125	4148	7509	8832	9462	17869	13934	3151
4880	5407	4632	7530	9472	9870	18431	14899	3374
5409	6013	3782	6987	10081	10493	19344	15830	3575
5971	6343	6062	7051	10854	11616	20555	16999	3813
7554	6850	6441	7038	11758	12630	22129	18141	4074
10160	7398	7118	6902	12706	13793	23888	19253	4243
12699	8300	7588	6780	13785	14857	25383	19764	4343
14073	8462	8587	6600	14762	16224	26186	20381	4652
15384	8062	9269	6587	14363	16089	25094	19461	4343
16154	8443	10274	6521	14686	17172	25922	20036	4615
16176	8946	11484	5979	15112	17918	26725	20886	4892
16365	9497	12522	4444	15315	18245	26483	20732	4914
17146	9857	12877	3488	15314	18456	26468	21164	5414
17451	10308	12560	3384	15550	18805	27030	22119	5744
17464	10630	12575	3464	16186	18123	28438	22993	5784
16771	10945	14174	3381	16796	17702	29098	23569	6103
16647	11479	13404	3422	17386	18193	30531	24641	6426
16753	12036	13725	3419	17982	18809	31405	25996	6760
17030	12632	14204	3385	18841	19130	32227	27272	7078
16190	11771	11717	2904	18204	19053	30365	26091	6561
16972	12995	12356	2682	19754	19542	31453	28086	7557
17683	14360	13072	2495	20592	18700	32247	29452	6675

Polandgd	Romaniag	Russiagdp	Slovakiaç	Ukrainegd	Estoniagd	Lithuaniaç	Latviagdp	Albaniagd
9022	6629	12475	11815	9387	16541	13287	14273	3846
9240	6718	12694	12242	9408	16866	13667	14539	3870
9059	6552	12597	12473	9648	16898	13831	14360	3894
9228	6512	12720	12659	9704	17661	14851	14875	3920
9060	6281	12766	12734	10082	18779	14693	15661	3948
8150	5596	12400	12374	9607	17247	13809	15806	3983
7626	5033	12012	10537	8899	15351	12999	13775	2942
7874	4818	10488	9839	8130	13214	10241	9411	2842
8230	5056	9789	10481	7090	12207	8621	8439	3137
8722	5427	8744	11064	5585	12070	7830	8704	3400
9408	6015	8586	11874	5024	12881	8299	8709	3667
10066	6476	8479	12623	4637	13548	8794	8947	3966
10811	6446	8813	13335	4612	15075	9593	9773	3527
11413	6407	8557	13822	4640	15593	10382	10430	3873
12058	6614	9340	13750	4751	15335	10336	10708	4432
12732	7090	10553	13905	5166	16807	10807	11310	4808
13018	7860	11391	14362	5794	17697	11607	12104	5286
13416	8674	12259	14985	6263	18589	12491	13015	5609
14036	9369	13521	15773	7030	19772	13918	14142	6005
14906	10531	14881	16571	8049	20804	14996	15371	6420
15581	11314	16243	17650	8497	22519	16418	17070	6858
16711	12823	18008	19099	9356	24579	17916	19129	7347
18067	14419	20018	21110	10356	26178	20138	21042	7866
19012	16347	21563	22232	10824	24429	20880	20342	8522
19718	15866	20336	20953	9381	20538	17983	17582	8859
20609	16377	21737	21941	9601	20713	18664	17140	9223
21837	17174	23130	22483	10333	21997	20243	18428	9484
22170	17581	24020	22745	10382	22784	21305	19970	9634
22378	17691	24390	22861	10405	23204	22286	20582	9749
23248	18487	24135	23453	9817	23983	23273	21169	9942
24285	19162	23608	24648	8959	24474	23969	22179	10193
25021	19826	23610	25088	9222	25196	24887	22913	10547
26310	23004	24014	25789	9478	26662	26318	23894	10958
27881	24000	24690	26791	9827	27596	27631	25040	11427
29142	23268	25244	27429	10140	28506	28986	25859	11715
28577	24800	24625	26468	9770	28262	28972	25453	11373
30688	26103	26120	27366	10134	30499	30557	26739	12456
32468	26198	25437	27519	7749	30067	30863	27220	12978

BAHgdpp	Serbiagdpp	Croatiagdpp	Monteneg	Macedoni	Sloveniag	Eggdppc	Cubagdpp	Chinagdpp
6875	10839	14067	7804	9894	20663	16318	5120	2.75E+09
7066	11180	14416	8021	10551	21122	16883	5075	2.93E+09
6886	10973	14362	7592	10385	20851	17494	4887	3.18E+09
6668	10793	14207	7299	10034	20083	18228	4994	3.35E+09
6711	10963	13959	7278	10206	19837	18760	4991	3.38E+09
5893	9754	12948	6605	9173	18093		4713	3.39E+09
5129	8858	10296	6068	8616	16402		3982	3.51E+09
3830	6568	9332	4605	7992	15463		3400	3.76E+09
2831	4674	8740	2982	7318	15872		2657	4.13E+09
3017	4911	9353	3074	7154	16665		2683	4.44E+09
3237	5325	10180	3571	7039	17817		2861	4.81E+09
4915	5710	11030	4663	7095	18376		3250	5.13E+09
5913	6288	11998	5122	7179	19259		3426	5.29E+09
6590	6604	12476	5522	7412	19851		3416	5.34E+09
7026	5533	12580	5143	7732	20794		3624	5.58E+09
7082	6060	13245	6046	8084	21501		4009	5.95E+09
7129	6543	13818	6430	7835	22007		4231	6.33E+09
7392	7156	14684	6898	7964	22725		4369	6.81E+09
7551	7638	15740	7445	8160	23257		4507	7.25E+09
7857	8512	16629	8194	8561	24151		4561	7.83E+09
8058	9198	17581	9002	8985	24967		4714	8.6E+09
8411	9902	18707	10308	9466	26183		5416	9.49E+09
8812	10769	19984	12027	10099	27833		5636	1.04E+10
9199	11650	20717	13550	10674	28474		6005	1.08E+10
9019	11577	19502	13459	10658	25906		6248	1.16E+10
8982	11964	19511	14510	11038	26001		6578	1.29E+10
8961	12463	19813	15747	11321	26004		6928	1.37E+10
8895	12438	19414	15421	11256	25269		7138	1.48E+10
9113	12860	19403	16064	11568	24975		7379	1.59E+10
9229	12718	19404	16437	11968	25639		7628	1.71E+10
9639	13014	20026	17249	12413	26186		7889	1.83E+10
9966	13519	20915	17777	12753	27005		7928	1.95E+10
10306	13877	21852	18640	12878	28289		8075	2.09E+10
10719	14580	22678	19617	13238	29461		8263	2.23E+10
11048	15293	23590	20449	13751	30229		8258	2.36E+10
10736	15256	21672	17356	13126	28733		7364	2.42E+10
11554	16564	25060	19657	13692	31020		7486	2.62E+10
12143	17098	26987	20942	14051	32674		7649	2.7E+10

NK	gdp	Mongolia	gdp	Kyrgyzstan	Kazakhstan	Tajikistan	gdp	Turkmenistan	Uzbekistan	Cambodia	Laos	gdp
45476865	3725436	22495180	1.82E+08	24508116	20045560	1.31E+08	12046065	5053974				
51459393	4076226	22136344	1.85E+08	25084404	20620067	1.29E+08	12748870	5300658				
54379404	4221096	21884309	1.83E+08	24519600	21125901	1.28E+08	12486216	5223150				
50470257	4436184	24099774	1.99E+08	27608090	23051920	1.39E+08	12803095	5118280				
54479130	4620240	24601434	1.98E+08	25544376	21128380	1.42E+08	13115360	5846232				
50223946	4502883	25164235	1.95E+08	25321416	21201768	1.39E+08	13119771	6235010				
48202052	4287970	22948277	1.74E+08	23089123	20437990	1.37E+08	13798588	6550550				
44917408	4068638	19580869	1.66E+08	15578962	17569988	1.2E+08	14439550	6965636				
43046073	4139238	16865849	1.52E+08	13768161	18037537	1.15E+08	14667327	7432922				
42085991	4442844	13389775	1.33E+08	10806548	15091316	1.07E+08	15501963	8095762				
40146511	4952893	12537053	1.23E+08	9460376	14166397	1.05E+08	16125090	8730282				
38689713	5318281	13289582	1.24E+08	8989080	13366623	1.05E+08	16613604	9400685				
36018760	5799570	14463585	1.27E+08	9105133	11995996	1.08E+08	17142179	10121303				
35521587	6297035	14624138	1.25E+08	9509161	12945785	1.11E+08	17586612	10598968				
37610925	6816930	15009645	1.29E+08	9907047	15251900	1.14E+08	19228281	11458883				
37758036	7225650	15667564	1.43E+08	10735321	18299208	1.17E+08	20435399	12209286				
39161384	7799421	16336535	1.63E+08	11697951	22282744	1.2E+08	21587210	13005703				
39634147	8569503	16170495	1.8E+08	12792792	26092439	1.23E+08	22474402	13872426				
40352083	9616040	17135753	1.98E+08	14027449	30907678	1.26E+08	23833222	14843719				
41171426	11156312	18155687	2.18E+08	15391087	35857575	1.33E+08	25705927	16000772				
42708228	12547452	17942169	2.4E+08	16469330	41000508	1.4E+08	28474469	17208611				
42264171	14288906	18316467	2.67E+08	17531184	46020000	1.48E+08	30776292	17176980				
41765544	16520107	19684075	2.92E+08	18731098	51693700	1.6E+08	33022341	20454812				
43163304	18865760	20960395	3.04E+08	20202936	60001293	1.72E+08	34271377	22210960				
42791212	19534583	21348199	3.09E+08	20969202	64409214	1.83E+08	34042366	24050843				
42605439	21791069	21037149	3.33E+08	22275913	71115150	1.95E+08	34365989	26198017				
42961430	26811268	22069322	3.6E+08	23719353	82520294	2.08E+08	35746714	28505277				
43545175	30114910	22049906	3.78E+08	25498308	87581260	2.23E+08	38360866	30756620				
44033757	33619040	24456653	4.01E+08	27385179	87286115	2.39E+08	41183074	33213895				
44518472	36271699	25440783	4.18E+08	29219995	90639649	2.56E+08	44124787	35727804				
44032954	37107214	26426867	4.22E+08	30973187	93346148	2.74E+08	47229402	38423808				
45739742	37322351	27572738	4.26E+08	33110337	92434152	2.9E+08	50470756	41128977				
44160500	39053687	28879685	4.42E+08	35461178	96785031	3.03E+08	54002197	43958737				
42349472	41646693	29878639	4.61E+08	38156221	97638673	3.21E+08	58035620	46694967				
42499101	43979843	31253356	4.81E+08	40979786	94329703	3.4E+08	62129453	49243305				
40572893	41975350	28560253	4.69E+08	42778796	91559236	3.47E+08	60205931	50856672				
40550429	42662390	29628979	4.88E+08	46800010	95789277	3.72E+08	62027759	52626359				
40488847	44809465	31703001	5.04E+08	50544009	97502947	3.94E+08	65136593	54953906				

Myanmar	Vietnam	Armenia	Azerbaijan	Georgia	Iraq	Syria	Bulgaria	Belarus
54736507	88396903	35371483	57181877	72015928	98367610	1.05E+08	88756018	1.1E+08
54201446	90992382	35568039	58704586	72225367	97354061	99884050	91110691	1.13E+08
51998274	94239737	33956641	61831432	71824269	1E+08	1.02E+08	91271729	1.14E+08
46103010	99912856	32698726	65070244	78021437	78971519	1.15E+08	90700488	1.14E+08
47782108	1.05E+08	35006066	59803710	76442579	71995361	1.05E+08	89073465	1.21E+08
49171474	1.1E+08	32650115	53235602	65874153	71051762	1.13E+08	79352518	1.17E+08
48063926	1.16E+08	27959801	53426622	51732417	26538332	1.21E+08	72983249	1.14E+08
51891173	1.26E+08	15777769	41795294	28371172	36550622	1.32E+08	67935673	1.02E+08
54198382	1.35E+08	13153115	32486437	19963505	39244450	1.37E+08	67190225	93050205
57338247	1.47E+08	13441640	26372096	17806250	39993905	1.39E+08	68680749	81212665
60452841	1.6E+08	14085547	23188275	18187119	39049963	1.42E+08	70965737	71373341
63378627	1.74E+08	14363162	24026380	19998468	47130675	1.4E+08	65116241	72504135
65935168	1.88E+08	14402223	26436934	22010932	48594364	1.34E+08	64936505	79868073
68698887	1.98E+08	14841976	28328461	22536773	59869190	1.36E+08	67715866	85602472
75075474	2.07E+08	14850694	31898574	23091342	69697267	1.27E+08	64098092	87455012
84093546	2.2E+08	15244825	34255708	23550476	83019213	1.25E+08	67719768	91419112
92175089	2.35E+08	16184137	36862176	24543620	97219513	1.26E+08	68877646	94624118
1.02E+08	2.51E+08	18018115	40295217	25794087	1.12E+08	1.29E+08	73450038	98263264
1.14E+08	2.68E+08	19928772	44994639	28539726	94480598	1.23E+08	77722333	1.04E+08
1.27E+08	2.88E+08	21350959	50107479	30055113	1.56E+08	1.27E+08	83228670	1.14E+08
1.42E+08	3.09E+08	23626538	64019174	32780624	1.71E+08	1.3E+08	89684786	1.24E+08
1.58E+08	3.29E+08	25937836	87026412	35705813	1.9E+08	1.33E+08	96404595	1.35E+08
1.74E+08	3.52E+08	28610367	1.1E+08	40006943	2.01E+08	1.39E+08	1.04E+08	1.45E+08
1.78E+08	3.71E+08	29672469	1.23E+08	40781280	2.32E+08	1.4E+08	1.11E+08	1.58E+08
1.83E+08	3.9E+08	24704738	1.36E+08	39112877	2.6E+08	1.42E+08	1.07E+08	1.56E+08
1.9E+08	4.14E+08	24485670	1.44E+08	41359834	2.98E+08	1.44E+08	1.09E+08	1.66E+08
1.97E+08	4.38E+08	24860520	1.46E+08	44138043	3.44E+08	1.34E+08	1.09E+08	1.73E+08
2.1E+08	4.62E+08	26634067	1.49E+08	46970387	3.92E+08	98907695	1.1E+08	1.76E+08
2.27E+08	4.88E+08	27542824	1.58E+08	48668830	4.22E+08	72894800	1.1E+08	1.78E+08
2.45E+08	5.2E+08	28536291	1.62E+08	50830708	4.25E+08	65379102	1.11E+08	1.81E+08
2.64E+08	5.56E+08	29464865	1.64E+08	52369352	4.36E+08	63295278	1.14E+08	1.74E+08
2.81E+08	5.93E+08	29522320	1.59E+08	53891203	5.02E+08	59731472	1.18E+08	1.69E+08
2.97E+08	6.34E+08	31741803	1.59E+08	56501157	4.85E+08	59296820	1.21E+08	1.74E+08
3.16E+08	6.82E+08	33400319	1.62E+08	59237503	5.08E+08	60171980	1.24E+08	1.79E+08
3.37E+08	7.32E+08	35949093	1.66E+08	62188712	5.38E+08	62427098	1.3E+08	1.82E+08
3.48E+08	7.53E+08	33378376	1.59E+08	57984757	4.53E+08	55872252	1.24E+08	1.8E+08
2.85E+08	7.72E+08	35275597	1.68E+08	64053442	4.88E+08	54251959	1.34E+08	1.85E+08
2.91E+08	8.34E+08	39721735	1.75E+08	70528609	5.28E+08	52353140	1.38E+08	1.76E+08

Czechgdp	Hungaryg	Moldovag	Polandgdp	Romaniag	Russiagdp	Sloveniagdp	Slovakia	Ukraine	Estonia
1.33E+08	1.11E+08	38984520	3.36E+08	1.49E+08	1.8E+09	60783827	4.78E+08	25443763	
1.36E+08	1.14E+08	41569283	3.47E+08	1.52E+08	1.84E+09	63316652	4.81E+08	26143514	
1.39E+08	1.15E+08	41542751	3.42E+08	1.49E+08	1.84E+09	64851093	4.94E+08	26398461	
1.42E+08	1.17E+08	41742672	3.49E+08	1.48E+08	1.87E+09	66120767	4.99E+08	27713412	
1.45E+08	1.14E+08	44850495	3.44E+08	1.44E+08	1.88E+09	66785526	5.19E+08	29500419	
1.46E+08	1.07E+08	43214990	3.11E+08	1.28E+08	1.83E+09	65119610	4.96E+08	27066097	
1.3E+08	95825229	35018365	2.91E+08	1.15E+08	1.78E+09	55297379	4.6E+08	23940083	
1.31E+08	94567783	24417046	3.01E+08	1.1E+08	1.56E+09	51864849	4.22E+08	20201099	
1.33E+08	95740694	23693283	3.15E+08	1.15E+08	1.45E+09	55439230	3.68E+08	18206161	
1.39E+08	1E+08	16084201	3.34E+08	1.23E+08	1.3E+09	58762734	2.88E+08	17713297	
1.5E+08	1.05E+08	15569831	3.61E+08	1.36E+08	1.27E+09	63261330	2.57E+08	18632898	
1.58E+08	1.07E+08	14391438	3.86E+08	1.47E+08	1.26E+09	67378062	2.36E+08	19286856	
1.59E+08	1.13E+08	14366532	4.15E+08	1.45E+08	1.3E+09	71311500	2.32E+08	21171919	
1.6E+08	1.2E+08	13191426	4.38E+08	1.44E+08	1.26E+09	74015464	2.32E+08	21698758	
1.65E+08	1.26E+08	12519935	4.63E+08	1.48E+08	1.38E+09	73695873	2.35E+08	21126869	
1.74E+08	1.34E+08	12554043	4.88E+08	1.58E+08	1.55E+09	74595102	2.53E+08	23525731	
1.82E+08	1.42E+08	13087723	4.99E+08	1.74E+08	1.66E+09	76744255	2.81E+08	24618993	
1.87E+08	1.51E+08	13857814	5.14E+08	1.9E+08	1.78E+09	80059373	3.01E+08	25689833	
1.96E+08	1.6E+08	14513550	5.37E+08	2.05E+08	1.96E+09	84281955	3.35E+08	27160246	
2.09E+08	1.72E+08	15302842	5.7E+08	2.29E+08	2.14E+09	88594622	3.81E+08	28392968	
2.25E+08	1.83E+08	16157452	5.96E+08	2.44E+08	2.33E+09	94444702	3.99E+08	30566893	
2.44E+08	1.94E+08	16628254	6.39E+08	2.75E+08	2.58E+09	1.02E+08	4.37E+08	33162230	
2.61E+08	1.99E+08	16819758	6.9E+08	3.05E+08	2.86E+09	1.13E+08	4.81E+08	35115521	
2.72E+08	2.05E+08	17815143	7.26E+08	3.4E+08	3.08E+09	1.19E+08	5E+08	32659893	
2.62E+08	1.95E+08	16449235	7.61E+08	3.27E+08	2.9E+09	1.13E+08	4.31E+08	27402480	
2.71E+08	2E+08	17299250	7.96E+08	3.35E+08	3.11E+09	1.18E+08	4.39E+08	27586969	
2.79E+08	2.08E+08	18145357	8.43E+08	3.5E+08	3.31E+09	1.21E+08	4.71E+08	29217295	
2.77E+08	2.06E+08	18038303	8.56E+08	3.57E+08	3.44E+09	1.22E+08	4.72E+08	30160489	
2.77E+08	2.1E+08	19669686	8.64E+08	3.58E+08	3.5E+09	1.23E+08	4.72E+08	30600421	
2.83E+08	2.18E+08	20653170	8.97E+08	3.73E+08	3.53E+09	1.26E+08	4.41E+08	31521829	
2.98E+08	2.26E+08	20583362	9.36E+08	3.84E+08	3.46E+09	1.33E+08	3.98E+08	32106015	
3.06E+08	2.31E+08	21490883	9.64E+08	3.95E+08	3.46E+09	1.35E+08	4.08E+08	33119084	
3.22E+08	2.41E+08	22393499	1.01E+09	4.56E+08	3.53E+09	1.39E+08	4.17E+08	35037377	
3.32E+08	2.54E+08	23311636	1.07E+09	4.73E+08	3.63E+09	1.45E+08	4.32E+08	36363238	
3.42E+08	2.67E+08	24150850	1.12E+09	4.57E+08	3.71E+09	1.49E+08	4.46E+08	37723176	
3.23E+08	2.55E+08	22146329	1.1E+09	4.83E+08	3.61E+09	1.44E+08	4.29E+08	37515273	
3.35E+08	2.73E+08	25234639	1.17E+09	5.05E+08	3.81E+09	1.48E+08	4.43E+08	40521485	
3.43E+08	2.85E+08	23827555	1.23E+09	5.05E+08	3.73E+09	1.5E+08	3.09E+08	39999859	

Lithuaniagdp	Latviagdp	Albaniagdp	BAHgdp	Serbiagdp	Croatiagdp	Monteneg	Macedoni	Sloveniagdp
47660894	37201219	11371457	29393066	83673828	62708914	4374626	18251858	39553528
49468978	38189519	11669745	30438102	86531389	64464777	4528320	19526514	40810429
50529332	38039137	11975300	29878196	85105491	64403660	4323439	19278486	40573335
54561831	39628012	12296425	29138453	83849587	63841655	4192553	18652403	39398327
54088606	41772726	12617018	29517139	85287251	62835001	4213824	19013849	39202217
51021990	42100704	12947849	26068546	75948516	58374077	3852049	17074264	36026963
47786759	36517763	9375857	22759464	68073340	46751296	3585538	16091740	32795389
37159350	24594370	8745674	16213892	50213316	41906544	2862750	15099005	30913205
30736278	21610171	9636341	11077008	35705126	39138895	1856251	14032387	31743129
27661451	21908377	10596115	11242945	37491186	42079801	1915099	13843335	33346743
29216541	21602326	11600732	11770817	40606795	45617096	2226639	13754792	35684024
30721761	21944517	12468868	17934743	43633612	48751848	2909672	13985283	36825997
33267370	23733282	11020137	21766880	47961715	52770093	3194862	14249362	38526558
35744384	25089221	12026243	24470420	50248497	54840833	3442988	14789522	39631444
35335380	25549228	13673694	26775071	42108006	55058850	3205304	15508167	41542297
36684798	26727292	14736836	27629711	45972178	57398793	3766656	16307995	43078788
39077071	28242589	16103905	27937500	49551965	57633270	4003585	15872668	44148370
41716676	30014856	17037279	28997438	54169286	61317063	4290323	16017433	45659924
46106059	32298350	18171077	29748999	57670216	65731220	4622736	16465999	46758349
49123583	34720424	19346600	31229543	64122161	69483829	5084312	17324217	48578175
52913078	38143685	20562850	32159582	69083772	73522779	5587146	18219130	50309131
56827411	42361247	21889974	33580207	74079356	78256427	6395709	19227781	52944722
63122229	46228817	23259285	35158269	80235728	83591430	7462960	20547244	56590132
64776246	44217430	25006230	36676628	86433158	86620962	8414771	21752222	57981922
55174533	37590731	25821550	35937395	85543119	81451700	8371378	21758910	53266138
56073777	35881179	26747820	35774435	88047937	81303797	9033098	22583126	53649780
59460371	37869724	27431047	35664063	91032743	82302806	9801452	23201484	53761450
61745947	40536566	27819813	35371051	90412314	80383804	9593080	23095686	52342435
63937851	41350513	28098562	36202158	93027614	80064089	9985415	23771198	51803647
66199375	42137029	28597168	36619950	91548934	79720251	10205668	24633991	53237653
67539657	43774191	29231664	38199900	93202358	81731108	10689585	25583815	54414239
69240913	44810999	30200694	39438505	96314000	84640160	10988666	26312536	56151176
72206128	46295493	31348975	40717914	98337716	87528487	11487748	26597106	58855036
75089599	48143399	32609002	42277004	1.03E+08	89979864	12049855	27363281	61476693
78562637	49380620	33289746	43497124	1.07E+08	93057983	12539435	28433277	63598212
78545277	48292864	32130728	42185626	1.06E+08	85073362	10620038	27100246	60849951
83242850	50255493	34866975	45303798	1.14E+08	96194359	12005262	28165634	65846343
84814258	51248482	36171005	47069138	1.17E+08	1.02E+08	12774399	28770256	69385345

Eggdp	Cubagdp	Chinagdp	piNKgdp	pir	Mongolia	çKyrgyzstar	Kazakhsta	Tajikistan	çTurkmenis
2.72E+08	51630080	8.63%	-1.28%	5.74%	-3.67%	0.95%	1.13%	0.29%	
2.81E+08	51678725	6.50%	13.16%	9.42%	-1.60%	1.26%	2.35%	2.87%	
2.9E+08	50233473	8.38%	5.67%	3.55%	-1.14%	-0.61%	-2.25%	2.45%	
3.02E+08	51847708	5.58%	-7.19%	5.10%	10.12%	8.26%	12.60%	9.12%	
3.08E+08	52350599	0.78%	7.94%	4.15%	2.08%	-0.48%	-7.48%	-8.34%	
	49943661	0.13%	-7.81%	-2.54%	2.29%	-1.38%	-0.87%	0.35%	
	42598296	3.66%	-4.03%	-4.77%	-8.81%	-10.52%	-8.82%	-3.60%	
	36651620	7.01%	-6.81%	-5.12%	-14.67%	-4.81%	-32.53%	-14.03%	
	28833506	9.88%	-4.17%	1.74%	-13.87%	-8.72%	-11.62%	2.66%	
	29226992	7.58%	-2.23%	7.33%	-20.61%	-12.11%	-21.51%	-16.33%	
	31242330	8.36%	-4.61%	11.48%	-6.37%	-7.71%	-12.46%	-6.13%	
	35625752	6.60%	-3.63%	7.38%	6.00%	1.04%	-4.98%	-5.65%	
	37729137	3.19%	-6.90%	9.05%	8.83%	2.24%	1.29%	-10.25%	
	37789593	0.90%	-1.38%	8.58%	1.11%	-1.38%	4.44%	7.92%	
	40250897	4.49%	5.88%	8.26%	2.64%	3.25%	4.18%	17.81%	
	44692112	6.71%	0.39%	6.00%	4.38%	10.39%	8.36%	19.98%	
	47301969	6.33%	3.72%	7.94%	4.27%	14.11%	8.97%	21.77%	
	48976942	7.66%	1.21%	9.87%	-1.02%	10.38%	9.36%	17.10%	
	50659543	6.37%	1.81%	12.21%	5.97%	9.89%	9.65%	18.45%	
	51357048	8.03%	2.03%	16.02%	5.95%	10.19%	9.72%	16.02%	
	53114043	9.86%	3.73%	12.47%	-1.18%	10.26%	7.01%	14.34%	
	61018247	10.31%	-1.04%	13.88%	2.09%	11.28%	6.45%	12.24%	
	63463485	9.15%	-1.18%	15.61%	7.47%	9.48%	6.84%	12.33%	
	67603095	4.26%	3.35%	14.20%	6.48%	3.85%	7.86%	16.07%	
	70355907	7.16%	-0.86%	3.55%	1.85%	1.74%	3.79%	7.35%	
	74083582	11.11%	-0.43%	11.55%	-1.46%	7.87%	6.23%	10.41%	
	78036992	6.48%	0.84%	23.04%	4.91%	8.07%	6.48%	16.04%	
	80388156	7.90%	1.36%	12.32%	-0.09%	5.00%	7.50%	6.13%	
	83013750	7.80%	1.12%	11.64%	10.91%	5.90%	7.40%	-0.34%	
	85723464	7.40%	1.10%	7.89%	4.02%	4.30%	6.70%	3.84%	
	88467246	7.00%	-1.09%	2.30%	3.88%	1.00%	6.00%	2.99%	
	88635598	6.80%	3.88%	0.58%	4.34%	0.90%	6.90%	-0.98%	
	90020493	6.90%	-3.45%	4.64%	4.74%	3.90%	7.10%	4.71%	
	91854652	6.70%	-4.10%	6.64%	3.46%	4.10%	7.60%	0.88%	
	91703239	6.00%	0.35%	5.60%	4.60%	4.50%	7.40%	-3.39%	
	81663256	2.20%	-4.53%	-4.56%	-8.62%	-2.60%	4.39%	-2.94%	
	82687389	8.40%	-0.06%	1.64%	3.74%	4.10%	9.40%	4.62%	
	84154969	3.00%	-0.15%	5.03%	7.00%	3.21%	8.00%	1.79%	

Uzbekistan	Cambodia	Laos	Georgia	Myanmar	Vietnam	Armenia	Azerbaijan	Georgia	Iraq	Georgia
2.79%	6.32%	5.13%	2.85%	5.94%	4.60%	3.97%	6.27%	-1.59%		
-1.20%	5.83%	4.88%	-0.98%	2.94%	0.56%	2.66%	0.29%	-1.03%		
-0.65%	-2.06%	-1.46%	-4.06%	3.57%	-4.53%	5.33%	-0.56%	2.84%		
8.55%	2.54%	-2.01%	-11.34%	6.02%	-3.70%	5.24%	8.63%	-21.12%		
1.96%	2.44%	14.22%	3.64%	4.69%	7.06%	-8.09%	-2.02%	-8.83%		
-1.90%	0.03%	6.65%	2.91%	5.11%	-6.73%	-10.98%	-13.83%	-1.31%		
-2.02%	5.17%	5.06%	-2.25%	5.45%	-14.37%	0.36%	-21.47%	-62.65%		
-12.50%	4.65%	6.34%	7.96%	8.29%	-43.57%	-21.77%	-45.16%	37.73%		
-3.84%	1.58%	6.71%	4.45%	7.70%	-16.64%	-22.27%	-29.63%	7.37%		
-6.68%	5.69%	8.92%	5.79%	8.45%	2.19%	-18.82%	-10.81%	1.91%		
-2.45%	4.02%	7.84%	5.43%	9.21%	4.79%	-12.07%	2.14%	-2.36%		
0.13%	3.03%	7.68%	4.84%	8.93%	1.97%	3.61%	9.96%	20.69%		
3.57%	3.18%	7.67%	4.03%	7.80%	0.27%	10.03%	10.06%	3.11%		
2.68%	2.59%	4.72%	4.19%	5.43%	3.05%	7.15%	2.39%	23.20%		
2.69%	9.33%	8.11%	9.28%	4.43%	0.06%	12.60%	2.46%	16.42%		
2.18%	6.28%	6.55%	12.01%	6.43%	2.65%	7.39%	1.99%	19.11%		
2.58%	5.64%	6.52%	9.61%	6.56%	6.16%	7.61%	4.22%	17.10%		
2.39%	4.11%	6.66%	10.22%	6.74%	11.33%	9.31%	5.09%	15.35%		
2.58%	6.05%	7.00%	12.00%	6.99%	10.60%	11.66%	10.64%	-15.75%		
5.72%	7.86%	7.79%	11.72%	7.44%	7.14%	11.36%	5.31%	65.05%		
5.34%	10.77%	7.55%	11.74%	7.23%	10.66%	27.76%	9.07%	9.59%		
5.82%	8.08%	-0.18%	11.26%	6.61%	9.78%	35.94%	8.92%	11.23%		
7.81%	7.30%	19.08%	10.18%	6.79%	10.30%	26.33%	12.05%	5.50%		
7.31%	3.78%	8.59%	1.95%	5.37%	3.71%	11.99%	1.94%	15.49%		
6.42%	-0.67%	8.28%	3.26%	5.12%	-16.74%	10.47%	-4.09%	12.16%		
6.81%	0.95%	8.93%	3.51%	6.16%	-0.89%	6.08%	5.74%	14.54%		
6.61%	4.02%	8.81%	4.02%	5.97%	1.53%	1.17%	6.72%	15.70%		
7.10%	7.31%	7.90%	6.49%	5.50%	7.13%	2.20%	6.42%	13.94%		
7.30%	7.36%	7.99%	7.90%	5.55%	3.41%	5.84%	3.62%	7.63%		
6.87%	7.14%	7.57%	8.20%	6.42%	3.61%	2.80%	4.44%	0.74%		
7.22%	7.04%	7.55%	7.47%	6.99%	3.25%	1.05%	3.03%	2.54%		
5.93%	6.86%	7.04%	6.41%	6.69%	0.19%	-3.06%	2.91%	15.20%		
4.39%	7.00%	6.88%	5.75%	6.94%	7.52%	0.15%	4.84%	-3.40%		
5.88%	7.47%	6.22%	6.41%	7.46%	5.23%	1.50%	4.84%	4.70%		
5.98%	7.05%	5.46%	6.75%	7.36%	7.63%	2.48%	4.98%	5.81%		
2.00%	-3.10%	3.28%	3.19%	2.87%	-7.15%	-4.20%	-6.76%	-15.70%		
7.40%	3.03%	3.48%	-17.94%	2.56%	5.68%	5.60%	10.47%	7.68%		
5.67%	5.01%	4.42%	1.97%	8.02%	12.60%	4.65%	10.11%	8.14%		

Syriagdp	Bulgariagdp	Belarusgdp	Czechgdp	Hungarygdp	Moldovagdp	Polandgdp	Romaniagdp	Russiangdp
13.03%	-3.00%	1.08%	2.25%	-2.46%	-9.80%	1.05%	-0.15%	1.49%
-4.94%	2.65%	2.73%	2.76%	1.99%	6.63%	3.18%	1.71%	2.48%
1.92%	0.18%	1.13%	2.32%	1.54%	-0.06%	-1.34%	-2.10%	-0.08%
13.27%	-0.63%	-0.36%	2.07%	1.53%	0.48%	2.21%	-0.25%	1.60%
-8.96%	-1.79%	6.57%	1.72%	-2.24%	7.45%	-1.55%	-3.19%	0.69%
7.64%	-10.91%	-3.42%	1.03%	-6.66%	-3.65%	-9.69%	-10.85%	-2.46%
7.30%	-8.03%	-2.54%	-10.72%	-10.27%	-18.97%	-6.24%	-10.22%	-2.85%
9.24%	-6.92%	-10.64%	0.76%	-1.31%	-30.27%	3.43%	-4.39%	-12.61%
3.53%	-1.10%	-8.67%	1.40%	1.24%	-2.96%	4.65%	4.79%	-6.71%
1.70%	2.22%	-12.72%	4.24%	4.80%	-32.11%	6.07%	7.19%	-10.71%
1.53%	3.33%	-12.12%	7.58%	4.40%	-3.20%	7.93%	10.61%	-1.82%
-0.86%	-8.24%	1.58%	5.59%	2.04%	-7.57%	7.03%	7.37%	-1.39%
-4.81%	-0.28%	10.16%	0.59%	5.42%	-0.17%	7.41%	-0.75%	3.77%
1.63%	4.28%	7.18%	0.97%	6.34%	-8.18%	5.55%	-0.81%	-3.06%
-6.63%	-5.34%	2.16%	2.72%	5.31%	-5.09%	5.61%	2.78%	8.81%
-1.25%	5.65%	4.53%	5.82%	6.32%	0.27%	5.55%	6.47%	12.51%
0.33%	1.71%	3.51%	4.27%	5.88%	4.25%	2.23%	10.09%	7.49%
2.68%	6.64%	3.85%	2.91%	6.63%	5.88%	3.01%	9.47%	7.13%
-5.03%	5.82%	5.77%	4.96%	5.94%	4.73%	4.52%	7.51%	9.79%
3.28%	7.08%	10.15%	6.31%	7.15%	5.44%	6.16%	11.76%	9.61%
2.48%	7.76%	8.19%	7.94%	6.50%	5.58%	4.48%	6.77%	8.74%
2.48%	7.49%	8.73%	8.30%	5.97%	2.91%	7.17%	12.67%	10.51%
4.56%	7.94%	7.38%	6.84%	2.49%	1.15%	8.07%	10.79%	10.97%
0.81%	6.58%	8.96%	4.23%	2.94%	5.92%	5.23%	11.50%	7.67%
1.30%	-3.19%	-0.99%	-3.60%	-4.66%	-7.67%	4.72%	-3.75%	-5.67%
1.34%	1.56%	6.56%	3.56%	2.72%	5.17%	4.61%	2.61%	6.94%
-6.61%	0.36%	4.14%	2.89%	3.95%	4.89%	5.98%	4.35%	6.49%
-26.34%	0.75%	1.63%	-0.79%	-1.25%	-0.59%	1.55%	1.92%	4.02%
-26.30%	-0.56%	0.96%	-0.05%	1.80%	9.04%	0.86%	0.27%	1.76%
-10.31%	0.97%	1.69%	2.26%	4.23%	5.00%	3.84%	4.12%	0.74%
-3.19%	3.43%	-3.83%	5.39%	3.71%	-0.34%	4.38%	3.16%	-1.97%
-5.63%	3.04%	-2.54%	2.54%	2.20%	4.41%	2.95%	2.86%	0.19%
-0.73%	2.76%	2.54%	5.17%	4.27%	4.20%	5.14%	15.36%	1.83%
1.48%	2.68%	3.14%	3.22%	5.36%	4.10%	5.95%	3.75%	2.81%
3.75%	4.04%	1.45%	3.03%	4.86%	3.60%	4.45%	-3.50%	2.20%
-10.50%	-3.96%	-0.67%	-5.50%	-4.54%	-8.30%	-2.02%	5.90%	-2.65%
-2.90%	7.63%	2.27%	3.55%	7.20%	13.95%	6.85%	4.47%	5.62%
-3.50%	3.36%	-4.74%	2.46%	4.58%	-5.58%	4.87%	-0.07%	-2.05%

Slovakia	Ukraine	Estonia	Lithuania	Latvia	Albania	BAH	Serbia	Croatia
4.13%	-0.60%	0.07%	-2.82%	-2.11%	2.64%	1.27%	1.95%	0.10%
4.17%	0.52%	2.75%	3.79%	2.66%	2.62%	3.56%	3.42%	2.80%
2.42%	2.80%	0.98%	2.14%	-0.39%	2.62%	-1.84%	-1.65%	-0.09%
1.96%	0.98%	4.98%	7.98%	4.18%	2.68%	-2.48%	-1.48%	-0.87%
1.01%	4.05%	6.45%	-0.87%	5.41%	2.61%	1.30%	1.71%	-1.58%
-2.49%	-4.49%	-8.25%	-5.67%	0.79%	2.62%	-11.68%	-10.95%	-7.10%
-15.08%	-7.17%	-11.55%	-6.34%	-13.26%	-27.59%	-12.69%	-10.37%	-19.91%
-6.21%	-8.39%	-15.62%	-22.24%	-32.65%	-6.72%	-28.76%	-26.24%	-10.36%
6.89%	-12.76%	-9.88%	-17.29%	-12.13%	10.18%	-31.68%	-28.89%	-6.60%
5.99%	-21.60%	-2.71%	-10.00%	1.38%	9.96%	1.50%	5.00%	7.51%
7.66%	-10.73%	5.19%	5.62%	-1.40%	9.48%	4.70%	8.31%	8.41%
6.51%	-8.50%	3.51%	5.15%	1.58%	7.48%	52.37%	7.45%	6.87%
5.84%	-1.39%	9.77%	8.29%	8.15%	-11.62%	21.37%	9.92%	8.24%
3.79%	-0.26%	2.49%	7.45%	5.71%	9.13%	12.42%	4.77%	3.92%
-0.43%	1.47%	-2.64%	-1.14%	1.83%	13.70%	9.42%	-16.20%	0.40%
1.22%	7.68%	11.35%	3.82%	4.61%	7.78%	3.19%	9.18%	4.25%
2.88%	11.02%	4.65%	6.52%	5.67%	9.28%	1.11%	7.79%	0.41%
4.32%	7.11%	4.35%	6.75%	6.28%	5.80%	3.79%	9.32%	6.39%
5.27%	11.35%	5.72%	10.52%	7.61%	6.65%	2.59%	6.46%	7.20%
5.12%	13.67%	4.54%	6.54%	7.50%	6.47%	4.98%	11.19%	5.71%
6.60%	4.80%	7.66%	7.71%	9.86%	6.29%	2.98%	7.74%	5.81%
8.29%	9.37%	8.49%	7.40%	11.06%	6.45%	4.42%	7.23%	6.44%
10.64%	10.03%	5.89%	11.08%	9.13%	6.26%	4.70%	8.31%	6.82%
5.50%	3.96%	-6.99%	2.62%	-4.35%	7.51%	4.32%	7.72%	3.62%
-5.55%	-13.71%	-16.10%	-14.82%	-14.99%	3.26%	-2.02%	-1.03%	-5.97%
4.95%	1.94%	0.67%	1.63%	-4.55%	3.59%	-0.45%	2.93%	-0.18%
1.87%	7.23%	5.91%	6.04%	5.54%	2.55%	-0.31%	3.39%	1.23%
1.32%	0.22%	3.23%	3.84%	7.04%	1.42%	-0.82%	-0.68%	-2.33%
0.63%	0.00%	1.46%	3.55%	2.01%	1.00%	2.35%	2.89%	-0.40%
2.70%	-6.58%	3.01%	3.54%	1.90%	1.77%	1.15%	-1.59%	-0.43%
5.17%	-9.79%	1.85%	2.02%	3.89%	2.22%	4.31%	1.81%	2.52%
1.94%	2.42%	3.16%	2.52%	2.37%	3.32%	3.24%	3.34%	3.56%
2.94%	2.36%	5.79%	4.28%	3.31%	3.80%	3.24%	2.10%	3.41%
4.03%	3.49%	3.78%	3.99%	3.99%	4.02%	3.83%	4.50%	2.80%
2.52%	3.20%	3.74%	4.63%	2.57%	2.09%	2.89%	4.33%	3.42%
-3.37%	-3.75%	-0.55%	-0.02%	-2.20%	-3.48%	-3.02%	-0.90%	-8.58%
3.01%	3.35%	8.01%	5.98%	4.06%	8.52%	7.39%	7.55%	13.07%
1.67%	-30.26%	-1.29%	1.89%	1.98%	3.74%	3.90%	2.25%	6.33%

Montenegro	Macedonia	Slovenia	Croatia	Bosnia and Herzegovina
0.23%	-0.70%	1.01%	3.16%	2.39%
3.51%	6.98%	3.18%	3.28%	0.09%
-4.52%	-1.27%	-0.58%	3.37%	-2.80%
-3.03%	-3.25%	-2.90%	4.01%	3.21%
0.51%	1.94%	-0.50%	2.11%	0.97%
-8.59%	-10.20%	-8.10%		-4.60%
-6.92%	-5.75%	-8.97%		-14.71%
-20.16%	-6.17%	-5.74%		-13.96%
-35.16%	-7.06%	2.68%		-21.33%
3.17%	-1.35%	5.05%		1.36%
16.27%	-0.64%	7.01%		6.90%
30.68%	1.68%	3.20%		14.03%
9.80%	1.89%	4.62%		5.90%
7.77%	3.79%	2.87%		0.16%
-6.90%	4.86%	4.82%		6.51%
17.51%	5.16%	3.70%		11.03%
6.29%	-2.67%	2.48%		5.84%
7.16%	0.91%	3.42%		3.54%
7.75%	2.80%	2.41%		3.44%
9.98%	5.21%	3.89%		1.38%
9.89%	5.17%	3.56%		3.42%
14.47%	5.54%	5.24%		14.88%
16.69%	6.86%	6.89%		4.01%
12.75%	5.86%	2.46%		6.52%
-0.52%	0.03%	-8.13%		4.07%
7.90%	3.79%	0.72%		5.30%
8.51%	2.74%	0.21%		5.34%
-2.13%	-0.46%	-2.64%		3.01%
4.09%	2.92%	-1.03%		3.27%
2.21%	3.63%	2.77%		3.26%
4.74%	3.86%	2.21%		3.20%
2.80%	2.85%	3.19%		0.19%
4.54%	1.08%	4.82%		1.56%
4.89%	2.88%	4.45%		2.04%
4.06%	3.91%	3.45%		-0.16%
-15.31%	-4.69%	-4.32%		-10.95%
13.04%	3.93%	8.21%		1.25%
6.41%	2.15%	5.37%		1.77%